

787

Quick Reference Handbook

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**QRH Revision Record**
Model Identification**Chapter QRR**
Section ModID**General**

The airplanes listed in the table below are covered in this manual. The numbers are used to distinguish data peculiar to one or more, but not all of the airplanes. Where data applies to all airplanes listed, no reference is made to individual airplane numbers.

The table permits flight crew correlation of configuration differences by Registry Number in alpha/numeric order within an operator's fleet for airplanes covered in this manual. Configuration data reflects the airplane as delivered configuration and is updated for service bulletin incorporations in conformance with the policy stated in the introduction section of this chapter.

Airplane number is supplied by the operator. Registry number is supplied by the national regulatory agency. Serial and tabulation numbers are supplied by Boeing.

Airplane Number	Registry Number	Serial Number	Tabulation Number
787001	ZA78787	7878787	ZZ787



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**QRH Revision Record**
Revision Record**Chapter QRR**
Section RevRec**Revision Transmittal Letter**

To: All holders of The Boeing Company 787 Flight Crew Operations Manual, Boeing Document Number D615Z003 Boeing Proprietary.

Subject: Quick Reference Handbook Revision.

This revision reflects the most current information available to The Boeing Company 45 days before the subject revision date. The following revision highlights explain changes in this revision. General information below explains the use of revision bars to identify new or revised information.

Revision Record

No.	Revision Date	Date Filed
0	Preliminary 12/15/07	

No.	Revision Date	Date Filed

General

The Boeing Company issues flight crew operations manual revisions to provide new or revised procedures and information. Formal revisions also incorporate appropriate information from previously issued flight crew operations manual bulletins.

The revision date is the approximate date the manual is mailed to the customer.

Formal revisions include a Transmittal Letter, a new Revision Record, Revision Highlights, and a current List of Effective Pages. Use the information on the new Revision Record and List of Effective Pages to verify the manual content.

Pages containing revised technical material have revision bars associated with the changed text or illustration. Editorial revisions (for example, spelling corrections) may have revision bars with no associated highlight.

The record above should be completed by the person incorporating the revision into the manual.

Filing Instructions

Consult the List of Effective Pages (CI.3). Pages identified with an asterisk (*) are either replacement pages or new (original) issue pages. Remove corresponding old pages and replace or add new pages. Remove pages that are marked DELETED; there are no replacement pages for deleted pages.

Revision Highlights

This section (0.4) replaces the existing section 0.4 in your manual.

Be careful when inserting changes not to throw away pages from the manual that are not replaced. Using the List of Effective Pages (0.5) can help determine the correct content of the manual.

Throughout the manual, airplane effectivity may be updated to reflect coverage as listed on the Preface - Model Identification page, or to show service bulletin airplane effectivity. Highlights are not supplied.

This manual is published from a database; the text and illustrations are marked with configuration information. Occasionally, because the editors rearrange the database markers, or mark items with configuration information due to the addition of new database content, some customers may receive revision bars on content that appears to be unchanged. Pages may also be republished without revision bars due to slight changes in the flow of the document.



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Normal Checklists

Chapter NC

PREFLIGHT

Oxygen - - - - - Tested, 100%
Flight instruments - - - - - Heading __, Altimeter __
Parking brake - - - - - Set
FUEL CONTROL switches - - - - - CUTOFF

BEFORE START

Flight deck door - - - - - Closed and locked
Passenger signs - - - - - __
MCP - - - - - V2 __, HDG/TRK __, ALT __
Takeoff speeds - - - - - V1 __, VR __, V2 __
CDU preflight - - - - - Completed
Trim - - - - - __ Units, 0
Taxi and takeoff briefing - - - - - Completed
BEACON - - - - - ON

BEFORE TAXI

Anti-ice - - - - - __
Recall - - - - - Checked
Autobrake - - - - - RTO
Flight controls - - - - - Checked
Ground equipment - - - - - Clear



BEFORE TAKEOFF

Flaps ____

AFTER TAKEOFF

Landing gear Up

Flaps Up

DESCENT

Recall Checked

Notes Checked

Autobrake ____

Landing data VREF____, Minimums____

Approach briefing Completed

APPROACH

Altimeters ____

LANDING

Speedbrake Armed

Landing gear Down

Flaps ____



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SHUTDOWN

Hydraulic panel- - - - - Set
Fuel pumps- - - - -Off
Flaps- - - - -Up
Parking brake - - - - - ____
FUEL CONTROL switches- - - - - CUTOFF
Weather radar - - - - -Off

SECURE

IRS - - - - -Off
Emergency lights - - - - -Off
Packs - - - - -Off



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Checklist Introduction

Chapter CI

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Checklist Introduction

Normal Checklists

Chapter CI

Section 1

Introduction

This introduction gives guidelines for use of the printed version of the Normal Checklist (NC) as well as the Electronic Checklist (ECL).

The NC is organized by phase of flight.

The NC is used to verify that critical items have been done.

Normal Checklist Operation

Normal checklists are used after doing all respective procedural items.

The following table shows which pilot calls for the checklist and which pilot reads the checklist. Both pilots visually verify that each item is in the needed configuration or that the step is done. The far right column shows which pilot gives the response. This is different than the normal procedures where the far right column can show which pilot does the step.

Checklist	Call	Read	Verify	Respond
PREFLIGHT	Captain	First officer	Both	Area of responsibility
BEFORE START	Captain	First officer	Both	Area of responsibility
BEFORE TAXI	Captain	First officer	Both	Area of responsibility
BEFORE TAKEOFF	Pilot flying	Pilot monitoring	Both	Pilot flying
AFTER TAKEOFF	Pilot flying	Pilot monitoring	Both	Pilot monitoring
DESCENT	Pilot flying	Pilot monitoring	Both	Area of responsibility
APPROACH	Pilot flying	Pilot monitoring	Both	Area of responsibility
LANDING	Pilot flying	Pilot monitoring	Both	Pilot flying
SHUTDOWN	Captain	First officer	Both	Area of responsibility
SECURE	Captain	First officer	Both	Area of responsibility

If the airplane configuration does not agree with the needed configuration:

- stop the checklist
- complete the respective procedure steps
- continue the checklist

If it becomes apparent that an entire procedure was not done:

- stop the checklist
- complete the entire procedure
- do the checklist from the start

Try to do checklists before or after high work load times. The crew may need to stop a checklist for a short time to do other tasks. If the interruption is short, continue the checklist with the next step. If a pilot is not sure where the checklist was stopped, do the checklist from the start. If the checklist is stopped for a long time, also do the checklist from the start.

After completion of each checklist, the pilot reading the checklist calls,
"_____ CHECKLIST COMPLETE."

Electronic Checklist Operation

Operation with the electronic normal checklist is the same as the printed normal checklist except that, there is no need to read aloud or visually confirm items that are complete (green). For the BEFORE TAKEOFF and LANDING checklists, the PM announces "___ CHECKLIST COMPLETE," the PF visually confirms that the CHECKLIST COMPLETE indication is shown, and announces "CHECKLIST COMPLETE."

Closed loop (sensed) checklist items change from white to green when the action is taken. The PM is responsible to check off any open loop (not sensed) item and to verify that all closed loop items are green. See Chapter 10, Flight Instruments, Displays, for a complete description of the electronic checklist system.

Checklist Content

The checklist has the minimum items needed to operate the airplane safely.

Normal checklists have items that meet any of the following criteria:

- items essential to safety of flight that are not monitored by an alerting system, or
- items essential to safety of flight that are monitored by an alerting system but if not done, would likely result in a catastrophic event if the alerting system fails, or
- needed to meet regulatory requirements, or
- items needed to maintain fleet commonality between the 737, 747-400, 757, 767, 777, and 787, or
- items that enhance safety of flight and are not monitored by an alerting system (for example autobrakes), or
- during shutdown and secure, items that could result in injury to personnel or damage to equipment if not done

Checklist Construction

When a checklist challenge does not end with “switch or lever”, then the challenge refers to system status. For example, “Landing Gear...Down”, refers to the status of the landing gear, not just the position of the lever.

When a checklist challenge ends with “switch or lever”, then the challenge refers to the position of the switch or lever. For example, “FUEL CONTROL switches...CUTOFF” refers to the position of the switches.



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Checklist Introduction

Non-Normal Checklists

Chapter CI

Section 2

Introduction

The Non-Normal Checklists chapter contains checklists used by the flight crew to cope with non-normal situations. The checklists are grouped in logical sections which match the system description chapters in Volume 2. The checklists are in alphabetical order in each section.

Most checklists correspond to an EICAS alert message. An EICAS alert message annunciates a failure condition and is the cue to select and do the checklist.

Checklists without an EICAS alert message (such as DITCHING) are called unannunciated checklists. The title of an unannunciated checklist on the printed checklist is preceded by an italic *u* character. All unannunciated checklists are found in the first section of the printed Non-Normal Checklists chapter. Some unannunciated checklists also appear in the respective systems section (such as FUEL JETTISON in the Fuel section).

A rectangular icon symbol  precedes all EICAS alert messages that have procedural steps, notes, or other information of which the crew must be made aware. The icon is removed from the EICAS message when the checklist has been completed. The title of the printed non-normal checklists also has the  icon to agree with the EICAS message symbology. Alert messages without icons have obvious checklists (such as OVERSPEED). These messages are for information only, or are for ground use only to prevent dispatch.

A condition statement is given for all alert messages. The condition statement briefly describes the condition which caused the message to show. Unannunciated checklists also have condition statements to help in understanding the reason for the checklist.

Checklists can have both recall and reference items. Recall items are critical steps that must be done from memory. Recall items are preceded by the number or pound symbol (#) in the printed non-normal checklists. Reference items are actions to be done while reading the checklist. In the Table of Contents for each non-normal checklist section, the titles of checklists containing recall items are printed in bold type. No identification is made of recall items in the electronic checklist.

Some amplified information is included in brackets [] in the printed non-normal checklist when the reason for an item is not obvious. Information in brackets is also used to identify the checklist from which an item originated.

Electronic Checklist Operation

A non-normal menu is given for the electronic checklist. The menu is arranged in the same manner as the Non-Normal Checklists chapter.

The primary purpose of the non-normal menu is to access unannunciated checklists and condition statements for EICAS alert messages without icons.

The non-normal menu is also used to access checklists with icons to allow review of a specific checklist. Operation of the electronic checklist is described in Chapter 10, Flight Instruments, Displays.

Non-Normal Checklist Operation

Non-normal checklists start with steps to correct the situation or condition. Information for planning the rest of the flight is included. When special items are needed to configure the airplane for landing, the items are deferred to the Approach or Landing checklist. The electronic checklist automatically transfers deferred items to the Approach or Landing checklist. Flight patterns for some non-normal situations are located in the Maneuvers chapter, and show the sequence of configuration changes.

While every attempt is made to provide needed non-normal checklists, it is not possible to develop checklists for all conceivable situations, especially those involving multiple failures. In some unrelated multiple failure situations, the flight crew may combine elements of more than one checklist or exercise judgment to determine the safest course of action. The captain must assess the situation and use good judgment to determine the safest course of action.

There are some situations where the crew must always land at the nearest suitable airport. These situations include, but are not limited to, conditions where:

- the non-normal checklist has the words “Plan to land at the nearest suitable airport”
- cabin smoke or fire persists
- there is only one AC power source (this includes one main engine generator, the APU generator, or the backup power system [both generators])
- any other situation determined by the crew to have a significant adverse effect on safety if the flight is continued

It must be stressed that for persistent smoke or a fire that cannot be positively confirmed to be completely extinguished, the earliest possible descent, landing, and evacuation must be done.

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Checklists prescribing an engine shutdown must be evaluated by the captain to determine whether an actual shutdown, or operation at reduced thrust, is the safest course of action. Consideration must be given to probable effects if the engine is operated at the minimum needed thrust.

There are no non-normal checklists associated with the loss of an engine indication, or with an automatic display of the secondary engine indications. Operate the engine normally unless an EICAS message shows or a limit is exceeded.

Non-normal checklists also assume:

- During engine start and before takeoff, the respective non-normal checklist is done if an EICAS alert message is shown. Upon completion of the checklist, the Dispatch Deviations Guide or operator equivalent is consulted to determine if Minimum Equipment List relief is available
- System controls are in the normal configuration for the phase of flight before the start of the non-normal checklists
- Aural alerts are silenced and the system reset by the flight crew as soon as the cause of the alert is recognized
- The EMERGENCY position of the oxygen regulator is used when needed to supply positive pressure in the masks and goggles to evacuate contaminants. The 100% position of the oxygen regulator is used when positive pressure is not needed, but contamination of flight deck air exists. The NORMAL position of the oxygen regulator is used if prolonged use is needed and the situation allows. Normal boom mic operation is restored when oxygen use is no longer needed
- Indicator lights are tested to verify suspected faults
- Flight crew reset of a tripped circuit breaker in flight is not recommended. However, a tripped circuit breaker may be reset once, after a short cooling period (approximately 2 minutes), if in the judgment of the captain, the situation resulting from the circuit breaker trip has a significant adverse effect on safety. A ground reset of a tripped circuit breaker by the flight crew should only be done after maintenance has determined that it is safe to reset the circuit breaker
- Flight crew cycling (pulling and resetting) of circuit breakers to clear non-normal conditions is not recommended unless directed by a non-normal checklist

Non-Normal Checklist Use

Non-normal checklist use starts when the airplane flight path and configuration are correctly established. Only a few situations need an immediate response (such as a stall warning, ground proximity PULL UP and WINDSHEAR warnings, or a rejected takeoff). Usually, time is available to assess the situation before corrective action is started. All actions must then be coordinated under the captain's supervision and done in a deliberate, systematic manner. Flight path control must never be compromised.

When a non-normal situation occurs, at the direction of the pilot flying, both crewmembers systematically and without delay do all recall items in their areas of responsibility.

The pilot flying calls for the checklist when:

- the flight path is in control
- the airplane is not in a critical phase of flight (such as takeoff or landing)
- all recall items are complete.

For those checklists with only recall items or a combination of recall and reference items, the pilot monitoring first verifies each recall item has been done. The checklist is normally read aloud during this verification. The pilot flying does not need to respond except for items not in agreement with the checklist. With the electronic checklist, there is no need to read aloud or verify items that are complete (green).

Checklist reference items, including the response or action, and any amplifying information, are read aloud by the pilot monitoring. Read aloud as much of the condition statement as needed to verify the selection of the correct checklist. Information appearing in brackets does not need to be read aloud. The pilot flying need not repeat these items, but must acknowledge that the items were heard and understood. Action is taken by the crewmember if the control is in the crewmember's area of responsibility. After moving the control, the crewmember taking the action also states the checklist response.

Both pilots must agree before moving critical controls in flight, such as:

- the thrust lever of a failed engine
- a fuel control switch
- an engine or APU fire switch or cargo fire arming switch
- a generator drive disconnect switch

This does not apply to the DUAL ENG FAIL/STALL checklist

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After moving the control, the crewmember taking the action also states the checklist response. Action is taken by the pilot flying and the pilot monitoring based on the crewmember's area of responsibility. With the airplane stationary on the ground, action is taken by the captain as pilot flying and the first officer as pilot monitoring based on preflight areas of responsibility.

Closed loop (sensed) checklist items change from white to green when the action is taken. The pilot monitoring is responsible to "check off" any open loop (not sensed) item and to verify that all closed loop items are green.

The pilot flying may also direct reference procedures to be done by recall if no hazard is created by such action, or if the situation does not allow reference to a checklist.

Checklists show lists of inoperative equipment only when knowledge of the condition of such equipment is needed for planning the rest of the flight and the condition is not shown on EICAS.

The pilot flying is to be made aware when there are deferred items. These items may be delayed until the usual point during approach or landing.

Consequential EICAS alert messages can show as the result of a primary failure condition (such as AUTO SPEEDBRAKE from HYD PRESS SYS C) or the result of a non-normal checklist crew action (such as PACK L from SMOKE AIR CONDITIONING). Consequential messages are shown for crew awareness with icons automatically removed. The checklists for the consequential messages can have procedural steps, notes, or other information. The crew does not do these checklists. If consequential checklist steps, notes, and information are applicable to the primary failure condition, then these are included in the primary checklist. The statement "Do not accomplish the following checklists:" followed by a list of the checklists, shows in the printed primary checklist to inform the crew of consequential checklists. All consequential alert messages may not show while doing the primary checklist, depending on operational circumstances. The statement "Inhibited checklists:" followed by the list, shows in the electronic checklist. The statement need not be read aloud when doing the electronic checklist.

Following completion of the applicable non-normal checklist items, normal checklists are used to verify that the configuration is correct for each phase of flight.

Pilots must be aware that checklists cannot be made for all conceivable situations and are not intended to replace good judgment. In some conditions, deviation from checklists may, at the captain's discretion, be needed.

The following symbol shows that the checklist is complete.



Each checklist has a checklist complete symbol at the end.

The checklist complete symbol can also be in the body of the checklist. This occurs only when a checklist divides into two or more paths. Each path can have a checklist complete symbol. The checklist complete symbol shows the end of the applicable path. The crew need not continue the checklist after that point.

Following completion of each non-normal checklist, the pilot monitoring states:

“_____ CHECKLIST COMPLETE.” When a non-normal checklist is complete except for the deferred items, and the normal checklist to which the items have been deferred has not yet been done, the pilot monitoring states:

“_____ CHECKLIST COMPLETE EXCEPT FOR DEFERRED ITEMS.”



Non-Normal Checklists
Unannunciated Checklists

Chapter NNC
Section 0

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***U* ABORTED ENGINE START L, R**

Condition: **During a ground start, an abort engine start condition occurs.**

FUEL CONTROL switch. CUTOFF

START selector. START

Motor the engine for 30 seconds.

START selector. NORM



U DITCHING

Condition: **Ditching is needed.**

Transponder. 7700

Mayday. Transmit
Report: aircraft type, position, heading, altitude and
airspeed

Advise the cabin to prepare for ditching.

If at least one engine is running, plan to do the following:

Jettison fuel as needed to increase buoyancy.

[Improved ability to control touchdown.]

Ditch before fuel starvation.

Ditch with landing gear up.

Use flaps 30 and VREF30 on final approach.

Touchdown on the upwind side and parallel to waves and swells.

If both engines are failed, plan to do the following:

Ditch with landing gear up.

Maintain flaps up maneuvering speed + 20 knots until required to extend flaps for ditching.

If flaps are extended, use flaps 20 and VREF30 + 20 knots on final approach.

[Flaps 20 required for controllability in flare when both engines are out.]

Touchdown on the upwind side and parallel to waves and swells.

Life vests. On

Seat belts and shoulder harnesses. On

GND PROX GEAR OVRD OVRD

Continued on next page

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GND PROX TERR OVRD. OVRD**Review the following touchdown techniques:**

- Use thrust, if available, to maintain a 200-300 fpm descent rate until the start of flare
- Flare for a minimum descent rate at touchdown
- Rotate smoothly to a 10-12 degree attitude at touchdown

Notes. Check**Do not accomplish the following checklists:**

- Descent
- Approach
- Landing

When below 5000 feet:**Passenger signs. On****PACK switches (Both). OFF**

[Depressurizes the airplane which allows the exits to open.]

OUTFLOW VALVE switches (Both). MAN**Outflow valve MANUAL switches (Both). CLOSE****Hold until the outflow valves are fully closed.**

[Prevents water from entering airplane through outflow valve.]

Ditching position. Transmit**Review the after touchdown procedure:**

- Set both FUEL CONTROL switches to CUTOFF
- Override and pull the APU fire switch
- Advise the cabin to evacuate.

Do not accomplish the following checklists:

- OUTFLOW VLV L
- OUTFLOW VLV R
- PACK L
- PACK R
- CABIN ALTITUDE AUTO
- PACK L+R

Continued on next page

Continued from previous page

When on final approach:

Landing gear lever. UP

Flaps. 20 or 30

Advise the cabin to brace for touchdown.



U DUAL ENG FAIL/STALL

Condition: **Engine speed for both engines is below idle.**

**# FUEL CONTROL switches
(Both). CUTOFF, then RUN**

[Attempts to clear stall condition and allows engines to be put into start mode.]

**# RAM AIR TURBINE switch. PUSH
Push and hold for 1 second.**

[Backs up automatic deployment of the RAT.]

Airspeed. Above 250 KTS

[Ensures best windmill start capability.]

**APU selector
(If APU available). START, then ON**

[Backs up automatic APU start.]

Engines may accelerate to idle very slowly, especially at high altitudes. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N3 is steadily increasing and EGT remains within limits, the start is progressing normally. Any further cycling of the fuel control switches will result in longer start times.

When HEAT PITOT L+R+C message no longer displayed:

**PRIMARY FLIGHT COMPUTERS
DISCONNECT switch. DISC, then AUTO**

[Restores flight control normal mode following reversion to secondary mode caused by loss of pitot heat.]

Autopilot can be re-engaged when flight control normal mode is restored.



U ENG IN-FLIGHT START L, R

Condition: **Engine start is needed after a shutdown with no fire or apparent damage.**

If X-START is shown:

START selector.START

[Allows electrical power to starter.]

FUEL CONTROL switch.RUN

If an abort start condition as listed in normal procedures occurs:

FUEL CONTROL switch. CUTOFF

START selector. NORM

If X-START is not shown:

FUEL CONTROL switch.RUN

If an abort start condition as listed in normal procedures occurs:

FUEL CONTROL switch. CUTOFF

Engine may accelerate to idle very slowly, especially at high altitudes. The time from fuel control switch to RUN to stabilized idle may be as long as two and a half minutes. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N2 is steadily increasing and EGT remains within limits, the start is progressing normally.

If engine starts and runs normally:

GND PROX FLAP OVRD.OFF

TRANSPONDER MODE selector. TA/RA



U ENG LIM/SURGE/STALL L, R

Condition: **(TBD)**





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U

ENG SVR DAMAGE/SEP L, R

Condition: **Engine has severe damage, vibration, or has separated.**

Autothrottle ARM switch.OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever.Close

[Assists in recognition of affected engine.]

FUEL CONTROL switch. CUTOFF

ENGINE FIRE switch.PULL

[Shuts off fuel and hydraulic fluid.]

If high airframe vibration occurs and continues after engine shutdown:

Without delay, reduce airspeed and descend to a safe altitude which results in an acceptable vibration level. If high vibration returns and further airspeed reduction and descent is not practical, increasing the airspeed may reduce the vibration.

**APU selector
(If APU available).START, then ON**

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector.TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Continued on next page

Continued from previous page

If landing using flaps 30 (performance permitting): (continued)

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



***U* FIRE ENG TAILPIPE L, R**

Condition: **An engine tailpipe fire is reported on the ground with no engine fire warning.**

FUEL CONTROL switch (Affected engine). CUTOFF

Advise the cabin.

If electrical is available:

START selector (Affected engine). START

Advise the tower.

If the engine is being motored:

Continue to motor until the tailpipe fire is extinguished.

START selector (Affected engine). NORM



U FUEL JETTISON

Condition: **Fuel jettison is required.**

FUEL JETTISON ARM switch. ARMED

If desired FUEL TO REMAIN different than displayed FUEL TO REMAIN:

FUEL TO REMAIN selector. Pull, set manually

FUEL JETTISON NOZZLE switches (Both). ON

Do not accomplish the following checklists:

- FUEL PUMP CENTER L**
- FUEL PUMP CENTER R**

When fuel jettison complete:

FUEL JETTISON NOZZLE switches (Both). OFF

FUEL JETTISON ARM switch. OFF





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U FUEL LEAK

Condition: An in flight fuel leak is suspected or confirmed.

One or more of the following may be evidence of a fuel leak:

- visual observation of fuel spray from strut/engine
- excessive engine fuel flow
- total fuel quantity decreasing at an abnormal rate
- FUEL DISAGREE message on EICAS
- FUEL IMBALANCE message on EICAS
- FUEL QTY LOW message on EICAS
- INSUFFICIENT FUEL message on EICAS.

CENTER FUEL PUMP switches (Both). OFF

CROSSFEED switch. OFF

BALANCE switch. OFF

Identify an engine fuel leak by observing a left or right main fuel tank quantity decreasing faster than the other.

An increase in fuel imbalance of approximately 1000 lbs or more in 30 minutes should be considered a fuel leak.

Conditions permitting, visually check for engine fuel leak.

If left and right main tank quantities decrease at same rate:

Fuel leak may be from center tank to center wing area. Do not use center tank fuel.

If FUEL DISAGREE message displayed:

PROGRESS page 2/4. Select

CALCULATED. Select Use

Use CALCULATED to determine gross weight.

Destination fuel estimate. Check

Verify adequate fuel available in left and right main tanks to complete flight.

Continued on next page

Continued from previous page

If engine fuel leak confirmed:

Autothrottle ARM switch
(Affected engine). **OFF**

[Allows thrust lever to remain where manually positioned.]

Thrust lever (Affected engine). **Close**

[Assists in recognition of affected engine.]

FUEL CONTROL switch
(Affected engine). **CUTOFF**

APU selector
(If APU available). **START, then ON**

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. **TA ONLY**

[Prevents climb commands which can exceed single engine performance capability.]

If FUEL DISAGREE message displayed:

PROGRESS page 2/4. **Select**

TOTALIZER. **Select Use**

Use TOTALIZER to determine fuel remaining.

Note: After engine shutdown, all remaining fuel can be used for the operating engine. Resume normal fuel management procedures. Plan to balance fuel when the FUEL IMBALANCE message is displayed.

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. **OVRD**

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Continued on next page

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If FUEL QTY LOW message displayed:

CROSSFEED switch.ON

[From FUEL QTY LOW checklist. Ensures fuel is available to both engines if the low tank empties.]

FUEL PUMP switches (All).ON

[From FUEL QTY LOW checklist. Ensures all fuel is available for use.]

Plan to land at the nearest suitable airport.

[From FUEL QTY LOW checklist.]

Note: Use flaps 20 and VREF20 for landing.

[From FUEL QTY LOW checklist. Provides improved elevator authority in event of dual engine flameout.]

GND PROX FLAP OVRD. OVRD

Note: Avoid high nose up attitude and excessive acceleration or deceleration.

[From FUEL QTY LOW checklist. Prevents forward pumps from uncovering.]

Do not complete the following checklist:

FUEL QTY LOW



***U* GEAR LEVER LOCKED DN**

Condition: **Landing gear lever cannot be positioned to UP.**

Landing gear lever lock

OVERRIDE switch. PUSH and hold

Landing gear lever. UP





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U LOSS OF ALL DISPLAYS

Condition: **All 5 forward displays and both HUDs are blank.**

Autopilot. Engage

Note: LNAV and VNAV modes are not operational. Use Heading/Track Select and HOLD, VS/FPA and Altitude HOLD.

Note: Use the ISFD to maintain the flight plan.

CCR RESET R switch. Push
[Attempts to restore partial display operation.]

Wait 1 minute.

If partial display operation restored:

Note: Four forward displays restored. Lower display and both HUDs remain blank.



If all forward displays remain blank:

CCR RESET switches (both). Push

Wait 1 minute.

If partial or full display operation restored:

Note: The lower display unit and both HUDs remain blank.



If all forward displays remain blank:

Avoid icing conditions.

[Window heat and wing anti-ice are inoperative.]

Use alternate navigation.

Left TCP. Enter desired waypoint

Note: Alternate NAV function resides in left TCP.

HDG REF switch. TRUE

[TCP backup navigation uses a true track reference.]

Continued on next page

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If all forward displays remain blank: (continued)

HDG/TRK reference switch.TRK

Heading/Track selector. Enter desired track

Heading/Track Select (SEL) switch. Push

Note: Monitor the left TCP and update MCP selected track as necessary.

Note: The following systems or displays may be inoperative:

- Primary displays are all inoperative, including crew alert messages
- Autothrottle
- Flight Management Computer, LNAV and VNAV
- Normal landing gear extension and retraction
- Center tank fuel pumps and center tank fuel scavenge leading to trapped fuel
- Automatic cabin pressure control - defaults to 8000 ft cabin
- Passenger oxygen deployment
- First officer's and observer's audio control
- Voice communication on HF or SATCOM
- Window heat and wing anti-ice

-----**DEFERRED ITEMS**-----

==> APPROACH CHECKLIST

Landing gear lever.DN

**ALTN GEAR switch.DOWN and hold
Push and hold for 1 second.**



***U* OVERWEIGHT LANDING**

Condition: **A landing at greater than maximum landing weight is required.**

Refer to Landing Climb Limit Weight table in Performance-Inflight chapter.

If landing gross weight greater than LANDING CLIMB LIMIT WEIGHT:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.





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U SMOKE/FUMES AIR COND

Condition: **A concentration of air conditioning smoke/fumes are identified.**

Oxygen masks and smoke goggles
(If required).ON

Crew communications
(If required). Establish

Recirculation FANS switches (Both). Off
[Removes fans as a possible source of smoke/fumes. Stops recirculation of smoke/fumes and increases fresh air flow.]

If smoke/fumes continue:

Left PACK switch. OFF
[Removes left side of the air conditioning system as a possible source of smoke/fumes.]

Left TRIM AIR switch Off
[Removes left side of the trim air system as a possible source of smoke/fumes.]

Do not accomplish the following checklists:
PACK L
TRIM AIR L

If smoke/fumes continue:

Left PACK switch. AUTO
[Restores left side of the air conditioning system.]

Continued on next page

Continued from previous page

If smoke/fumes continue: (continued)

If smoke/fumes continue: (continued)

Left TRIM AIR switchON

[Restores left side of the trim air system.]

Right PACK switch.OFF

[Removes right side of the air conditioning system as a possible source of smoke/fumes.]

Right TRIM AIR switchOff

[Removes right side of the trim air system as a possible source of smoke/fumes.]

Do not accomplish the following checklists:

PACK R

TRIM AIR R

If smoke/fumes are persistent:

Plan to land at the nearest suitable airport.



U

SMOKE/FUMES/FIRE ELEC

Condition: **Electrical smoke/fumes/fire is identified.**

Oxygen masks and smoke goggles
(If required). **On**

Crew communications
(If required). **Establish**

Recirculation FANS switches (Both). **Off**
[Removes fans as a possible source of smoke/fumes. Stops recirculation of smoke/fumes and increases fresh air flow.]

If smoke/fumes/fire source known:

Electrical power (Affected equipment). **Remove**
If practical, remove power from affected equipment
by switch or circuit breaker in flight deck or cabin.

If smoke/fumes/fire persists or source unknown:

IFE/PASS SEATS switch. **OFF**

CABIN/UTILITY power switch. **OFF**

Plan to land at the nearest suitable airport.





Intentionally
Blank

U

SMOKE/FUMES REMOVAL

Condition: **Smoke/fumes removal is required.**

Oxygen masks and smoke goggles
(If required). **On**

Crew communications
(If required). **Establish**

Flight deck door. **Close**
[Prevents smoke/fumes from penetrating onto the flight deck.]

Recirculation FANS switches (Both). **Off**
[Stops recirculation of smoke/fumes and increases fresh air flow.]

EQUIPMENT COOLING switch. **OFF**
[Attempts to discharge smoke/fumes overboard by using the equipment cooling override mode.]

Note: After 30 minutes of operation at low altitude and low cabin differential pressure, non-critical electronic equipment may fail.
[From EQUIP CLG OVRD FWD checklist.]

Do not accomplish the following checklist:
EQUIP CLG OVRD FWD

If most smoke/fumes are in cabin forward of mid-wing and outflow valve manual control available:

AFT OUTFLOW VALVE switch. **MAN**

AFT OUTFLOW VALVE MANUAL switch. **CLOSE**

Position outflow valve fully closed.
[Concentrates smoke/fumes in the forward part of the airplane and attempts to remove it through the forward outflow valve.]

Do not accomplish the following checklist:
OUTFLOW VALVE AFT



Continued on next page

Continued from previous page

If most smoke/fumes are in cabin aft of mid-wing and outflow valve manual control available:

FORWARD OUTFLOW VALVE switch. MAN

**FORWARD OUTFLOW VALVE MANUAL
switch. CLOSE**

Position outflow valve fully closed.

[Concentrates smoke/fumes in the aft part of the airplane and attempts to remove it through the aft outflow valve.]

Do not accomplish the following checklist:

OUTFLOW VALVE FWD



U TAT PROBE ICING

Condition: Airplane TAT probe or engine TAT probe icing is suspected.

One or more of the following may be evidence of TAT probe icing:

- autothrottle disconnects and Reference/Target TPR and Reference TPR are blanked
- thrust lever stagger with TPR aligned
- TPR stagger with thrust levers aligned
- decrease or increase in Reference/Target TPR at constant altitude and speed
- unable to achieve maximum continuous or maximum climb rating with thrust levers fully advanced
- TAT remains constant and near 0 degrees C during climb, cruise, or descent.

Autothrottle DISCONNECT switch. PUSH

[Allows thrust levers to remain where manually positioned.]

Thrust levers (Both). Retard to MID position

[Prevents exceeding thrust limits when switching to alternate mode.]

EEC MODE switches (Both). OFF

Push one switch at a time.

[Ensures both engines operate in alternate mode.]

Thrust levers (Both). Set

[Maintains desired airspeed and rate of climb or descent.]

Note: Maximum thrust limiting is not available in alternate mode. Alternate thrust setting information is displayed on the N1 indications.

If TAT remains constant and near 0 degrees C:

Airplane TAT probe icing is suspected:

- Reference/Target N1 and Reference N1 may be temporarily unreliable
- Observe EGT caution limits
- Maximum N1 may be temporarily unreliable.

Continued on next page

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If TAT remains constant and near 0 degrees C: (continued)

Do not accomplish the following checklists:

ENG LIMIT PROT L

ENG LIMIT PROT R

If TAT is not near 0 degrees C:

Engine TAT probe icing is suspected:

- Airplane TAT is reliable
- Reference/Target N1 and Reference N1 are reliable if displayed
- Maximum N1 is reliable if displayed.

Autothrottle may be re-engaged when:

- Reference/Target N1 and Reference N1 are displayed
- TAT is not near 0 degrees C.

Do not accomplish the following checklists:

ENG EEC MODE L

ENG EEC MODE R



U VOLCANIC ASH

Condition: **Static discharge around the windshield, bright glow in the engine inlets, smoke or dust on the flight deck, or acrid odor indicates the airplane is in volcanic ash.**

Exit volcanic ash as quickly as possible. Consider a 180 degree turn.

Oxygen masks and smoke goggles
(If required). On

Crew communications (If required). Establish

Autothrottle DISCONNECT switch. PUSH
[Allows thrust levers to remain where manually positioned.]

Thrust levers.Close
Conditions permitting, operate engines at idle.
[Reduces possible engine damage and/or flameout by decreasing EGT.]

ENGINE ANTI-ICE selectors (Both).ON
[Increases bleed air extraction to improve engine stall margin.]

Recirculation FANS switches (Both).OFF
[Increases bleed air extraction to improve engine stall margin by putting packs into high flow.]

APU selector
(If APU available).START, then ON
[Provides an electrical power source in the event one or both engines flame out.]

Note: Volcanic ash can cause non-normal system reactions such as:

- **engine malfunctions, increasing EGT, engine stall or flameout**
- **decrease or loss of airspeed indications**
- **EQUIP COOLING OVRD indications**
- **FIRE CARGO FWD or AFT indications.**

Continued on next page

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If engines flamed out or stalled, or EGT rapidly approaching or exceeding limit:

FUEL CONTROL switches

(Both). CUTOFF, then RUN

[Attempts to clear stall condition and allows engines to be put into start mode.]

RAM AIR TURBINE switch. PUSH

Push and hold for 1 second.

[Backs up automatic deployment of the RAT.]

For a dual engine in-flight start, do not manually abort the start if EGT indication turns red. Autostart allows EGT to exceed the in-flight start limit, however autostart aborts the start before EGT becomes too high for continued engine operation.

Airspeed. Above (TBD) KTS

[Ensures best windmill start capability.]

Engines may accelerate to idle very slowly, especially at high altitudes. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N3 is steadily increasing and EGT remains within limits, the start is progressing normally. Any further cycling of the fuel control switches will result in longer start times.

When HEAT PITOT L+C+R message no longer displayed:

PRIMARY FLIGHT COMPUTERS

DISCONNECT switch. DISC, then AUTO

[Restores flight control normal mode following reversion to secondary mode caused by loss of pitot heat.]

Autopilot can be re-engaged when flight control normal mode is restored.

Plan to land at the nearest suitable airport.



U WINDOW DAMAGE L, R

Condition: **Arcing, delamination, shattered, or cracked condition of any flight deck window is observed.**

If forward window arcing, shattered, or cracked:

Forward WINDOW HEAT switch (Affected window). OFF

[Removes electrical power to prevent arcing.]

BACKUP WINDOW HEAT switch (Affected side). . . . OFF

Do not accomplish the following checklist:

WINDOW HEAT FWD

If damaged window deforms, or air leak is observed:

Plan to land at the nearest suitable airport.

If airplane altitude above 10,000 feet:

Descend to lowest safe altitude or 10,000 feet, whichever is higher.

[Minimizes forces on the window. Outward cabin pressure differential counters inward dynamic air load on the window.]

Sustained flight below 10,000 feet is not recommended due to greater risk of bird strike.





Non-Normal Checklists

Chapter NNC

Airplane General, Emer. Equip., Doors, Windows

Section 1

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Intentionally
Blank

U AUTOMATIC UNLOCK

Condition: **Flight deck door auto unlock light illuminated indicates correct emergency access code has been entered and flight deck door is programmed to automatically unlock after a time delay.**

FLIGHT DECK DOOR ACCESS selector. DENY

Rotate and hold for 1 second.

[Prohibits unauthorized access to flight deck.]



CREW OXYGEN LOW

Condition: **Crew oxygen pressure is low.**



DOOR AFT, FWD CARGO

Condition: **Cargo door is not closed and latched and locked.**

LANDING ALTITUDE

selector.PULL, set 8000

[Reduces cabin differential pressure to decrease risk of door separation.]

If airplane altitude at or below 8000 feet:

LEVEL OFF.Initiate

Level off at lowest safe altitude.

If airplane altitude above 8000 feet:

Descent.Initiate

**Descend to lowest safe altitude or 8000 feet,
whichever is higher.**

[Reduces cabin differential pressure.]

After level off, allow sufficient time for cabin altitude to stabilize.

[Minimizes passenger discomfort from effects of cabin depressurization.]

OUTFLOW VALVE switches (Both). MAN

OUTFLOW VALVE MANUAL switches (Both). OPEN

**Position outflow valves fully open to depressurize
airplane.**

Do not accomplish the following checklists:

CABIN ALTITUDE AUTO

LANDING ALTITUDE



DOOR AFT, FWD E/E ACCESS

Condition: **Electrical and electronic access door is not closed and latched.**

Note: The door is in a safe configuration as long as cabin pressurization is normal.

[Positive cabin differential pressure ensures door remains in place.]



DOOR BULK CARGO

Condition: **Bulk cargo door is not closed and latched.**

Note: The door is in a safe configuration as long as cabin pressurization is normal.

[Positive cabin differential pressure ensures door remains in place.]



DOOR ENTRY 1–4L, R

Condition: **Entry door is not closed and latched and locked.**

Note: The door is in a safe configuration as long as cabin pressurization is normal.

[Positive cabin differential pressure ensures door remains in place.]



DOOR FD OVHD

Condition: **Flight deck overhead door is not closed and latched.**

Note: The door is in a safe configuration as long as cabin pressurization is normal.

[Positive cabin differential pressure ensures door remains in place.]



DOOR FWD ACCESS

Condition: **Forward access door is not closed and latched.**

Note: The door is in a safe configuration as long as cabin pressurization is normal.

[Positive cabin differential pressure ensures door remains in place.]



DOORS

Condition: **Two or more doors are not closed and latched and locked.**



ELT ON

Condition: **Emergency locator transmitter is on.**



EMER LIGHTS

Condition: **Emergency lighting system has been manually activated or emergency lights switch is OFF.**



EVAC COMMAND

Condition: **Evacuation command signal has been activated.**



[J]FD DOOR AUTO UNLOCK

Condition: **Flight deck door emergency access code has been entered. Door will unlock after a time delay.**

FLIGHT DECK DOOR ACCESS selector. DENY

Rotate and hold for 1 second.

[Prevents unauthorized access to flight deck.]



[J]FD DOOR LOCK FAIL

Condition: **Flight deck door lock has failed, or lock bolt is not engaged, or door power is OFF.**

FLIGHT DECK DOOR POWER switch

(Conditions permitting). OFF

[Removes electrical power to prevent possible lock overheat.]

Door can be locked with deadbolt.



FD DOOR OPEN

Condition: **Flight deck security door is open.**



U LOCK FAIL

Condition: **Flight deck door lock fail light illuminated indicates flight deck door lock has failed, or flight deck access system switch is OFF.**

FLIGHT DECK DOOR POWER switch
(Conditions permitting). OFF

[Removes electrical power to prevent possible lock overheat.]

Door can be locked with deadbolt.



PASS OXYGEN ON

Condition: **Passenger oxygen system is activated.**



U WINDOW DAMAGE L, R

Condition: **Arcing, delamination, shattered, or cracked condition of any flight deck window is observed.**

If forward window arcing, shattered, or cracked:

FORWARD WINDOW HEAT

switch (Affected window). OFF

[Removes electrical power to prevent arcing.]

BACKUP WINDOW HEAT switch (Affected side). . . . OFF

Do not accomplish the following checklist:

WINDOW HEAT FWD

If damaged window deforms, or air leak is observed:

Plan to land at the nearest suitable airport.

If airplane altitude above 10,000 feet:

Descend to lowest safe altitude or 10,000 feet, whichever is higher.

[Minimizes forces on the window. Outward cabin pressure differential counters inward dynamic air load on the window.]

Sustained flight below 10,000 feet is not recommended due to greater risk of bird strike.



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CABIN ALTITUDE

Condition: **Cabin altitude is excessive.**

OXYGEN masks. On

Crew communications. Establish

CABIN ALTITUDE and RATE. Check

[Confirms pressurization problem.]

If cabin altitude uncontrollable:

Passenger OXYGEN switch. PUSH

Push and hold for 1 second.

[Backs up automatic activation of the passenger oxygen system.]

Descent. Accomplish

Without delay, close thrust levers, extend speedbrakes, and descend at VMO/MMO. Level off at lowest safe altitude or 10,000 feet, whichever is higher. If structural integrity is in doubt, limit airspeed and avoid high maneuvering loads.





CABIN ALTITUDE AUTO

Condition: Automatic pressurization control has failed or both outflow valve switches are in manual.

OUTFLOW VALVE switches (Both). MAN

OUTFLOW VALVE MANUAL switches
(Both). OPEN/CLOSE

Position as required to control desired cabin rate and altitude.

Note: Recommended cabin rate is approximately 500 FPM for climbs and descents.

Note: Recommended cabin altitude in cruise is:

FLIGHT LEVEL	CABIN ALTITUDE
Up to 230	Landing Field Elevation
260	2000
300	4000
350	6000
400 and above	8000

-----DEFERRED ITEMS-----

==> APPROACH CHECKLIST

When at pattern altitude:

OUTFLOW VALVE MANUAL switches
(Both). OPEN

Push and hold for 30 seconds or until outflow valve positions indicate fully open.



CABIN TEMPERATURE

Condition: Flight deck or cabin temperature zone(s) are or may become dangerously hot or cold.



CARGO A/C FWD

Condition: **Forward cargo air conditioning is inoperative.**



CARGO HEAT BULK

Condition: **Bulk cargo heat is inoperative or BULK CARGO TEMP switch is OFF.**

Note: Temperature in bulk cargo compartment will remain above freezing.



CARGO HEAT FWD

Condition: **Forward cargo heat is inoperative.**

Note: Temperature in forward cargo compartment will remain above freezing.



EQUIP CLG OVRD AFT, FWD

Condition: **Equipment cooling system is in override mode.**

Note: After 30 minutes of operation at low altitude and low cabin differential pressure, non-critical electronic equipment may fail.



EQUIP COOLING AFT

Condition: **Aft equipment cooling system is inoperative.**

Plan to land at the nearest suitable airport.

Note: After 30 minutes, non-critical electronic equipment may fail.



EQUIP COOLING FWD

Condition: **Forward equipment cooling system is inoperative.**

Plan to land at the nearest suitable airport.

Note: After 30 minutes, non-critical electronic equipment and EFB displays may fail.



EQUIP OVBD VLV AFT

Condition: **Aft equipment cooling system overboard exhaust valve is failed in the open position on the ground.**



EQUIP OVRD VLV AFT, FWD

Condition: **Equipment cooling system override valve is failed in the open position on the ground.**



[]LANDING ALTITUDE

Condition: **FMC has failed to provide a landing altitude or landing altitude selector is pulled.**

Landing altitude selector. PULL, set manually



LIQUID CLG QTY L, R

Condition: **Liquid cooling system quantity is low.**



LIQUID COOLING L

Condition: A fault is detected in the left liquid cooling system.

Note: Cooling for the large motor power system may not be available. The following items may be inoperative:

- left pack
- L center fuel pump
- R demand hydraulic pump
- C1 hydraulic pump

Note: Equipment may become operative or inoperative later in the flight. Leave all controls in the normal positions.

Do not accomplish the following checklists:

PACK L

FUEL PUMP CTR L

HYD PRESS C1

HYD PRESS DEM R



LIQUID COOLING R

Condition: A fault is detected in the right liquid cooling system.

Note: Cooling for the large motor power system may not be available. The following items may be inoperative:

- right pack
- R center fuel pump
- L demand hydraulic pump
- C2 hydraulic pump

Note: Equipment may become operative or inoperative later in the flight. Leave all controls in the normal positions.

Do not accomplish the following checklists:

PACK R

FUEL PUMP CTR R

HYD PRESS C2

HYD PRESS DEM L



OUTFLOW VALVE AFT, FWD

Condition: **Automatic control has failed or outflow valve switch is in MAN.**

OUTFLOW VALVE switch. MAN

OUTFLOW VALVE MANUAL switch. CLOSE

Push and hold for 30 seconds or until outflow valve position indicates fully closed.

[Allows the other outflow valve to take full control of cabin pressure.]



PACK L, R

Condition: **Pack is shut down.**

Wait 2 minutes.

[Allows time for any overheat condition to cool.]

Air conditioning RESET switch. PUSH

Push and hold for 1 second.



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[PACK L+R]

Condition: **Both packs are shut down.**

Oxygen Masks. On

Crew Communications. Establish

Descent. Accomplish

Level off at lowest safe altitude or 10,000 feet, whichever is higher. If flight conditions require, accomplish descent with speedbrakes extended at VMO/MMO to prevent excessive cabin altitude.

Plan to land at the nearest suitable airport.

After message displayed, wait minimum of 2 minutes.

Air conditioning RESET switch. PUSH

Push and hold for 1 second.

Wait 2 minutes.

If PACK L+R message displayed again:

Continue with descent.

In-flight entertainment system/passenger SEATS power switch. OFF

[Removes electrical power to minimize cabin heating.]

CABIN/UTILITY power switch. OFF

[Removes electrical power to minimize cabin heating.]

Flight deck lighting. Set

Minimize flight deck lighting intensity to reduce heat.

Flight deck sunshades/sunvisors. Install

[Reduces flight deck heating during daylight operations.]

Continued on next page

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If PACK L+R message displayed again: (continued)

When at level off:

VENTILATION switch.ALTN

[Provides source of outside air ventilation.]

OUTFLOW VALVE switches (Both). MAN

FORWARD OUTFLOW VALVE MANUAL switch. OPEN

Position outflow valve to 9 o'clock position.

[Maximizes air ventilation to cabin.]

AFT OUTFLOW VALVE MANUAL switch. OPEN

Position outflow valve fully open.

[Maximizes air ventilation to cabin.]

Flight deck door.Open

[Improves flight deck ventilation.]

Flight deck overhead door vent valve. Open (if required)

[Reduces flight deck temperature.]

Instruct flight attendants to adjust cabin windows to full dark during daylight operations.

Do not accomplish the following checklist:

CABIN ALTITUDE AUTO

DOOR AJAR



[PACK MODE L, R

Condition: **Pack is operating in standby mode.**

Note: At lower altitudes and/or higher outside air temperatures, the pack may shut down.

[Standby mode is not capable of providing cool conditioned air during low altitude, high outside air temperature conditions.]



RECIRC FAN LWR, UPR OFF

Condition: **Recirculation fan switch is OFF.**



[TRIM AIR L, R

Condition: **Trim air is shut off.**

Wait 2 minutes.

[Allows time for any overheat condition to cool.]

Air conditioning RESET switch. PUSH
Push and hold for 1 second.



VENTILATION ALTN

Condition: **Ventilation system is in the alternate mode.**





Non-Normal Checklists

Chapter NNC

Anti-Ice, Rain

Section 3

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[] ANTI-ICE DET WING

Condition: **Wing ice detection has failed.**

Note: Operate wing anti-ice systems manually.



[] ANTI-ICE ENG L, R

Condition: **Engine anti-ice valve remains closed when commanded open.**

ENGINE ANTI-ICE selector. OFF, then ON

[Attempts to manually reset controller and open the valve.]

If ANTI-ICE ENG message no longer displayed:

Operate affected engine anti-ice system manually.

[Automatic anti-ice is not available for affected engine.]

If ANTI-ICE ENG message remains displayed:

ENGINE ANTI-ICE selector. OFF

[Enables display of ICING ENG message if encountering icing conditions.]

Note: Avoid icing conditions.

[Anti-ice is not available to affected engine because the engine anti-ice system has failed.]



ANTI-ICE LEAK ENG L, R

Condition: High temperature anti-ice bleed air leak is detected in the affected engine.

The engine anti-ice system automatically isolates heat source within approximately 2 minutes by closing the engine anti-ice valve.

Note: Avoid icing conditions.

[Anti-ice is not available to affected engine.]

If, after 2 minutes, ANTI-ICE LEAK ENG message remains displayed:

Autothrottle ARM switch. OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever. Retard

Retard slowly until the ANTI-ICE LEAK ENG message is no longer displayed or the thrust lever is closed.

[Reduces flow of bleed air through the leak.]

Do not accomplish the following checklist:

AUTOTHROTTLE

If ANTI-ICE LEAK ENG message remains displayed:

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Note: Operate engine at idle for remainder of flight.

If landing using flaps 20:

GND PROX FLAP OVRD OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.



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If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.



ANTI-ICE LOSS ENG L, R

Condition: Anti-ice bleed air for the affected engine is no longer available.



[] ANTI-ICE ON

Condition: Any anti-ice selector is ON, air temperature is above 10 degrees C, and ice is not detected.

ENGINE ANTI-ICE selectors (Both). AUTO or OFF

WING ANTI-ICE selector. AUTO



[] ANTI-ICE PACKS

Condition: Dual failure. Avoid icing conditions.



[] ANTI-ICE WING

Condition: More than one zone on an individual wing has failed. WAI is degraded.

WING ANTI-ICE selector. OFF

[Enables ICING WING message to be displayed if icing conditions are encountered.]



[] HEAT PITOT C

Condition: Center pitot probe heat is inoperative.

Note: Standby air data is unreliable in icing conditions.



[]HEAT PITOT L

Condition: **Left pitot probe heat is inoperative.**

Note: ADRF air data is not affected for a single pitot heat failure. Ensure that right air data/attitude source switch remains AUTO.

**[]HEAT PITOT L+C+R**

Condition: **All pitot probe heats are inoperative.**

Note: Air data is unreliable in icing conditions.

**[]HEAT PITOT R**

Condition: **Right pitot probe heat is inoperative.**

Note: ADRF air data is not affected for a single pitot heat failure. Ensure that left air data/attitude source switch remains AUTO.

**[]ICE DETECTORS**

Condition: **Ice detection has failed.**

Note: Operate engine and wing anti-ice systems manually.

**ICING ENG**

Condition: **Ice is detected and one or both engine anti-ice selectors are OFF.**



OVERHEAT WING

Condition: **Overheat is detected in the wing anti-ice system.**

WING ANTI-ICE selector. OFF

Note: Avoid icing conditions.

[Anti-ice is not available to the wing because the wing anti-ice system has failed.]



U TAT PROBE ICING

Condition: Airplane TAT probe or engine TAT probe icing is suspected.

One or more of the following may be evidence of TAT probe icing:

- autothrottle disconnects and Reference/Target N1 and Reference N1 are blanked
- thrust lever stagger with N1 aligned
- N1 stagger with thrust levers aligned
- decrease or increase in Reference/Target N1 at constant altitude and speed
- unable to achieve maximum continuous or maximum climb rating with thrust levers fully advanced
- TAT remains constant and near 0 degrees C during climb, cruise, or descent.

One or more of the following may be evidence of TAT probe icing:

- autothrottle disconnects and Reference/Target TPR and Reference TPR are blanked
- thrust lever stagger with EPR aligned
- EPR stagger with thrust levers aligned
- decrease or increase in Reference/Target TPR at constant altitude and speed
- unable to achieve maximum continuous or maximum climb rating with thrust levers fully advanced
- TAT remains constant and near 0 degrees C during climb, cruise, or descent.

Autothrottle DISCONNECT switch. PUSH

[Allows thrust levers to remain where manually positioned.]

Thrust levers (Both). Retard to MID position

[Prevents exceeding thrust limits when switching to alternate mode.]

EEC MODE switches (Both). OFF

Push one switch at a time.

[Ensures both engines operate in alternate mode.]

Continued on next page

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Thrust levers (Both).Set

[Maintains desired airspeed and rate of climb or descent.]

Note: Maximum thrust limiting is not available in alternate mode. Alternate thrust setting information is displayed on the N1 indications.

If TAT remains constant and near 0 degrees C:

Airplane TAT probe icing is suspected:

- Reference/Target N1 and Reference N1 may be temporarily unreliable
- Observe EGT caution limits
- Maximum N1 may be temporarily unreliable.

Do not accomplish the following checklists:

ENG LIMIT PROT L

ENG LIMIT PROT R

If TAT is not near 0 degrees C:

Engine TAT probe icing is suspected:

- Airplane TAT is reliable
- Reference/Target N1 and Reference N1 are reliable if displayed
- Maximum N1 is reliable if displayed.

Autothrottle may be re-engaged when:

- Reference/Target N1 and Reference N1 are displayed
- TAT is not near 0 degrees C.

Do not accomplish the following checklists:

ENG EEC MODE L

ENG EEC MODE R



WINDOW HEAT

Condition: Two or more window heats are off.



WINDOW HEAT L, R FWD

Condition: **Primary window heat for the affected forward window is off.**

FORWARD WINDOW HEAT

switch.OFF 10 seconds, then ON

[Attempts to reset system.]

If WINDOW HEAT FWD message remains displayed:

FORWARD WINDOW HEAT switch.OFF

[Removes power to prevent arcing. Window is defogged by the backup system.]



WINDOW HEAT L, R SIDE

Condition: **Primary window heat for the affected side window is off.**

SIDE WINDOW HEAT

switch.OFF 10 seconds, then ON

[Attempts to reset system.]

If WINDOW HEAT SIDE message remains displayed:

SIDE WINDOW HEAT switch.OFF

[Removes power to prevent arcing.]



WING ANTI-ICE OFF

Condition: **Wing anti-ice selector is in the off position.**





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AUTOPILOT

Condition: **Autopilot is operating in a degraded mode. Engaged roll and/or pitch mode may have failed, or the autopilot has entered envelope protection.**

AUTOPILOT DISC

Condition: **Autopilot has disconnected.**

AUTOTHROTTLE DISC

Condition: **Both autothrottles have disconnected.**

AUTOTHROTTLE L, R

Condition: **Affected autothrottle has failed or is OFF.**

Autothrottle ARM switch (Affected side). OFF

Other autothrottle may be re-engaged if desired.

NO AUTOLAND

Condition: **Autoland is not available.**

NO AUTOLAND GLS

Condition: **Autoland is not available for a GLS approach.**

NO AUTOLAND ILS

Condition: **Autoland is not available for an ILS approach.**

NO LAND 3

Condition: **Autoland system does not have redundancy for triple channel autoland.**

T/O THRUST DISAGREE

Condition: **Thrust management takeoff thrust calculation disagrees with EEC takeoff thrust calculation.**



**Non-Normal Checklists
Communications**

**Chapter NNC
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DATALINK LOST

Condition: **Datalink is temporarily lost.**



DATALINK SYS

Condition: **Datalink system has failed.**



HF DATALINK

Condition: **HF datalink has failed.**



[] RADIO TRANSMIT CAPT, F/O, OBS

Condition: **One of the affected crewmember's MIC switches is commanding a radio to transmit for 30 seconds or more.**

TRANSMITTER SELECT switches (Affected panel) FLT

Select flight interphone on the affected crewmember's audio control panel.

[Affected crewmember's stuck microphone will be transmitting on the flight interphone, rather than broadcasting over a radio.]

The affected audio control panel transmitter selectors should remain on FLT.



SATCOM

Condition: **SATCOM system has failed.**



SATCOM DATALINK

Condition: **SATCOM datalink has failed.**



SATCOM VOICE

Condition: **SATCOM** voice communication has failed.



SATVOICE LOST

Condition: **SATCOM** voice communication is temporarily lost.



VHF DATALINK

Condition: **VHF** datalink has failed.





Non-Normal Checklists

Chapter NNC

Electrical

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ELEC AC BUS L1, L2, R1, R2

Condition: **AC bus is unpowered.**

Generator control switch. OFF, then ON
Attempt only one reset.

Do not accomplish the following checklist:
ELEC GEN OFF

If ELEC AC BUS message remains displayed:

APU selector
(If APU available). START, then ON
 [Provides an additional source of electrical power for wing anti-ice.]

Do not accomplish the following checklist:
TBD



ELEC BATTERY OFF

Condition: **Battery switch is OFF.**



ELEC CABIN/UTIL OFF

Condition: **Cabin/utility power switch is OFF.**



[ELEC GEN DRIVE L1, L2, R1, R2]

Condition: **Generator drive oil pressure is low.**

DRIVE disconnect switch. PUSH

Push and hold for 1 second.

[Prevents engine gearbox damage.]

APU selector

(If APU available). START, then ON

[Provides an additional source of electrical power.]

Do not accomplish the following checklist:

ELEC GEN OFF



[ELEC GEN OFF APU L, R]

Condition: **APU generator control breaker is open when commanded closed.**

APU generator switch. OFF, then ON

Attempt only one reset.



[ELEC GEN OFF L1, L2, R1, R2]

Condition: **Generator control breaker is open.**

Generator control switch. OFF, then ON

Attempt only one reset.

If ELEC GEN OFF message remains displayed:

APU selector

(If APU available). START, then ON

[Provides an additional source of electrical power.]



ELEC IFE/SEATS OFF

Condition: **In-flight entertainment system/passenger seats power switch is OFF.**



ELEC STANDBY SYS

Condition: **A fault is detected in the standby power system.**



MAIN BATTERY DISCH

Condition: **Main battery is discharging or hot battery bus is unpowered.**



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U ABORTED ENGINE START L, R

Condition: **During a ground start, an abort engine start condition occurs.**

FUEL CONTROL switch. CUTOFF

START selector. START

Motor the engine for 30 seconds.

START selector. NORM



[APU LIMIT

Condition: **APU operation has exceeded a limit.**

APU selector. OFF



[APU SHUTDOWN

Condition: **APU has automatically shut down.**

APU selector. OFF

[Attempts to reset APU controller.]

APU selector. START, then ON



U DUAL ENG FAIL/STALL

Condition: **Engine speed for both engines is below idle.**

FUEL CONTROL switches

(Both). CUTOFF, then RUN

[Attempts to clear stall condition and allows engines to be put into start mode.]

RAM AIR TURBINE switch. PUSH

Push and hold for 1 second.

[Backs up automatic deployment of the RAT.]

Airspeed. Above 250 KTS

[Ensures best windmill start capability.]

APU selector

(If APU available). START, then ON

[Backs up automatic APU start.]

Engines may accelerate to idle very slowly, especially at high altitudes. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N3 is steadily increasing and EGT remains within limits, the start is progressing normally. Any further cycling of the fuel control switches will result in longer start times.

When HEAT PITOT L+R+C message no longer displayed:

PRIMARY FLIGHT COMPUTERS

disconnect switch. DISC, then AUTO

[Restores flight control normal mode following reversion to secondary mode caused by loss of pitot heat.]

Autopilot can be re-engaged when flight control normal mode is restored.



[]ENG AUTOSTART L, R

Condition: **Autostart has failed to start the engine.**

FUEL CONTROL switch. CUTOFF
START selector. NORM



ENG CONTROL L, R

Condition: **Fault is detected in the affected engine control system.**



[]ENG EEC MODE L, R

Condition: **Control for the affected engine is operating in alternate mode.**

Autothrottle DISCONNECT switch. PUSH
 [Allows thrust levers to remain where manually positioned.]

Thrust lever (Both). Retard to MID position
 [Prevents exceeding thrust limits when switching to alternate mode.]

EEC MODE switches (Both). OFF
Push one switch at a time.
 [Ensures both engines operate in alternate mode.]

Autothrottle ENGAGE switch. PUSH

Note: Maximum thrust limiting is not available with autothrottle disconnected. Alternate thrust setting information is displayed on N1 indications.

Do not accomplish the following checklist:
ENG EEC MODE (for other engine)



[ENGINE FAIL L, R]

Condition: **Engine speed is below idle.**

Autothrottle ARM switch.OFF
[Allows thrust lever to remain where manually positioned.]

Thrust lever.Close
[Assists in recognition of affected engine.]

Engine will automatically attempt to relight and recover to running.

If N2 is less than 30% or is not steadily increasing after 1 minute:

FUEL CONTROL switch. CUTOFF
[Shuts off fuel and allows engine to be put into start mode.]

Restart may be attempted if no apparent damage.

If restart desired:

If X-START is shown:

START selector.START
[Allows electrical power to starter.]

FUEL CONTROL switch.RUN

If an abort start condition as listed in normal procedures occurs:

FUEL CONTROL switch. CUTOFF
START selector. NORM

If X-START is not shown:

FUEL CONTROL switch.RUN

If an abort condition as listed in normal procedures occurs:

FUEL CONTROL switch. CUTOFF

Continued on next page

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If restart desired: (continued)

Engine may accelerate to idle very slowly, especially at high altitudes. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N3 is steadily increasing and EGT remains within limits, the start is progressing normally. Any further cycling of the fuel control switches will result in longer start times.

If engine remains failed or is damaged:

FUEL CONTROL switch. CUTOFF

[Returns switch to proper position in case of unsuccessful start attempt.]

START selector. NORM

[Returns selector to proper position in case of unsuccessful start attempt.]

APU selector

(If APU available). START, then ON

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



ENG FUEL FILTER L, R

Condition: **An impending fuel filter bypass condition exists on the affected engine.**

Note: Erratic engine operation and flameout may occur due to fuel contamination.



ENG FUEL VALVE L, R

Condition: **Engine fuel or spar valve position disagrees with commanded position.**

If ENG FUEL VALVE message is displayed when the fuel control switch is positioned to CUTOFF, the engine may continue to run for approximately 1 minute.

If on the ground:

Engine START selector. NORM

[Prevents overheat of starter because it may not be receiving fuel for cooling if engine fuel valve failed to open.]

Do not attempt engine start.

[Prevents possibility of not being able to shutdown the engine if both valves subsequently fail open.]



ENG IDLE DISAGREE

Condition: **One engine is at approach idle and the other engine is at minimum idle.**



U ENG IN-FLIGHT START L, R

Condition: **Engine start is needed after a shutdown with no fire or apparent damage.**

If X-START is shown:

START selector. START

[Allows electrical power to starter.]

FUEL CONTROL switch. RUN

If an abort start condition as listed in normal procedures occurs:

FUEL CONTROL switch. CUTOFF

START selector. NORM

If X-START is not shown:

FUEL CONTROL switch. RUN

If an abort start condition as listed in normal procedures occurs:

FUEL CONTROL switch. CUTOFF

Engine may accelerate to idle very slowly, especially at high altitudes. The time from fuel control switch to RUN to stabilized idle may be as long as two and a half minutes. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N2 is steadily increasing and EGT remains within limits, the start is progressing normally.

If engine starts and runs normally:

GND PROX FLAP OVRD. OFF

TRANSPONDER MODE selector. TA/RA



ENG LIMIT EXCEED L, R

Condition: **Engine indications are exceeding limits.**

Autothrottle ARM switch. OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever. Retard

Retard until ENG LIMIT EXCEED message is no longer displayed or the thrust lever is closed.

[Stabilizes air flow through engine.]

If ENG LIMIT EXCEED message remains displayed:

FUEL CONTROL switch. CUTOFF

APU selector

(If APU available). START, then ON

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If indications stabilized/EGT decreasing:

Thrust lever. Advance

Advance slowly. Check that RPM and EGT follow thrust lever movement.

[Attempts to restore normal control of engine operation.]

Operate engine normally or at a reduced thrust level which is surge and stall free.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

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If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



[] ENG LIMIT PROT L, R

Condition: **Engine control is operating in alternate mode and commanded N1 exceeds maximum N1.**

Thrust lever.Retard

Retard until N1 remains within appropriate limits.



***U* ENG LIM/SURGE/STALL L, R**

Condition: **(TBD)**



[] ENG LPC ANTI-ICE L, R

Condition: **Engine anti-ice capability is degraded.**

Note: Avoid icing conditions.



ENG OIL FILTER L, R

Condition: Affected engine oil filter contamination has caused filter bypass.

Autothrottle ARM switch. OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever. Retard

Retard slowly until ENG OIL FILTER message is no longer displayed or the thrust lever is closed.

[Decrease oil pressure in an attempt to stop filter bypass.]

If ENG OIL FILTER message remains displayed:

FUEL CONTROL switch. CUTOFF

APU selector

(If APU available). START, then ON

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



[] ENG OIL PRESS L, R

Condition: **Engine oil pressure is low.**

Autothrottle ARM switch. OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever. Retard

Retard slowly until ENG OIL PRESS message is no longer displayed or the thrust lever is closed.

[Stabilizes air flow through engine.]

If ENG OIL PRESS message remains displayed:

FUEL CONTROL switch. CUTOFF

APU selector

(If APU available). START, then ON

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



ENGINE OIL TEMP L, R

Condition: **Engine oil temperature is high.**

Autothrottle ARM switch. OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever. Move to MID Position

[Allows oil to cool.]

If oil temperature above red line limit:

Thrust lever. Close

FUEL CONTROL switch. CUTOFF

**APU selector
(If APU available). START, then ON**

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



ENG REV AIR/GND

Condition: **Thrust reverser control system has lost air/ground protection to prevent inadvertent deployment in flight.**



[]ENG REV COMMANDED L, R

Condition: **Reverse thrust lever is not stowed in flight.**



ENG REV LIMITED L, R

Condition: **Fault is detected that may prevent normal deployment of the affected reverser.**

Note: Engine thrust reverser will not deploy or reverse thrust will be limited to idle on landing.



ENG REVERSER L, R

Condition: **Fault is detected in the affected engine reverser system.**



ENG RPM LIMITED L, R

Condition: **Engine control is limiting affected engine thrust to prevent N1, N2 or N3 from exceeding the RPM operating limit.**



ENG SHUTDOWN

Condition: **Both engines were shutdown on the ground by the fuel control switches or fire switches.**



ENG SHUTDOWN L, R

Condition: **Engine was shutdown by the fuel control switch or fire switch.**



□ENG START CUTOFF L, R

Condition: **Start selector remains in START when engine is running.**

START selector. NORM



□ENG START INHIBIT

Condition: **Engine starter operation is inhibited because the starter duty cycle has expired.**

FUEL CONTROL switch. CUTOFF

START selector. NORM

Engine starters are inhibited for a 30 minute cooldown period.

Do not accomplish the following checklist:

ENG AUTOSTART



ENG STARTERS L, R

Condition: **Engine starters are inoperative.**

Ground or in-flight start using electric starters may be unsuccessful.

If on the ground:

FUEL CONTROL switch. CUTOFF

START selector. NORM

[Prevents engine from unexpectedly rotating if starter function is regained.]

If in flight:

START selector. NORM

[Prevents engine from unexpectedly rotating if starter function is regained.]

Increase airspeed until X-START is no longer displayed.

[Assures sufficient air flow for windmill start.]



ENG SURGE L, R

Condition: **(TBD)**



U ENG SVR DAMAGE/SEP L, R

Condition: **Engine has severe damage, vibration, or has separated.**

Autothrottle ARM switch. OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever. Close

[Assists in recognition of affected engine.]

FUEL CONTROL switch. CUTOFF

ENGINE FIRE switch. PULL

[Shuts off fuel and hydraulic fluid.]

If high airframe vibration occurs and continues after engine shutdown:

Without delay, reduce airspeed and descend to a safe altitude which results in an acceptable vibration level. If high vibration returns and further airspeed reduction and descent is not practical, increasing the airspeed may reduce the vibration.

**APU selector
(If APU available). START, then ON**

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

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Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



ENG THRUST L, R

Condition: Engine is producing greater than or less than commanded thrust.





U VOLCANIC ASH

Condition: **Static discharge around the windshield, bright glow in the engine inlets, smoke or dust on the flight deck, or acrid odor indicates the airplane is in volcanic ash.**

Exit volcanic ash as quickly as possible. Consider a 180 degree turn.

**Oxygen masks and smoke goggles
(If required).ON**

Crew communications (If required). Establish

Autothrottle DISCONNECT switch. PUSH
[Allows thrust levers to remain where manually positioned.]

Thrust levers.Close
Conditions permitting, operate engines at idle.
[Reduces possible engine damage and/or flameout by decreasing EGT.]

ENGINE ANTI-ICE selectors (Both).ON
[Increases bleed air extraction to improve engine stall margin.]

Recirculation FANS switches (Both).OFF
[Increases bleed air extraction to improve engine stall margin by putting packs into high flow.]

**APU selector
(If APU available).START, then ON**
[Provides an electrical power source in the event one or both engines flame out.]

- Note: Volcanic ash can cause non-normal system reactions such as:**
- engine malfunctions, increasing EGT, engine stall or flameout
 - decrease or loss of airspeed indications
 - EQUIP COOLING OVRD indications
 - FIRE CARGO FWD or AFT indications.

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If engines flamed out or stalled, or EGT rapidly approaching or exceeding limit:

FUEL CONTROL switches

(Both). CUTOFF, then RUN

[Attempts to clear stall condition and allows engines to be put into start mode.]

RAM AIR TURBINE switch. PUSH

Push and hold for 1 second.

[Backs up automatic deployment of the RAT.]

For a dual engine in-flight start, do not manually abort the start if EGT indication turns red. Autostart allows EGT to exceed the in-flight start limit, however autostart aborts the start before EGT becomes too high for continued engine operation.

Airspeed. Above (TBD) KTS

[Ensures best windmill start capability.]

Engines may accelerate to idle very slowly, especially at high altitudes. Slow acceleration may be incorrectly interpreted as a hung start or engine malfunction. If N3 is steadily increasing and EGT remains within limits, the start is progressing normally. Any further cycling of the fuel control switches will result in longer start times.

When HEAT PITOT L+C+R message no longer displayed:

PRIMARY FLIGHT COMPUTERS

DISCONNECT switch. DISC, then AUTO

[Restores flight control normal mode following reversion to secondary mode caused by loss of pitot heat.]

Autopilot can be re-engaged when flight control normal mode is restored.

Plan to land at the nearest suitable airport.



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BOTTLE 1, 2 DISCH ENG

Condition: **Engine fire extinguisher bottle 1 or bottle 2 pressure is low.**



BOTTLE DISCH APU

Condition: **APU fire extinguisher bottle pressure is low.**



BOTTLE DISCH CARGO

Condition: **Both rapid discharge cargo fire extinguisher bottle pressures are low.**



[]DET FIRE APU

Condition: **APU fire detection is inoperative.**

If APU not running:

Note: Do not start the APU unless use is required.



If APU running:

Plan to shut down the APU as soon as practical.



DET FIRE CARGO AFT, FWD

Condition: **Affected cargo compartment smoke detection is inoperative.**



[]DET FIRE ENG L, R

Condition: **A fault is detected in the engine fire detection system.**

Note: Engine fire detection may not be available. An engine fire warning which occurs is valid.



▣ FIRE APU

Condition: **Fire is detected in the APU.**

APU fire switch. Pull, rotate, and hold
Rotate to the stop and hold for 1 second.

If the FIRE APU message stays shown:

Plan to land at the nearest suitable airport.

Do not accomplish the following checklist:

APU SHUTDOWN



▣ FIRE CARGO AFT

Condition: **Smoke is detected in the aft cargo compartment.**

AFT CARGO FIRE ARM switch. ARMED

CARGO FIRE discharge switch. PUSH
Push and hold for 1 second.

Landing altitude selector. PULL, set 8000

[Minimizes extinguisher agent leakage out of the compartment.]

Plan to land at the nearest suitable airport.

Note: Aft equipment cooling normal mode is inoperative.
After 30 minutes of operation at low altitude and low
cabin differential pressure, non-critical electronic
equipment may fail.

[From EQUIP CLG OVRD AFT checklist.]

Do not accomplish the following checklist:

EQUIP CLG OVRD AFT

LANDING ALTITUDE

When at top of descent:

Landing altitude selector. PUSH

[Restores automatic selection of the FMC landing altitude.]

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When at top of descent: (continued)

Note: After landing, inform ground personnel NOT to open the cargo door until all passengers and crew have exited the airplane and fire fighting equipment is nearby.



[FIRE CARGO FWD]

Condition: **Smoke is detected in the forward cargo compartment.**

FORWARD CARGO FIRE ARM switch. ARMED

CARGO FIRE discharge switch. PUSH

Push and hold for 1 second.

Landing altitude selector. PULL, set 8000

[Minimizes extinguisher agent leakage out of the compartment.]

Plan to land at the nearest suitable airport.

Note: Forward equipment cooling normal mode is inoperative. After 30 minutes of operation at low altitude and low cabin differential pressure, non-critical electronic equipment may fail.

[From EQUIP CLG OVRD FWD checklist.]

Do not accomplish the following checklists:

EQUIP CLG OVRD FWD

LANDING ALTITUDE

When at top of descent:

LANDING ALTITUDE selector. PUSH

[Restores automatic selection of the FMC landing altitude.]

Note: After landing, inform ground personnel NOT to open the cargo door until all passengers and crew have exited the airplane and fire fighting equipment is nearby.





[FIRE ENG L, R

Condition: **Fire is detected in the engine.**

Autothrottle ARM switch. OFF
[Allows thrust lever to remain where manually positioned.]

Thrust lever. Close
[Assists in recognition of affected engine.]

FUEL CONTROL switch. CUTOFF

ENGINE FIRE switch. Pull
[Shuts off fuel and hydraulic fluid and arms the fire extinguishing system.]

If FIRE ENG message remains displayed:

ENGINE FIRE switch. Rotate
Rotate to the stop and hold for 1 second.

If, after 30 seconds, FIRE ENG message remains displayed:

ENGINE FIRE switch. Rotate to Other Bottle
Rotate to the stop and hold for 1 second.

APU selector (If APU available). START, then ON
[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY
[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

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If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Do not accomplish the following checklist:

AUTOTHROTTLE



***U* FIRE ENG TAILPIPE L, R**

Condition: An engine tailpipe fire is reported on the ground with no engine fire warning.

FUEL CONTROL switch (Affected engine). CUTOFF

Advise the cabin.

If electrical is available:

START selector (Affected engine). START

Advise the tower.

If the engine is being motored:

Continue to motor until the tailpipe fire is extinguished.

START selector (Affected engine). NORM



[FIRE WHEEL WELL]

Condition: **Fire is detected in a main wheel well.**

Observe gear EXTEND limit speed (270K/.82M).

Landing gear lever. DN
[Attempts to remove and extinguish the fire source.]

Plan to land at the nearest suitable airport.

Flight with gear down increases fuel consumption and decreases climb performance. Refer to Gear Down performance tables in Performance–Inflight chapter for flight planning.

If landing gear must be retracted for airplane performance:

When FIRE WHEEL WELL message no longer displayed:

Wait 20 minutes.
[Attempts to ensure that wheel well fire is extinguished.]

Landing gear lever. UP
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[OVERHEAT ENG L, R]

Condition: **Overheat is detected in the nacelle.**

ENGINE ANTI-ICE selector. OFF

[Attempts to stop the flow of engine anti-ice bleed air through the leak.]

Autothrottle ARM switch. OFF

[Allows thrust lever to remain where manually positioned.]

Thrust lever. Retard

Retard slowly until OVERHEAT ENG message is no longer displayed or the thrust lever is closed.

[Reduces temperature in the nacelle.]

If OVERHEAT ENG message no longer displayed:

Note: Operate engine at the reduced thrust level for remainder of flight.

If OVERHEAT ENG message remains displayed:

FUEL CONTROL switch. CUTOFF

APU selector

(If APU available). START, then ON

[Provides an additional source of electrical power.]

TRANSPONDER MODE selector. TA ONLY

[Prevents climb commands which can exceed single engine performance capability.]

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

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Do not accomplish the following checklist:

AUTOTHROTTLE



[] SMOKE EQUIP CLG AFT

Condition: **Smoke is detected in the aft equipment cooling system.**

Wait 2 minutes.

[Allows time for any smoke in the system to clear.]

AFT EQUIPMENT COOLING switch. OFF, then AUTO

[Attempts reset of the aft equipment cooling system.]

If SMOKE EQUIP CLG AFT message remains displayed:

Plan to land at the nearest suitable airport.

Note: After 30 minutes of operation at low altitude and low cabin differential pressure, non-critical electronic equipment may fail.



[] SMOKE EQUIP CLG FWD

Condition: **Smoke is detected in the forward equipment cooling system.**

Wait 2 minutes.

[Allows time for any smoke in the system to clear.]

FWD EQUIPMENT COOLING switch. OFF, then AUTO

[Attempts reset of the forward equipment cooling system.]

If SMOKE EQUIP CLG FWD message remains displayed:

Plan to land at the nearest suitable airport.

Note: After 30 minutes of operation at low altitude and low cabin differential pressure, non-critical electronic equipment may fail.





[SMOKE EQUIP CLG MISC

Condition: **Smoke is detected in the miscellaneous equipment cooling system in the cabin overhead area.**

Note: All power is removed from the affected equipment for the remainder of the flight. Broadband services and IFE are inoperative. Smoke detection is no longer accurate after SMOKE EQUIP CLG MISC is initially displayed.

If smoke increases or is persistent in the cabin:

Plan to land at the nearest suitable airport.



***U* SMOKE/FUMES AIR COND**

Condition: **A concentration of air conditioning smoke/fumes are identified.**

**Oxygen masks and smoke goggles
(If required).ON**

**Crew communications
(If required). Establish**

Recirculation FANS switches (Both).OFF
[Removes fans as a possible source of smoke/fumes. Stops recirculation of smoke/fumes and increases fresh air flow.]

If smoke/fumes continue:

Left PACK switch.OFF
[Removes left side of the air conditioning system as a possible source of smoke/fumes.]

Left TRIM AIR switchOFF
[Removes left side of the trim air system as a possible source of smoke/fumes.]

Do not accomplish the following checklists:

- PACK L**
- TRIM AIR L**

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If smoke/fumes continue: (continued)

If smoke/fumes continue:

Left PACK switch. AUTO

[Restores left side of the air conditioning system.]

Left TRIM AIR switch ON

[Restores left side of the trim air system.]

Right PACK switch. OFF

[Removes right side of the air conditioning system as a possible source of smoke/fumes.]

Right TRIM AIR switch OFF

[Removes right side of the trim air system as a possible source of smoke/fumes.]

Do not accomplish the following checklists:

PACK R

TRIM AIR R

If smoke/fumes are persistent:

Plan to land at the nearest suitable airport.





U SMOKE/FUMES/FIRE ELEC

Condition: **Electrical smoke/fumes/fire is identified.**

Oxygen masks and smoke goggles
(If required).ON

Crew communications
(If required). Establish

Recirculation FANS switches (Both).OFF
[Removes fans as a possible source of smoke/fumes. Stops recirculation of smoke/fumes and increases fresh air flow.]

If smoke/fumes/fire source known:

Electrical power (Affected equipment). Remove
If practical, remove power from affected equipment
by switch or circuit breaker in flight deck or cabin.

If smoke/fumes/fire persists or source unknown:

IFE/PASS SEATS switch.OFF

CABIN/UTILITY POWER switch.OFF

Plan to land at the nearest suitable airport.



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U SMOKE/FUMES REMOVAL

Condition: **Smoke/fumes removal is required.**

Oxygen masks and smoke goggles
(If required). **ON**

Crew communications
(If required). **Establish**

Flight deck door. **Close**
[Prevents smoke/fumes from penetrating onto the flight deck.]

Recirculation FANS switches (Both). **OFF**
[Stops recirculation of smoke/fumes and increases fresh air flow.]

Equipment COOLING switch. **OFF**
[Attempts to discharge smoke/fumes overboard by using the equipment cooling override mode.]

Note: After 30 minutes of operation at low altitude and low cabin differential pressure, non-critical electronic equipment may fail.

[From EQUIP CLG OVRD FWD checklist.]

Do not accomplish the following checklist:
EQUIP CLG OVRD FWD

If most smoke/fumes are in cabin forward of mid-wing and outflow valve manual control available:

AFT OUTFLOW VALVE switch. **MAN**

AFT OUTFLOW VALVE MANUAL switch. **CLOSE**

Position outflow valve fully closed.
[Concentrates smoke/fumes in the forward part of the airplane and attempts to remove it through the forward outflow valve.]

Do not accomplish the following checklist:
OUTFLOW VALVE AFT



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If most smoke/fumes are in cabin aft of mid-wing and outflow valve manual control available:

FORWARD OUTFLOW VALVE switch. MAN

FORWARD OUTFLOW VALVE MANUAL switch. CLOSE

Position outflow valve fully closed.

[Concentrates smoke/fumes in the aft part of the airplane and attempts to remove it through the aft outflow valve.]

Do not accomplish the following checklist:

OUTFLOW VALVE FWD



SMOKE LAVATORY

Condition: **Smoke is detected in one or more lavatories.**



SMOKE LAV/COMPT

Condition: **Smoke is detected in one or more lavatories, or other enclosed compartment.**



[] SMOKE REST UPR DR 1

Condition: **Smoke is detected in door 1 upper crew rest compartment.**

Cabin crew communications. Establish

If smoke is persistent:

Plan to land at the nearest suitable airport.



SMOKE REST UPR DR 4

Condition: **Smoke is detected in door 4 upper crew rest compartment.**

Cabin crew communications. Establish

If smoke is persistent:

Plan to land at the nearest suitable airport.



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[] AUTO SPEEDBRAKE

Condition: **A fault is detected in the automatic speedbrake system.**

Note: Do not arm speedbrake lever.

[Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.



[] CRUISE FLAPS SYS

Condition: **The cruise flaps system is in the incorrect position.**

Note: Do not use FMC fuel predictions.



[] FLAPS DRIVE

Condition: **Flap drive mechanism has failed.**

Do not use alternate flaps.

[Asymmetry protection is not provided in alternate mode.]

GND PROX FLAP OVRD OVRD

Note: Do not use FMC fuel predictions with flaps extended.

If flap position UP or 1:

Note: Position flap lever to 1 and use VREF30 + 40 for landing.

If flap position between 1 and 5 or at 5:

Note: Use current flaps and VREF30 + 40 for landing.

If flap position between 5 and 20:

Note: Use current flaps and VREF30 + 20 for landing.

If flap position 20 or greater:

Note: Use current flaps and VREF20 for landing.

Do not accomplish the following checklist:

FLAPS PRIMARY FAIL



□FLAPS PRIMARY FAIL

Condition: **Flaps are operating in secondary mode.**

GND PROX FLAP OVRD OVRD

Note: Plan additional time for slower flap operation.

Note: Use flaps 20 and VREF20 for landing.

[Provides improved go-around performance.]



□FLAP/SLAT CONTROL

Condition: **Flap/slat electronics units are inoperative.**

If flap retraction required:

Flap lever. UP

[Allows maximum speed indication on PFD to increase to VMO/MMO.]

Alternate FLAPS ARM switch. ALTN

Alternate FLAPS selector. RET

Monitor airspeed during retraction.

GND PROX FLAP OVRD OVRD

Note: Plan additional time for alternate slat and flap extension.

[Flap operation is slower in alternate mode.]

Note: Use flaps 20 and VREF20 for landing.

[Alternate mode is limited to a maximum of flaps 20.]

-----DEFERRED ITEMS-----

==> APPROACH CHECKLIST

Alternate FLAPS ARM switch. ALTN

Alternate FLAPS selector. EXT

Monitor airspeed during extension.

[Extends flaps to flaps 20 position.]



[FLIGHT CONTROL MODE

Condition: **Flight control system is operating in secondary mode.**

PRIMARY FLIGHT COMPUTERS

disconnect switch. DISC, then AUTO

[Attempts to restore flight control normal mode.]

If FLIGHT CONTROL MODE message remains displayed:

Avoid abrupt control inputs.

[Airplane response is changed by simplified elevator feel and rudder ratio systems.]

GND PROX FLAP OVRD OVRD

Note: Inoperative items:

- envelope protection functions
- autopilot.

Note: Yaw damper is degraded.

Note: Manual control inputs are required to compensate for asymmetric thrust conditions.

Note: Use flaps 20 and VREF20 for landing.

[Ensures sufficient pitch trim capability for landing.]

Note: Do not arm speedbrake lever.

[From AUTO SPEEDBRAKE checklist. Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.

[From AUTO SPEEDBRAKE checklist.]

Do not accomplish the following checklists:

AUTO SPEEDBRAKE



FLIGHT CONTROLS

Condition: **Multiple flight control surfaces are inoperative or other flight control system faults are detected.**

Handling qualities are degraded.

[Pitch and roll control capability is reduced with fewer operating control surfaces.]

Plan to land at the nearest suitable airport.

GND PROX FLAP OVRD OVRD

Note: Use flaps 20 and VREF30 + 20 for landing.

[Higher approach speeds improve airplane maneuvering characteristics.]

Note: Crosswind limit for landing is 20 knots.

[Less control authority decreases crosswind landing capability.]

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

[From SPOILERS or SPOILER PAIRS checklist.]

Do not accomplish the following checklist:

SPOILERS

SPOILER PAIRS



FLT CONTROLS LOCKED

Condition: **One or both flight control surface lock switches are in the locked position.**



PITCH DOWN AUTHORITY

Condition: **Pitch down authority is limited.**

Slower airspeeds assist nose down pitch control.

[Airplane is approaching its nose down pitch control limit.]

Note: Avoid speedbrake use and rapid thrust increases.

[Only limited elevator authority is available to counter nose up pitching.]



[PITCH UP AUTHORITY]

Condition: **Pitch up and flare authority are limited.**

Do not extend flaps any further until on approach.

[Airplane is approaching its nose up pitch control limit.]

GND PROX FLAP OVRD OVRD

Note: Do not use autoland.

If flap position 15 or less:

Note: Use flaps 5 and VREF30 + 40 for landing.

[Higher approach speeds provide better pitch up control authority.]



If flap position 20 or greater:

Note: Use flaps 20 and VREF30 + 20 for landing.

[Higher approach speeds provide better pitch up control authority.]



PRI FLIGHT COMPUTERS

Condition: **Flight control system is operating in direct mode.**

PRIMARY FLIGHT COMPUTERS

disconnect switch. DISC, then AUTO

[Attempts to restore flight control normal mode.]

If PRI FLIGHT COMPUTERS message remains displayed:

Avoid abrupt control inputs.

[Airplane response is changed by simplified elevator feel and rudder ratio systems.]

GND PROX FLAP OVRD OVRD

Note: Inoperative items:

- envelope protection functions
- autopilot

Note: Manual control inputs are required to compensate for asymmetric thrust conditions.

Note: Use flaps 20 and VREF20 for landing.

[Ensures sufficient pitch trim capability for landing.]

Note: Do not arm speedbrake lever.

[From AUTO SPEEDBRAKE checklist. Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.

[From AUTO SPEEDBRAKE checklist.]

Do not accomplish the following checklists:

AUTO SPEEDBRAKE



ROLL LEFT AUTHORITY

Condition: **There is limited authority remaining to roll the airplane to the left.**

The flight control system is commanding a large left wing down and/or left rudder input to control an airplane asymmetry.

Plan to land at the nearest suitable airport

[Unknown drag condition may decrease range.]

Note: Use LRC speed for diversion. Do not use FMC fuel predictions.

Note: Use flaps 20 and VREF30 + 20 for landing.

[Higher approach speed improves maneuvering characteristics.]

Note: Crosswind limit for landing is 10 knots.

[Maintains sufficient control authority for de-crab maneuver.]

Note: Left control wheel may be required after landing to control roll asymmetry.



[ROLL RIGHT AUTHORITY]

Condition: **There is limited authority remaining to roll the airplane to the right.**

The flight control system is commanding a large right wing down and/or right rudder input to control an airplane asymmetry.

Plan to land at the nearest suitable airport

[Unknown drag condition may decrease range.]

Note: Use LRC speed for diversion. Do not use FMC fuel predictions.

Note: Use flaps 20 and VREF30 + 20 for landing.

[Higher approach speed improves maneuvering characteristics.]

Note: Crosswind limit for landing is 10 knots.

[Maintains sufficient control authority for de-crab maneuver.]

Note: Right control wheel may be required after landing to control roll asymmetry.



[ROLL/YAW ASYMMETRY]

Condition: **A significant airplane roll or yaw asymmetry has been detected.**

Plan to land at the nearest suitable airport.

[Unknown drag condition may decrease range.]

Note: Use LRC speed for diversion. Do not use FMC fuel predictions.



SLATS DRIVE

Condition: **Slat drive mechanism has failed.**

Do not use alternate flaps.

[Asymmetry and uncommanded motion protection are not provided in alternate mode.]

GND PROX FLAP OVRD OVRD

Note: Do not use FMC fuel predictions with slats extended.

Note: Do not use autoland.

Note: Use flaps 20 and VREF30 + 30 for landing.

[Provides better handling qualities when slats are not fully extended.]

Do not accomplish the following checklist:

SLATS PRIMARY FAIL



SLATS PRIMARY FAIL

Condition: **Slats are operating in secondary mode.**

Note: Plan additional time for slower slat operation.

Note: Slats will extend beyond midrange when airspeed is below 215 knots. For go-around, do not exceed 215 knots until slats retract to midrange.

[From SLATS PRIMARY FAIL checklist.]



SPEEDBRAKE EXTENDED

Condition: **Speedbrake is extended when radio altitude is between 15 and 800 feet, or when the flap lever is in a landing position, or when either thrust lever is not closed.**



[] SPOILER DRAG

Condition: **Increased spoiler drag due to loss of pressure in center hydraulic system and low altitude cruise.**

RAM AIR TURBINE switch. Push and hold for 1 second

[Reduces drag by enabling center hydraulic system to power and stow two spoiler pairs.]

Fly LRC speed.

[Improves range.]



[] SPOILER PAIRS

Condition: **Two spoiler pairs are inoperative.**

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

If landing using flaps 25:

Note: Use flaps 25 and VREF25 +5 for landing.

[Flap effectiveness is reduced.]

If landing using flaps 30:

Note: Use flaps 30 and VREF30 +5 for landing.

[Flap effectiveness is reduced.]



[] SPOILERS

Condition: **One or more spoiler panels are inoperative.**

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.



[STAB GREENBAND]

Condition: **Nose gear pressure switch disagrees with computed stabilizer greenband.**

FMC WEIGHT and CG Entries. Check

[An incorrect entry may cause an incorrect stab greenband calculation.]



[STABILIZER]

Condition: **Uncommanded stabilizer motion is detected or stabilizer is inoperative.**

STABILIZER CUTOUT switches (Both). CUTOUT

[Prevents subsequent uncommanded or inappropriate stabilizer motion.]

Do not exceed current airspeed.

[Nose down elevator authority is limited.]

Stabilizer is inoperative. Pitch trim remains available in the normal flight control mode.

GND PROX FLAP OVRD OVRD

Note: Use flaps 20 and VREF30 + 20 for landing.

[Provides sufficient elevator authority for landing.]

Do not accomplish the following checklist:

FLIGHT CONTROLS



STABILIZER CUTOUT

Condition: **Both stabilizer cutout switches are in CUTOUT.**



[]STABILIZER L2

Condition: **Left stabilizer control path is inoperative.**

L2 stabilizer CUTOUT switch. CUTOUT

[Prevents subsequent uncommanded or inappropriate stabilizer motion.]

Note: Left control wheel pitch trim switches may be inoperative.



[]STABILIZER R2

Condition: **Right stabilizer control path is inoperative.**

R2 stabilizer CUTOUT switch. CUTOUT

[Prevents subsequent uncommanded or inappropriate stabilizer motion.]

Note: Right control wheel pitch trim switches may be inoperative.



[]STALL PROTECTION

Condition: **Some stall protection functions are inoperative.**

Note: Any stall indications should be considered valid.



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[AIRSPEED UNRELIABLE

Condition: (TBD)



[ALTN ATTITUDE CAPT, F/O

Condition: The AIR DATA/ATTITUDE source switch is in the ALTN position.

Note: The PFD is displaying attitude information from the Integrated Standby Flight Display.



BARO SET DISAGREE

Condition: Captain's and First Officer's barometric settings disagree.



CHKL INCOMPLETE NORM

Condition: A normal checklist must be completed for the next phase of flight.



CHKL NON- NORMAL

Condition: One or more checklists in the ECL non-normal checklist queue need to be completed but do not appear on EICAS.



[EFIS/DSP PANEL L, R

Condition: EFIS control panel and display select panel are inoperative or MFD backup control of the EFIS control panel and display select panel is active.

Note: Control of the EFIS/DSP panel is accessed from the MFD system display, using the PANEL BACKUP key.



[J] HUD SINGLE OPERATION

Condition: **Only one HUD can operate.**

Note: Only one HUD can operate. To enable flying pilot's HUD operation, stow the opposite HUD.



HUD SYS CAPT, F/O

Condition: **[Captain's] [First Officer's] HUD has failed.**



[J] HUD TAKEOFF

Condition: **HUD takeoff guidance is unavailable.**

**Visibility is acceptable for takeoff without HUD guidance:
Continue normal operation.**



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Blank

U LOSS OF ALL DISPLAYS

Condition: **All 5 forward displays and both HUDs are blank.**

Autopilot. Engage

Note: LNAV and VNAV modes are not operational. Use Heading/Track Select and HOLD, VS/FPA and Altitude HOLD.

Note: Use the ISFD to maintain the flight plan.

CCR RESET R switch. Push
[Attempts to restore partial display operation.]

Wait 1 minute.

If partial display operation restored:

Note: Four forward displays restored. Lower display and both HUDs remain blank.



If all forward displays remain blank:

CCR RESET switches (both). Push

Wait 1 minute.

If partial or full display operation restored:

Note: The lower display unit and both HUDs remain blank.



If all forward displays remain blank:

Avoid icing conditions.

[Window heat and wing anti-ice are inoperative.]

Use alternate navigation.

Left TCP. Enter desired waypoint

Note: Alternate NAV function resides in left TCP.

HDG REF switch. TRUE
[TCP backup navigation uses a true track reference.]

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HDG/TRK reference switch.TRK
Heading/Track selector. Enter desired track
Heading/Track Select (SEL) switch. Push
Note: Monitor the left TCP and update MCP selected track as necessary.
Note: The following systems or displays may be inoperative:

- Primary displays are all inoperative, including crew alert messages
- Autothrottle
- Flight Management Computer, LNAV and VNAV
- Normal landing gear extension and retraction
- Center tank fuel pumps and center tank fuel scavenge leading to trapped fuel
- Automatic cabin pressure control - defaults to 8000 ft cabin
- Passenger oxygen deployment
- First officer's and observer's audio control
- Voice communication on HF or SATCOM
- Window heat and wing anti-ice

-----DEFERRED ITEMS-----

==> APPROACH CHECKLIST

Landing gear lever.DN
**ALTN GEAR switch.DOWN and hold
Push and hold for 1 second.**


SGL SOURCE APPROACH

Condition: Both PFDs are using the same source for ILS information.


[SGL SOURCE ATTITUDE

Condition: **Both AIR DATA/ATTITUDE source switches are in the ALTN position.**

Note: Both PFDs are displaying attitude information from the Integrated Standby Flight Display.



SGL SOURCE RAD ALT

Condition: **Both PFDs are using the same source for radio altimeter information.**



SINGLE SOURCE FD

Condition: **Both PFDs are using the same source for flight director information.**



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FMC

Condition: **All 3 FMCs have failed.**

Select autopilot roll and pitch modes appropriate for the desired flight path.

[LNAV and VNAV modes fail.]

Note: Inoperative items:

- LNAV
- VNAV
- all maps
- all FMC pages
- all navigation radios except left ILS
- all navigation functions except GPS, latitude and longitude

LEFT TUNING and CONTROL PANEL NAV Key. PUSH

Note: Present latitude and longitude are shown on the ALTN NAV RADIO page.

Note: Tune the left ILS receiver by entering frequency and course on the left tuning and control panel. ILS will display on the Captain's PFD.

Note: Refer to Performance–Inflight chapter for VREF speed and other applicable performance information.

LANDING ALTITUDE selector. Pull, set manually

[From LANDING ALTITUDE checklist.]

Do not accomplish the following checklist:

LANDING ALTITUDE



FMC HOLD AIRSPACE

Condition: **The FMC calculated holding pattern exceeds the maximum protected airspace limits.**

Decrease airspeed or advise ATC you are unable to comply with required holding airspace limits.



FMC INTERCEPT HDG

Condition: **LNAV is armed and the airplane is not on an intercept heading to the active leg.**



FMC MESSAGE

Condition: **A message is in the FMC message window.**



[FMC PERF UNAVAIL

Condition: **VNAV is selected, and gross weight, cost index or cruise altitude are not entered.**

FMC PERF INIT page. Select
Ensure all required data is entered.



FMC RUNWAY DISAGREE

Condition: **Airplane is not on FMC origin runway when either engine's thrust is in the takeoff range.**



FMC UNABLE RTA

Condition: **The FMC is unable to achieve the required time of arrival within the allowable time error.**



FMC VERIFY POSITION

Condition: **The FMC calculation of airplane position is based on conflicting data.**



GPS

Condition: **GPS has failed.**



[INSUFFICIENT FUEL]

Condition: FMC estimated fuel at the destination is less than the entered RESERVES fuel.

The INSUFFICIENT FUEL message may be caused by an engine fuel leak. For indications of an engine fuel leak, check:

- total fuel remaining on EICAS compared to planned fuel remaining
- fuel flow indications, for an engine with excessive fuel flow
- individual tank quantities
- totalizer compared to calculated quantities (PROGRESS page 2/3). The TOTALIZER value is the sum of the individual tank quantities. The CALCULATED value is the totalizer value at engine start minus fuel used (calculated using fuel flow rates and time).

If there is an indication of an engine fuel leak:

Accomplish the FUEL LEAK checklist.

If FUEL DISAGREE message is displayed without any indication of an engine fuel leak:

Obtain a more direct route to the destination, or divert to an alternate with shorter routing.



[IRS MOTION]

Condition: During inertial reference system alignment, excessive airplane motion is detected.

Airplane must be stopped to allow IRS alignment to complete.

When IRS ALIGN MODE memo message is no longer displayed:

Continue normal operation.



[LNAV BANK ANGLE LIM

Condition: The airplane may not follow an LNAV turn because bank angle is limited by available thrust or buffet.

If unable to remain within protected airspace, advise ATC.



[NAV AIR DATA SYS

Condition: Backup sources for airspeed and altitude are being used instead of air data.

Avoid abrupt control inputs.

[From FLIGHT CONTROL MODE checklist. Airplane response is changed by simplified elevator feel and rudder ratio systems.]

Note: Altitude is from GPS and is approximately equal to MSL altitude. Altimeter cannot be used above transition level to accurately maintain flight levels.

Note: Airspeed is calculated using angle of attack. Airspeed errors of up to 15 KIAS may exist at approach speeds.

GRND PROX FLAP OVRD. OVRD

[From FLIGHT CONTROL MODE checklist.]

Note: Inoperative items:

- envelope protection functions
- autopilot
- flight directors
- autothrottles
- PFD flap maneuvering speeds; refer to Flap Maneuvering Speed table in Performance–Inflight chapter.

[Inoperative items envelope protection functions and autopilot from FLIGHT CONTROL MODE checklist.]

Note: Yaw damper is degraded.

[From FLIGHT CONTROL MODE checklist.]

Note: Manual control inputs are required to compensate for asymmetric thrust conditions.

[From FLIGHT CONTROL MODE checklist.]

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Note: Use flaps 20 and VREF20 for landing.

[From FLIGHT CONTROL MODE checklist. Ensures sufficient pitch control for landing.]

Note: Avoid speedbrake use in flight.

Note: Do not arm speedbrake lever.

[From AUTO SPEEDBRAKE checklist. Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.

[From AUTO SPEEDBRAKE checklist.]

Do not accomplish the following checklists:

AUTO SPEEDBRAKE

FLIGHT CONTROL MODE



[NAV AIRSPEED DATA]

Condition: **Backup source for airspeed is being used instead of air data.**

Avoid abrupt control inputs.

[From FLIGHT CONTROL MODE checklist. Airplane response is changed by simplified elevator feel and rudder ratio systems.]

Note: Airspeed is calculated using angle of attack. Airspeed errors of up to 15 KIAS may exist at approach speeds.

GND PROX FLAP OVRD OVRD

[From FLIGHT CONTROL MODE checklist.]

Note: Inoperative items:

- envelope protection functions
- autopilot
- flight directors
- autothrottles
- PFD flap maneuvering speeds; refer to Flap Maneuvering Speed table in Performance-Inflight chapter.

[Inoperative items envelope protection functions and autopilot from FLIGHT CONTROL MODE checklist.]

Note: Yaw damper is degraded.

[From FLIGHT CONTROL MODE checklist.]

Note: Manual control inputs are required to compensate for asymmetric thrust conditions.

[From FLIGHT CONTROL MODE checklist.]

Note: Use flaps 20 and VREF20 for landing.

[From FLIGHT CONTROL MODE checklist. Ensures sufficient pitch control for landing.]

Note: Avoid speedbrake use in flight.

Note: Do not arm speedbrake lever.

[From AUTO SPEEDBRAKE checklist. Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.

[From AUTO SPEEDBRAKE checklist.]

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Do not accomplish the following checklist:

AUTO SPEEDBRAKE

FLIGHT CONTROL MODE



[NAV INERTIAL SYS

Condition: **IRS is not capable of providing valid attitude, position, track and groundspeed.**

GPS continues to provide position and track information.

ND map display orientation changes from heading up to track up.

[ND Map Heading-Up.]

TRANSPONDER ALTITUDE SOURCE selectorALTN

TRANSPONDER selectorRIGHT

Note: Inoperative items:

- **FMC VNAV pages**
- **FMC performance predictions**
- **PFD flap maneuvering speeds; refer to Flap Maneuvering Speed table in Performance-Inflight chapter**
- **ND wind direction/speed and wind arrow**
- **autobrake.**

Note: Inoperative AFDS modes:

- **LNAV**
- **VNAV**
- **TO/GA**
- **LOC**
- **GS**
- **FPA**
- **TRK HOLD/SEL.**

Note: Crosscheck heading periodically for drift with the magnetic compass. If magnetic compass information is unreliable or unavailable, track information may be used.

Note: VOR course deviation is available in the ND VOR mode. ILS localizer and glideslope deviation raw data is available on both the PFD and ND.

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Note: If GPS is unavailable, the following additional items are inoperative:

- ND map mode
- active leg course and distance
- direct to waypoint function
- alternate page DIVERT NOW function
- navigation radio autotuning.



[NAV UNABLE RNP

Condition: **Navigation performance does not meet required accuracy.**

If in flight:

If on procedure or airway that has an RNP alerting requirement:

Select alternate procedure or airway, or initiate a go-around.

If on procedure or airway without RNP:

Verify position.



If on the ground:

Message may display with GPS disabled.



PILOT RESPONSE

Condition: **Recent pilot activity has not been detected.**



[]RWY/APP CRS ERROR

Condition: **The selected ILS course does not match the FMC active arrival runway.**

FMC NAV RAD page. Select

Note: Delete manually entered ILS frequency, or select correct course for manually entered frequency.



[]RWY/APP TUNE ERROR

Condition: **The tuned ILS frequency or GLS channel does not match the FMC active arrival runway.**

FMC NAV RAD page. Select

Note: Delete manually entered ILS frequency or GLS channel, or select correct approach on active route.



SINGLE FMC

Condition: **Only a single FMC is operating.**



SGL SOURCE APPROACH

Condition: **Both PFDs are using the same source for ILS information.**



TRANSPONDER

Condition: **Transponders have failed.**



TRANSPONDER PANEL

Condition: **Transponder panel is inoperative or TCP control of transponder panel is active.**

Note: TCP control of the glideslope inhibit, transponder mode, and aural cancel controls is accessed from the TCP menu page.



WEATHER RADAR SYS

Condition: **Weather radar system is inoperative.**



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[FUEL AUTO JETTISON]

Condition: **Total fuel quantity is less than selected TO REMAIN quantity and a nozzle valve is open.**

FUEL JETTISON NOZZLE switches (Both). OFF

[Closes fuel jettison nozzle valves.]

FUEL JETTISON ARM switch OFF

[Disarms fuel jettison system and removes FUEL TO REMAIN quantity from EICAS.]



[FUEL BALANCE SYS

Condition: **Fuel balance system is inoperative.**

FUEL BALANCE switch. PUSH

[Deactivates fuel balance command and removes message.]

CROSSFEED switch ON

[Allows fuel from the high tank to feed both engines.]

If left main tank quantity low:

Left forward and AFT FUEL PUMP switches. OFF

[Allows fuel from the high tank to feed both engines.]

Do not accomplish the following checklists:

FUEL PUMP L AFT

FUEL PUMP L FWD

If right main tank quantity low:

Right forward and AFT FUEL PUMP switches. OFF

[Allows fuel from the high tank to feed both engines.]

Do not accomplish the following checklists:

FUEL PUMP R AFT

FUEL PUMP R FWD

When fuel balancing complete:

Forward and AFT FUEL PUMP switches (All). ON

[Restores fuel feed from low tank.]

CROSSFEED switch. OFF

[Restores main tank-to-engine fuel feed.]



FUEL CROSSFEED

Condition: **Crossfeed valve is not in the commanded position.**

If crossfeed switch ON:

CROSSFEED switch.OFF

[Takes away command for crossfeed valve to open, which would cause unwanted crossfeed if it opened later in flight.]

Do not complete the following checklist:

FUEL BALANCE SYS



FUEL DISAGREE

Condition: **Totalizer fuel quantity and FMC calculated fuel quantity disagree.**

The FUEL DISAGREE message may be caused by an engine fuel leak. For indications of an engine fuel leak, check:

- total fuel remaining on EICAS compared to planned fuel remaining
- fuel flow indications, for an engine with excessive fuel flow
- individual tank quantities
- totalizer compared to calculated quantities (PROGRESS page 2). The TOTALIZER value is the sum of the individual tank quantities. The CALCULATED value is the totalizer value at engine start minus fuel used (calculated using fuel flow rates and time).

If there is an indication of an engine fuel leak:

Accomplish the FUEL LEAK checklist.



If FUEL DISAGREE message is displayed without any indication of an engine fuel leak:

PROGRESS page 2. Select

TOTALIZER or CALCULATED.Select use

Select the most accurate value.



FUEL IMBALANCE

Condition: **Fuel imbalance between the main tanks is excessive.**

The FUEL IMBALANCE message may be caused by an engine fuel leak. For indications of an engine fuel leak, check:

- total fuel remaining on EICAS compared to planned fuel remaining
- fuel flow indications, for an engine with excessive fuel flow
- individual tank quantities
- totalizer compared to calculated quantities (PROGRESS page 2). The TOTALIZER value is the sum of the individual tank quantities. The CALCULATED value is the totalizer value at engine start minus fuel used (calculated using fuel flow rates and time).

If there is an indication of an engine fuel leak:

Accomplish the FUEL LEAK checklist.



If FUEL IMBALANCE message displayed without any indication of an engine fuel leak:

FUEL BALANCE switch. PUSH

Push and hold for 1 second.

[Initiates automatic fuel balancing.]



FUEL IN CENTER

Condition: **Both center fuel pump switches are OFF with fuel in the center tank.**



FUEL JETT NOZZLE L, R

Condition: **Jettison nozzle valve is not in the commanded position.**



U FUEL JETTISON

Condition: **Fuel jettison is required.**

FUEL JETTISON ARM switch. ARMED

If desired FUEL TO REMAIN different than displayed FUEL TO REMAIN:

FUEL TO REMAIN selector. PULL, set manually

FUEL JETTISON NOZZLE switches (Both). ON

Do not accomplish the following checklists:

FUEL PUMP CENTER L

FUEL PUMP CENTER R

When fuel jettison complete:

FUEL JETTISON NOZZLE switches (Both). OFF

FUEL JETTISON ARM switch. OFF



□ FUEL JETTISON MAIN

Condition: **Fuel jettison from main tanks is inoperative.**

Fuel jettison can occur only from the center tank.

Do not complete the following checklist:

FUEL JETTISON

When center tank empty or FUEL TO REMAIN quantity reached:

FUEL JETTISON NOZZLE switches (Both). OFF

FUEL JETTISON ARM switch. OFF



FUEL JETTISON SYS

Condition: **Fuel jettison system is inoperative.**

FUEL JETTISON NOZZLE switches (Both). OFF

[Closes fuel jettison nozzle valves.]

FUEL JETTISON ARM switch. OFF

[Disarms fuel jettison system and removes TO REMAIN quantity from EICAS.]

Do not complete the following checklist:

FUEL JETTISON



U FUEL LEAK

Condition: **An in flight fuel leak is suspected or confirmed.**

One or more of the following may be evidence of a fuel leak:

- **visual observation of fuel spray from strut/engine**
- **excessive engine fuel flow**
- **total fuel quantity decreasing at an abnormal rate**
- **FUEL DISAGREE message on EICAS**
- **FUEL IMBALANCE message on EICAS**
- **FUEL QTY LOW message on EICAS**
- **INSUFFICIENT FUEL message on EICAS.**

CENTER FUEL PUMP switches (Both). OFF

CROSSFEED switch. OFF

BALANCE switch. OFF

Identify an engine fuel leak by observing a left or right main fuel tank quantity decreasing faster than the other.

An increase in fuel imbalance of approximately 1000 lbs or more in 30 minutes should be considered a fuel leak.

Conditions permitting, visually check for engine fuel leak.

If left and right main tank quantities decrease at same rate:

Fuel leak may be from center tank to center wing area. Do not use center tank fuel.

If FUEL DISAGREE message displayed:

PROGRESS page 2. Select

CALCULATED. Select use

Use CALCULATED to determine gross weight.

Destination fuel estimate. CHECK

Verify adequate fuel available in left and right main tanks to complete flight.

Continued on next page

Continued from previous page

If engine fuel leak confirmed:

AUTOTHROTTLE ARM switch
(Affected engine). **OFF**

[Allows thrust lever to remain where manually positioned.]

Thrust Lever (Affected engine). **Close**

[Assists in recognition of affected engine.]

FUEL CONTROL switch
(Affected engine). **CUTOFF**

APU selector
(If APU available). **START, then ON**

[Provides an additional source of electrical power.]

Transponder mode selector. **TA ONLY**

[Prevents climb commands which can exceed single engine performance capability.]

If FUEL DISAGREE message displayed:

PROGRESS page 2. **Select**

TOTALIZER. **Select use**

Use TOTALIZER to determine fuel remaining.

Note: After engine shutdown, all remaining fuel can be used for the operating engine. Resume normal fuel management procedures. Plan to balance fuel when the FUEL IMBALANCE message is displayed.

Plan to land at the nearest suitable airport.

If landing using flaps 20:

GND PROX FLAP OVRD. **OVRD**

Note: Use flaps 20 and VREF20 for landing and flaps 5 for go-around.

If landing using flaps 30 (performance permitting):

Note: Use flaps 30 and VREF30 for landing and flaps 20 for go-around.

Continued on next page

Continued from previous page

If FUEL QTY LOW message displayed:

CROSSFEED switch.ON

[From FUEL QTY LOW checklist. Ensures fuel is available to both engines if the low tank empties.]

FUEL PUMP switches (All).ON

[From FUEL QTY LOW checklist. Ensures all fuel is available for use.]

Plan to land at the nearest suitable airport.

[From FUEL QTY LOW checklist.]

Note: Use flaps 20 and VREF20 for landing.

[From FUEL QTY LOW checklist. Provides improved elevator authority in event of dual engine flameout.]

GND PROX FLAP OVRD. OVRD

Note: Avoid high nose up attitude and excessive acceleration or deceleration.

[From FUEL QTY LOW checklist. Prevents forward pumps from uncovering.]

Do not complete the following checklist:

FUEL QTY LOW



FUEL LOW CENTER

Condition: **Center tank is empty and one or both center fuel pump switches are ON.**



[FUEL PRESS ENG L, R]

Condition: **Engine is on suction feed.**

If unable to maintain required thrust on affected engine:

CROSSFEED switchON

[Allows fuel from operative pump side to feed both engines.]

If airplane altitude above FL200:

Descent.Initiate

Descend to FL200.

[Ensures that suction feed will supply adequate fuel to engine.]

When airplane altitude at or below FL200:

Note: Do not climb above FL200 for the remainder of the flight.

CROSSFEED switchOFF

[Restores main tank-to-engine fuel feed.]

Note: Continue suction feed operation. Do not balance fuel.



FUEL PRESS ENG L+R

Condition: **All fuel pump output pressures are low.**



[FUEL PUMP CENTER L, R]

Condition: **Center fuel pump output pressure is low.**

CROSSFEED switch.ON

[Prevents fuel imbalance by feeding both engines from the same center fuel pump.]

FUEL PUMP switch (Affected pump).OFF



[FUEL PUMP CTR L+R

Condition: **Both center fuel pump output pressures are low.**

CROSSFEED switch. OFF

[Prevents fuel imbalance.]

CENTER FUEL PUMP switches (Both). OFF

Center tank fuel is available. Scavenge system will transfer center fuel to main tanks.



[FUEL PUMP L AFT, FWD

Condition: **Left aft or forward fuel pump output pressure is low.**

FUEL PUMP switch (Affected pump). OFF



[FUEL PUMP R AFT, FWD

Condition: **Right aft or forward fuel pump output pressure is low.**

FUEL PUMP switch (Affected pump). OFF



□ FUEL QTY LOW

Condition: **Fuel quantity is low in either main tank.**

The FUEL QTY LOW message may be caused by an engine fuel leak. For indications of an engine fuel leak, check:

- total fuel remaining on EICAS compared to planned fuel remaining
- fuel flow indications, for an engine with excessive fuel flow
- individual tank quantities
- totalizer compared to calculated quantities (PROGRESS page 2). The TOTALIZER value is the sum of the individual tank quantities. The CALCULATED value is the totalizer value at engine start minus fuel used (calculated using fuel flow rates and time).

If there is an indication of an engine fuel leak:

Accomplish the FUEL LEAK checklist.



If FUEL DISAGREE message is displayed without any indication of an engine fuel leak:

CROSSFEED switch.ON

[Ensures fuel is available to both engines if the low tank empties.]

FUEL PUMP switches (All).ON

[Ensures all fuel is available for use.]

Plan to land at the nearest suitable airport.

GND PROX FLAP OVRD. OVRD

Note: Use flaps 20 and VREF 20 for landing.

[Increased speed at Flaps 20 provides improved elevator control for landing flare in the event of a dual engine flameout.]

Note: Avoid high nose up attitude and excessive acceleration or deceleration.

[Prevents forward pumps from uncovering.]



[FUEL TEMP HIGH]

Condition: **Fuel temperature is approaching maximum.**

Note: Maximum fuel temperatures are:

FUEL TYPE	DEG C
JET A, JET A-1, JP-5, JP-8	60
TS-1	57

C1 and C2 PUMP selectors. AUTO

[Allows one pump (both on ground) to shut down if demand is low, reducing fuel heating.]

LEFT and RIGHT DEMAND PUMP selectors. AUTO

[Allows pumps to shut down if demand is low, reducing fuel heating.]

If on the ground:

Taxi out time is limited. For takeoff, fuel temperature must be 2 deg C less than the fuel temperature limit.

Refer to Fuel Heating Schedule table in Performance-Inflight chapter for available taxi time.

If in flight:

Note: At low altitude and high outside air temperature, engines may flame out. Avoid extended operation at low altitude.

Note: Fuel pump low pressure indications may occur. Main tank fuel pump switches should remain ON.

Do not accomplish the following checklists:

- FUEL PUMP L AFT**
- FUEL PUMP L FWD**
- FUEL PUMP R AFT**
- FUEL PUMP R FWD**



■ FUEL TEMP LOW

Condition: **Fuel temperature is approaching minimum.**

Increase airspeed, change altitude, and/or deviate to a warmer air mass to achieve a TAT equal to or higher than the fuel temperature limit (3 degrees C above the fuel freeze point).

TAT increases approximately 0.5 to 0.7 degrees C for each .01 Mach increase in airspeed. In extreme conditions it may be necessary to descend as low as FL250.



■ FUEL VALVE APU

Condition: **APU fuel valve is not in the commanded position.**

Do not start the APU.

[Prevents a potential fire hazard.]

Note: APU is not available for remainder of flight.

Do not accomplish the following checklist:

APU SHUTDOWN



Intentionally
Blank



Non-Normal Checklists
Hydraulics

Chapter NNC
Section 13

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Blank

HYD OVERHEAT C1, C2

Condition: **Center pump temperature is high.**

CENTER PUMP selector (Affected side).OFF

[Attempts to eliminate source of overheat.]

Do not accomplish the following checklist:

HYD PRESS (Affected side)



HYD OVERHEAT DEM L, R

Condition: **Demand pump temperature is high.**

DEMAND PUMP selector.OFF

[Attempts to eliminate source of overheat.]

Do not accomplish the following checklist:

HYD PRESS DEM



HYD OVERHEAT PRI L, R

Condition: **Primary pump temperature is high.**

PRIMARY PUMP switch.OFF

[Attempts to eliminate source of overheat.]

Note: Thrust reverser on the affected side may be inoperative.

Do not accomplish the following checklist:

HYD PRESS PRI



[]HYD PRESS C1, C2

Condition: **Center pump output pressure is low when commanded on.**

CENTER PUMP selector (Affected side).ON

[Restores system pressure if AUTO function failed to operate.]

If HYD PRESS (affected side) message remains displayed:

CENTER PUMP selector (Affected side).OFF

[Avoids system contamination and/or pump damage.]



[]HYD PRESS DEM L, R

Condition: **Demand pump output pressure is low when commanded on.**

DEMAND PUMP selector.ON

[Restores system pressure if AUTO demand function failed to operate.]

If HYD PRESS DEM message remains displayed:

DEMAND PUMP selector.OFF

[Avoids system contamination and/or pump damage.]

Note: Thrust reverser deployment may be slower than normal.



[]HYD PRESS PRI L, R

Condition: **Primary pump output pressure is low.**

PRIMARY PUMP switch.OFF



Intentionally
Blank

[HYD PRESS SYS C

Condition: **Center hydraulic system pressure is low.**

C1 or C2 PUMP selector.ON

[Restores system pressure if AUTO position of selector is inoperative.]

If HYD PRESS SYS C message remains displayed:

C1 PUMP selector. AUTO

[selector must stay in AUTO in case reserve steering system has successfully isolated.]

C2 PUMP selector.OFF

[Avoids system contamination and/or pump damage.]

GND PROX FLAP OVRD. OVRD

[From FLAPS PRIMARY FAIL checklist.]

Note: Inoperative items:

- main landing gear hydraulic operation

Note: Plan additional time for slower slat and flap operation.

[From FLAPS PRIMARY FAIL and SLATS PRIMARY FAIL checklists.]

Note: Slats will extend beyond midrange when airspeed is below 215 knots. For go-around, do not exceed 215 knots until slats retract to midrange.

Note: Use flaps 20 and VREF20 for landing.

[From FLAPS PRIMARY FAIL checklist. Ensures adequate go-around performance due to slower slat/flap operation in secondary mode.]

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

[From SPOILERS checklist.]

Note: Do not arm speedbrake lever.

[From AUTO SPEEDBRAKE checklist. Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.

[From AUTO SPEEDBRAKE checklist.]

Continued on next page

Continued from previous page

If HYD PRESS SYS C message remains displayed: (continued)

Note: For go-around, observe gear EXTEND limit speed (270K/.82M).

[From GEAR DOOR checklist.]

Do not accomplish the following checklists:

AUTO SPEEDBRAKE

SPOILERS

-----DEFERRED ITEMS-----

==> APPROACH CHECKLIST

Initiate flap extension as required.

LANDING GEAR lever.DN

[Ensures nose gear extension if pressure still exists in that part of the system.]

ALTERNATE GEAR switch.DOWN and hold

Push and hold until all gear indicate in-transit.

[Releases gear uplocks and gear door locks allowing gear to free fall.]

Do not accomplish the following checklists:

FLAPS PRIMARY FAIL

SLATS PRIMARY FAIL

GEAR DOOR



[HYD PRESS SYS L

Condition: **Left hydraulic system pressure is low.**

LEFT DEMAND PUMP selector.ON

[Restores system pressure if AUTO position of selector is inoperative.]

If HYD PRESS SYS L message remains displayed:

LEFT PRIMARY PUMP switch.OFF

[Avoids system contamination and/or pump damage.]

LEFT DEMAND PUMP selector.OFF

[Avoids system contamination and/or pump damage.]

Note: Left thrust reverser is inoperative.

[From ENG REVERSER L checklist.]

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

[From SPOILERS checklist.]

Do not accomplish the following checklist:

ENG REVERSER L

SPOILERS



Intentionally
Blank

[HYD PRESS SYS L+C]

Condition: **Left and center hydraulic system pressures are low.**

LEFT DEMAND PUMP selector. ON

[Restores system pressure if AUTO position of selector is inoperative.]

C1 or C2 PUMP selector. ON

[Restores system pressure if AUTO position of selector is inoperative.]

If HYD PRESS SYS L+C message remains displayed:

LEFT PRIMARY PUMP switch. OFF

LEFT DEMAND PUMP selector. OFF

[Avoids system contamination and/or pump damage.]

C1 PUMP selector. AUTO

[Selector must stay in AUTO in case reserve steering has successfully isolated.]

C2 PUMP selector. OFF

[Avoids system contamination and/or pump damage.]

Handling qualities are degraded.

[From FLIGHT CONTROLS checklist. Pitch and roll control capability is reduced with fewer operating control surfaces.]

Plan to land at the nearest suitable airport.

[From FLIGHT CONTROLS checklist.]

GND PROX FLAP OVRD OVRD

[From FLIGHT CONTROLS and FLAPS PRIMARY FAIL checklists.]

Note: Inoperative items:

- multiple flight control surfaces
- main landing gear hydraulic operation
- left thrust reverser

Note: Plan additional time for slower slat and flap operation.

[From FLAPS PRIMARY FAIL and SLATS PRIMARY FAIL checklists.]

Continued on next page

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If HYD PRESS SYS L+C message remains displayed: (continued)

Note: Slats will extend beyond midrange when airspeed is below 215 knots. For go-around, do not exceed 215 knots until slats retract to midrange.

[From SLATS PRIMARY FAIL checklist.]

Note: Use flaps 20 and VREF30 + 20 for landing.

[From FLIGHT CONTROLS checklist. Higher approach speeds improve airplane maneuvering characteristics.]

Note: Crosswind limit for landing is 20 knots.

[From FLIGHT CONTROLS checklist. Less control authority decreases crosswind landing capability.]

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

[From SPOILERS checklist.]

Note: Do not arm speedbrake lever.

[From AUTO SPEEDBRAKE checklist. Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.

[From AUTO SPEEDBRAKE checklist.]

Note: For go-around, observe gear EXTEND limit speed (270K/.82M).

[From GEAR DOOR checklist.]

Do not accomplish the following checklists:

AUTO SPEEDBRAKE

FLIGHT CONTROLS

SPOILERS

Continued on next page

Continued from previous page

If HYD PRESS SYS L+C message remains displayed: (continued)

-----**DEFERRED ITEMS**-----

==> APPROACH CHECKLIST

Initiate flap extension as required.

LANDING GEAR lever.DN

[Ensures nose gear extension if pressure still exists in that part of the system.]

ALTERNATE GEAR switch.DOWN and hold

Push and hold until all gear indicate in-transit.

[Releases gear uplocks and gear door locks allowing gear to free fall.]

Do not accomplish the following checklists:

FLAPS PRIMARY FAIL

SLATS PRIMARY FAIL

GEAR DOOR



HYD PRESS SYS L+C+R

Condition: **All hydraulic system pressures are low.**





Intentionally
Blank

[HYD PRESS SYS L+R

Condition: **Left and right hydraulic system pressures are low.**

LEFT DEMAND PUMP selector. ON

[Restores system pressure if AUTO position of selector is inoperative.]

RIGHT DEMAND PUMP selector. ON

[Restores system pressure if AUTO position of selector is inoperative.]

If HYD PRESS SYS L+R message remains displayed:

LEFT PRIMARY PUMP switch. OFF

RIGHT PRIMARY PUMP switch. OFF

LEFT DEMAND PUMP selector. OFF

[Avoids system contamination and/or pump damage.]

RIGHT DEMAND PUMP selector. OFF

[Avoids system contamination and/or pump damage.]

Handling qualities are degraded.

[From FLIGHT CONTROLS checklist. Pitch and roll control capability is reduced with fewer operating control surfaces.]

Plan to land at the nearest suitable airport.

[From FLIGHT CONTROLS checklist.]

GND PROX FLAP OVRD OVRD

[From FLIGHT CONTROLS checklist.]

Note: Inoperative items:

- **multiple flight control surfaces**
- **left and right thrust reversers**
- **autobrake**
- **normal brakes.**

Note: Use flaps 20 and VREF30 + 20 for landing.

[From FLIGHT CONTROLS checklist. Higher approach speeds improve airplane maneuvering characteristics.]

Continued on next page

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If HYD PRESS SYS L+R message remains displayed: (continued)

Note: Crosswind limit for landing is 20 knots.

[From FLIGHT CONTROLS checklist. Less control authority decreases crosswind landing capability.]

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

[From SPOILERS checklist.]

Do not accomplish the following checklists:

FLIGHT CONTROLS

SPOILERS



[HYD PRESS SYS R

Condition: **Right hydraulic system pressure is low.**

RIGHT DEMAND PUMP selector.ON

[Restores system pressure if AUTO position of selector is inoperative.]

If HYD PRESS SYS R message remains displayed:

RIGHT PRIMARY PUMP switch.OFF

[Avoids system contamination and/or pump damage.]

RIGHT DEMAND PUMP selector.OFF

[Avoids system contamination and/or pump damage.]

Note: Right thrust reverser is inoperative.

[From ENG REVERSER R checklist.]

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

[From SPOILERS checklist.]

Do not accomplish the following checklist:

ENG REVERSER R

SPOILERS



[HYD PRESS SYS R+C

Condition: **Right and center hydraulic system pressures are low.**

C1 or C2 PUMP selector.ON

[Restores system pressure if AUTO position of selector is inoperative.]

RIGHT DEMAND PUMP selector.ON

[Restores system pressure if AUTO position of selector is inoperative.]

If HYD PRESS SYS R+C message remains displayed:

C1 PUMP selector. AUTO

[Selector must stay in AUTO in case reserve steering has successfully isolated.]

C2 PUMP selector.OFF

[Avoids system contamination and/or pump damage.]

RIGHT PRIMARY PUMP switch.OFF

RIGHT DEMAND PUMP selector.OFF

[Avoids system contamination and/or pump damage.]

Handling qualities are degraded.

[From FLIGHT CONTROLS checklist. Pitch and roll control capability is reduced with fewer operating control surfaces.]

Plan to land at the nearest suitable airport.

[From FLIGHT CONTROLS checklist.]

GND PROX FLAP OVRD OVRD

[From FLIGHT CONTROLS, and FLAPS PRIMARY FAIL checklists.]

Note: Inoperative items:

- multiple flight control surfaces
- main landing gear hydraulic operation
- right thrust reverser

Note: Plan additional time for slower slat and flap operation.

[From FLAPS PRIMARY FAIL and SLATS PRIMARY FAIL checklists.]

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If HYD PRESS SYS R+C message remains displayed: (continued)

Note: Slats will extend beyond midrange when airspeed is below 215 knots. For go-around, do not exceed 215 knots until slats retract to midrange.

[From SLATS PRIMARY FAIL checklist.]

Note: Use flaps 20 and VREF30 + 20 for landing.

[From STABILIZER and FLIGHT CONTROLS checklists. Higher approach speeds improve airplane maneuvering characteristics.]

Note: Crosswind limit for landing is 20 knots.

[From FLIGHT CONTROLS checklist. Less control authority decreases crosswind landing capability.]

Note: Roll rate may be reduced in flight. Speedbrake effectiveness may be reduced in flight and during landing.

[From SPOILERS checklist.]

Note: Do not arm speedbrake lever.

[From AUTO SPEEDBRAKE checklist. Prevents inadvertent inflight extension.]

Note: Manually extend speedbrakes after landing.

[From AUTO SPEEDBRAKE checklist.]

Note: For go-around, observe gear EXTEND limit speed (270K/.82M).

[From GEAR DOOR checklist.]

Do not accomplish the following checklists:

AUTO SPEEDBRAKE

FLIGHT CONTROLS

SPOILERS

Continued on next page

Continued from previous page

If HYD PRESS SYS R+C message remains displayed: (continued)

-----**DEFERRED ITEMS**-----

==> APPROACH CHECKLIST

Initiate flap extension as required.

LANDING GEAR lever.DN

[Ensures nose gear extension if pressure still exists in that part of the system.]

ALTERNATE GEAR switch.DOWN and hold

Push and hold until all gear indicate in-transit.

[Releases gear uplocks and gear door locks allowing gear to free fall.]

Do not accomplish the following checklists:

FLAPS PRIMARY FAIL

SLATS PRIMARY FAIL

GEAR DOOR



HYD QTY LOW C, L, R

Condition: **Hydraulic quantity is low.**



RAT UNLOCKED

Condition: **Ram air turbine is not stowed and locked.**





Non-Normal Checklists

Chapter NNC

Landing Gear

Section 14

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■ ANTISKID

Condition: **A fault is detected in the antiskid system.**

Note: Autobrake system is inoperative. Use minimum braking consistent with runway conditions to reduce the possibility of tire blowout.



AUTOBRAKE

Condition: **Autobrake is disarmed or inoperative.**



BRAKE PEDALS CAPT, F/O

Condition: **One or both brake pedals are inoperative for the affected crewmember.**



■ BRAKE TEMP

Condition: **Temperature of one or more brakes is excessive.**

If in flight:

Observe gear EXTEND limit speed (270K/.82M).

LANDING GEAR lever. DN

[Allows cooling air to flow around brakes.]

When BRAKE TEMP message no longer displayed:

Wait 8 minutes.

[Ensures sufficient cooling time.]

LANDING GEAR lever. UP



If on the ground:

Refer to Brake Cooling Schedule table in Performance–Inflight chapter for required cooling time.



▣BRAKES

Condition: **Two or more wheel brakes are inoperative.**

Note: Wheel braking effectiveness is reduced. Refer to Reduced Braking table in Performance-Inflight chapter for flight planning.



▣GEAR DISAGREE

Condition: **Gear position disagrees with landing gear lever position.**

If landing gear lever UP:

Note: Observe gear EXTEND limit speed (270K/.82M).

[From GEAR DOOR checklist.]

Note: Flight with gear down increases fuel consumption and decreases climb performance. Refer to Gear Down performance tables in Performance-Inflight chapter for flight planning.

Do not accomplish the following checklists:

AUTO SPEEDBRAKE

GEAR DOOR



If landing gear lever DN:

Note: Observe gear EXTEND limit speed (270K/.82M).

[From GEAR DOOR checklist.]

ALTERNATE GEAR switch.DOWN and hold
Push and hold until all gear indicate in-transit.

[Releases gear uplocks and gear door locks allowing gear to free fall.]

Do not accomplish the following checklist:

GEAR DOOR

Wait 30 seconds.

Continued on next page

Continued from previous page

If landing gear lever DN: (continued)

If any gear remains UP or in transit:

Plan to land on available gear.

GND PROX GEAR OVRD OVRD

Note: Use flaps 30 for landing.

[Provides slowest landing speed.]

Note: Do not arm speedbrake lever.

Note: When stopping distance is critical, extend the speedbrakes after all gear, or the nose, or engine nacelle have contacted the runway.

Note: Do not use the thrust reversers unless stopping distance is critical.

When at pattern altitude:

OUTFLOW VALVE switches (Both). MAN

OUTFLOW VALVE MANUAL switches (Both). OPEN

Position outflow valves fully open to depressurize airplane.

FUEL PUMP switches (All). OFF

[Reduces possibility of fire.]

Do not accomplish the following checklists:

CABIN ALTITUDE AUTO

FUEL PRESS ENG L

FUEL PRESS ENG R



[] GEAR DOOR

Condition: One or more gear doors are not closed.

Note: Observe gear EXTEND limit speed (270K/.82M).



GEAR DRAG BRACE L, R

Condition: **Affected main gear is down with side brace locked and drag brace unlocked.**

Increase airspeed until GEAR DRAG BRACE message is no longer displayed (270K/ .82M maximum).

If GEAR DRAG BRACE message remains displayed:

GND PROX GEAR OVRD OVRD

Note: Use flaps 25 or 30 for landing.

[Provides slowest landing speed.]

Note: Do not arm speedbrake lever.

Note: Manually extend speedbrakes after landing.

[Allows coordinated speedbrake extension.]

When at pattern altitude:

FUEL PUMP switches (All). OFF

[Reduces possibility of fire.]

Do not accomplish the following checklists:

FUEL PRESS ENG L

FUEL PRESS ENG R



U GEAR LEVER LOCKED DN

Condition: **Landing gear lever cannot be positioned to UP.**

Landing gear lever lock

OVERRIDE switch. PUSH and hold

Landing gear lever. UP



[] GEAR SIDE BRACE L, R

Condition: **Affected main gear is down with drag brace locked and side brace unlocked.**

Decrease airspeed until GEAR SIDE BRACE message is no longer displayed (VREF minimum).

If GEAR SIDE BRACE message remains displayed:

GND PROX GEAR OVRD. OVRD

Note: Use flaps 25 or 30 for landing.

[Provides slowest landing speed.]

Note: Do not arm speedbrake lever.

Note: Manually extend speedbrakes after landing.

[Allows coordinated speedbrake extension.]

When at pattern altitude:

FUEL PUMP switches (All). OFF

[Reduces possibility of fire.]

Do not accomplish the following checklists:

FUEL PRESS ENG L

FUEL PRESS ENG R



[] NOSE WHEEL STEERING

Condition: **Nose wheel steering is not available.**

Note: Do not taxi with loss of steering.



TILLER L

Condition: **Tiller is inoperative on the affected side.**



TILLER R

Condition: **(TBD)**



TIRE PRESS

Condition: **Tire pressures are not normal.**



**Non-Normal Checklists****Chapter NNC****Warning Systems****Section 15****Table of Contents**

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AIRSPPEED LOW

Condition: **Airspeed is below minimum maneuvering speed.**



ALTITUDE ALERT

Condition: **Airplane has deviated from the MCP selected altitude.**



ALTITUDE CALLOUTS

Condition: **Altitude and minimums voice annunciations during approach are no longer provided.**



CONFIG DOORS

Condition: **An entry, forward cargo, or aft cargo door is not closed and latched and locked when either engine's thrust is in the takeoff range on the ground.**



CONFIG FLAPS

Condition: **Flaps are not in a takeoff position when either engine's thrust is in the takeoff range on the ground.**



CONFIG GEAR

Condition: **Any landing gear is not down and locked when either thrust lever is closed below 800 feet radio altitude or when flaps are in a landing position.**



CONFIG PARKING BRAKE

Condition: **Parking brake is set when either engine's thrust is in the takeoff range on the ground.**



CONFIG RUDDER

Condition: **Rudder trim is not centered when either engine's thrust is in the takeoff range on the ground.**



CONFIG SPOILERS

Condition: **Speedbrake lever is not DOWN when either engine's thrust is in the takeoff range on the ground.**



CONFIG STABILIZER

Condition: **Stabilizer is not within the greenband when either engine's thrust is in the takeoff range on the ground.**



[]CONFIG WARNING SYS

Condition: **A fault is detected in the configuration warning system.**

Note: Radio altitude voice callouts and other aural alerts may not be available.



GPWS FLAP OVRD

Condition: **GPWS FLAP OVERRIDE is in the OVRD position.**



GPWS GEAR OVRD

Condition: **GPWS GEAR OVERRIDE is in the OVRD position.**



GPWS TERR OVRD

Condition: **GPWS TERR OVERRIDE is in the OVRD position.**



GND PROX SYS

Condition: **Ground proximity alerts may not be provided.**

Note: Some or all ground proximity alerts are not available.
Ground proximity alerts which occur are valid.



OVERSPEED

Condition: **Airspeed has exceeded Vmo/Mmo.**



PILOT RESPONSE

Condition: **Recent pilot activity has not been detected.**



TAIL STRIKE

Condition: **A tail strike has been detected.**

OUTFLOW VALVE switches (Both). MAN

OUTFLOW VALVE MANUAL switches (Both). OPEN

**Position outflow valves fully open to depressurize
airplane.**

Plan to land at the nearest suitable airport.

Do not accomplish the following checklist:

CABIN ALTITUDE AUTO



TCAS

Condition: **TCAS has failed.**



TCAS OFF

Condition: **TCAS is in standby mode.**



TCAS RA CAPTAIN, F/O

Condition: **TCAS cannot display RA guidance on the affected PFD.**



[]TERR POS

Condition: **Terrain position data has been lost.**

Note: Position data for the ND terrain map and look-ahead terrain alerts are lost. Ground proximity alerts which occur are valid.



[]WINDSHEAR SYS

Condition: **Windshear alerts may not be provided.**

Note: Some or all windshear alerts are not available. Windshear alerts which occur are still valid.



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**Maneuvers****Chapter MAN****Non-Normal Maneuvers****Section 1**

Approach to Stall Recovery

The following is immediately accomplished at the first indication of stall buffet or stick shaker.

Pilot Flying	Pilot Monitoring
• Advance the thrust levers to maximum thrust. • Smoothly adjust the pitch attitude* to avoid ground contact or obstacles. • Level the wings (do not change flaps or landing gear configuration). • Retract the speedbrakes.	• Verify maximum thrust. • Monitor altitude and airspeed. • Call out any trend toward terrain contact.
When ground contact is no longer a factor: • Adjust the pitch attitude to accelerate while minimizing altitude loss. • Return to a speed appropriate for the configuration.	

Note: *At high altitudes it may be necessary to decrease pitch attitude below the horizon to achieve acceleration.

Rejected Takeoff

The captain has the sole responsibility for the decision to reject the takeoff. The decision must be made in time to start the rejected takeoff maneuver by V1. If the decision is to reject the takeoff, the captain must clearly announce “REJECT,” immediately start the rejected takeoff maneuver, and assume control of the airplane. If the first officer is making the takeoff, the first officer must maintain control of the airplane until the captain makes a positive input to the controls.

Prior to 80 knots, the takeoff should be rejected for any of the following:

- activation of the master caution system
- system failure(s)
- unusual noise or vibration
- tire failure
- abnormally slow acceleration
- unsafe takeoff configuration warning
- fire or fire warning
- engine failure
- predictive windshear warning
- if the airplane is unsafe or unable to fly.

Above 80 knots and prior to V1, the takeoff should be rejected for any of the following:

- fire or fire warning
- engine failure
- predictive windshear warning
- if the airplane is unsafe or unable to fly.

During the takeoff, the crew member observing the non-normal situation will immediately call it out as clearly as possible.

787 Flight Crew Operations Manual

Captain	First Officer
Without delay: Simultaneously close the thrust levers, disengage the autothrottles, and apply maximum manual wheel brakes or verify operation of RTO autobrakes. If RTO autobrakes is selected, monitor system performance and apply manual wheel brakes if the AUTOBRAKE message is displayed or deceleration is not adequate. Raise speedbrake lever. Apply maximum reverse thrust consistent with conditions. Continue maximum braking until certain the airplane will stop on the runway.	Verify actions as follows: Thrust levers closed. Autothrottles disengaged. Maximum brakes applied. Verify speedbrake lever UP and call "SPEEDBRAKES UP." If speedbrake lever not UP call "SPEEDBRAKES NOT UP." Reverse thrust applied. Call out any omitted action items.
Field length permitting: Initiate movement of the reverse thrust levers to reach the reverse idle detent by taxi speed.	Call out 60 knots. Communicate the reject decision to the control tower and cabin as soon as practical.
When the airplane is stopped, perform procedures as required. Review Brake Cooling Schedule for brake cooling time and precautions (refer to the Performance Inflight chapter). Consider the following: • the possibility of wheel fuse plugs melting • the need to clear the runway • the requirement for remote parking • wind direction in case of fire • alerting fire equipment • not setting the parking brake unless passenger evacuation is necessary • advising the ground crew of the hot brake hazard • advising passengers of the need to remain seated or evacuate • completion of Non-Normal checklist (if appropriate) for conditions which caused the RTO.	



Terrain Avoidance

Ground Proximity Caution

Accomplish the following maneuver for any of these aural alerts*:

- CAUTION TERRAIN
- SINK RATE
- TERRAIN
- DON'T SINK
- TOO LOW FLAPS
- TOO LOW GEAR
- TOO LOW TERRAIN
- GLIDESLOPE
- BANK ANGLE.

Pilot Flying	Pilot Monitoring
Correct the flight path or the airplane configuration.	

The below glideslope deviation alert may be cancelled or inhibited for:

- localizer or backcourse approach
- circling approach from an ILS
- when conditions require a deliberate approach below glideslope
- unreliable glideslope signal.

Note: If a terrain caution occurs when flying under daylight VMC, and positive visual verification is made that no obstacle or terrain hazard exists, the alert may be regarded as cautionary and the approach may be continued.

Note: *As installed, some repeat.

Ground Proximity Warning

Accomplish the following maneuver for any of these conditions:

- activation of the “PULL UP”, or “TERRAIN TERRAIN PULL UP” warning
- other situations resulting in unacceptable flight toward terrain.



787 Flight Crew Operations Manual

Pilot Flying	Pilot Monitoring
• Disconnect autopilot. • Disconnect autothrottle(s). • Aggressively apply maximum* thrust. • Simultaneously roll wings level and rotate to an initial pitch attitude of 20°. • Retract speedbrakes. • If terrain remains a threat, continue rotation up to the pitch limit indicator or stick shaker or initial buffet.	• Assure maximum* thrust. • Verify all required actions have been completed and call out any omissions.
• Do not change gear or flap configuration until terrain separation is assured. • Monitor radio altimeter for sustained or increasing terrain separation. • When clear of the terrain, slowly decrease pitch attitude and accelerate.	• Monitor vertical speed and altitude (radio altitude for terrain clearance and barometric altitude for a minimum safe altitude). • Call out any trend toward terrain contact.

Note: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be required to obtain positive terrain separation. Smooth, steady control will avoid a pitch attitude overshoot and stall.

Note: Do not use flight director commands.

Note: * Maximum thrust can be obtained by advancing the thrust levers full forward if the EECs are in the normal mode. If terrain contact is imminent, advance thrust levers full forward.

Note: If positive visual verification is made that no obstacle or terrain hazard exists when flying under daylight VMC conditions prior to a terrain or obstacle (as installed) warning, the alert may be regarded as cautionary and the approach may be continued.

Traffic Avoidance

Immediately accomplish the following by recall whenever a TCAS traffic advisory (TA) or resolution advisory (RA) occurs.

WARNING: Comply with RA if there is conflict between RA and air traffic control.

WARNING: Once an RA has been issued, safe separation could be compromised if current vertical speed is changed, except as necessary to comply with the RA. This is because TCAS II-to-TCAS II coordination may be in progress with the intruder aircraft, and any change in vertical speed that does not comply with the RA may negate the effectiveness of the other aircraft's compliance with the RA.

Note: If stick shaker or initial buffet occurs during the maneuver, immediately accomplish the APPROACH TO STALL RECOVERY procedure.

Note: If high speed buffet occurs during the maneuver, relax pitch force as necessary to reduce buffet, but continue the maneuver.

Note: Do not use flight director pitch commands until clear of conflict.

For TA:

Pilot Flying	Pilot Monitoring
Look for traffic using traffic display as a guide. Call out any conflicting traffic	
If traffic is sighted, maneuver as required.	

WARNING: A DESCEND (fly down) RA issued below 1000 feet AGL should not be followed.

For RA, except a climb in landing configuration:

Pilot Flying	Pilot Monitoring
If maneuvering is required, disengage the autopilot and autothrottle. Smoothly adjust pitch and thrust to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action.	
Attempt to establish visual contact. Call out any conflicting traffic.	

For a climb RA in landing configuration:

Pilot Flying	Pilot Monitoring
Disengage the autopilot and autothrottle. Advance thrust levers forward to ensure maximum thrust is attained and call for FLAPS 20. Smoothly adjust pitch to satisfy the RA command. Follow the planned lateral flight path unless visual contact with the conflicting traffic requires other action.	Verify maximum thrust set. Position flap lever to 20 detent.
After positive rate of climb established, call for GEAR UP.	Position gear lever up.
Attempt to establish visual contact. Call out any conflicting traffic.	

Upset Recovery

An upset can generally be defined as unintentionally exceeding the following conditions:

- pitch attitude greater than 25 degrees nose up, or
- pitch attitude greater than 10 degrees nose down, or
- bank angle greater than 45 degrees, or
- within above parameters but flying at airspeeds inappropriate for the conditions.

The following techniques represent a logical progression for recovering the airplane. The sequence of actions is for guidance only and represents a series of options to be considered and used depending on the situation. Not all the actions may be necessary once recovery is underway. If needed, use pitch trim sparingly. Careful use of rudder to aid roll control should be considered only if roll control is ineffective and the airplane is not stalled.

These techniques assume that the airplane is not stalled. A stalled condition can exist at any attitude and may be recognized by continuous stick shaker activation accompanied by one or more of the following:

- buffeting, which could be heavy at times
- lack of pitch authority and/or roll control
- inability to arrest descent rate.

Nose High Recovery

Pilot Flying	Pilot Monitoring
• Recognize and confirm the situation.	
• Disconnect autopilot and autothrottle. • Apply as much as full nose down elevator. • *Apply appropriate nose down stabilizer trim. • Reduce thrust. • *Roll (adjust bank angle) to obtain a nose down pitch rate. • Complete the recovery: • when approaching the horizon, roll to wings level • check airspeed and adjust thrust • establish pitch attitude.	• Call out attitude, airspeed and altitude throughout the recovery. • Verify all required actions have been completed and call out any omissions.

Nose Low Recovery

Pilot Flying	Pilot Monitoring
Recognize and confirm the situation.	
- Disconnect autopilot and autothrottle. - Recover from stall, if required. *Roll in the shortest direction to wings level (unload and roll if bank angle is more than 90 degrees). - Recover to level flight: - apply nose up elevator *apply nose up trim, if required - adjust thrust and drag as required.	- Call out attitude, airspeed and altitude throughout the recovery. - Verify all required actions have been completed and call out any omissions.

WARNING: * EXCESSIVE USE OF PITCH TRIM OR RUDDER MAY AGGRAVATE AN UPSET SITUATION OR MAY RESULT IN LOSS OF CONTROL AND/OR HIGH STRUCTURAL LOADS.

Windshear

Windshear Caution

For predictive windshear caution alert: (“MONITOR RADAR DISPLAY” aural)

Pilot Flying	Pilot Monitoring
Maneuver as required to avoid the windshear.	

Windshear Warning

Predictive windshear warning during takeoff roll: (“WINDSHEAR AHEAD, WINDSHEAR AHEAD” aural)

- Prior to V1, reject takeoff.
- After V1, perform the Windshear Escape Maneuver.

Windshear encountered during takeoff roll:

- If windshear is encountered prior to V₁, there may not be sufficient runway remaining to stop if an RTO is initiated at V₁. At VR, rotate at a normal rate toward a 15 degree pitch attitude. Once airborne, perform the Windshear Escape Maneuver.
- If windshear is encountered near the normal rotation speed and airspeed suddenly decreases, there may not be sufficient runway left to accelerate back to normal takeoff speed. If there is insufficient runway left to stop, initiate a normal rotation at least 2,000 feet before the end of the runway even if airspeed is low. Higher than normal attitudes may be required to lift off in the remaining runway. Ensure maximum thrust is set.

Predictive windshear warning during approach: (“GO-AROUND, WINDSHEAR AHEAD” aural)

- perform Windshear Escape Maneuver or, at pilot’s discretion, perform a normal go-around.

Windshear encountered in flight:

- perform the Windshear Escape Maneuver.

Note: The following are indications the airplane is in windshear:

- windshear warning (two-tone siren followed by “WINDSHEAR, WINDSHEAR, WINDSHEAR”) or
- unacceptable flight path deviations.

Note: Unacceptable flight path deviations are recognized as uncontrolled changes from normal steady state flight conditions below 1000 feet AGL, in excess of any of the following:

- 15 knots indicated airspeed
- 500 FPM vertical speed
- 5 degrees pitch attitude
- 1 dot displacement from the glideslope
- unusual thrust lever position for a significant period of time.



Windshear Escape Maneuver

MANUAL FLIGHT

Pilot Flying	Pilot Monitoring
• Disconnect autopilot. • Push either TO/GA switch. • Aggressively apply maximum* thrust. • Disconnect autothrottle(s). • Simultaneously roll wings level and rotate toward an initial pitch attitude of 15°. • Retract speedbrakes. • Follow flight director TO/GA guidance (if available).	• Assure maximum* thrust. • Verify all required actions have been completed and call out any omissions.

AUTOMATIC FLIGHT

Pilot Flying	Pilot Monitoring
• Press either TO/GA switch.** • Verify TO/GA mode annunciation. • Verify thrust advances to GA power. • Retract speedbrakes. • Monitor system performance***.	• Assure maximum* thrust. • Verify all required actions have been completed and call out any omissions.
• Do not change gear or flap configuration until windshear is no longer a factor. • Monitor vertical speed and altitude. • Do not attempt to regain lost airspeed until windshear is no longer a factor.	• Monitor vertical speed and altitude. • Call out any trend toward terrain contact, descending flight path, or significant airspeed changes.

Note: Aft control column force increases as the airspeed decreases. In all cases, the pitch attitude that results in intermittent stick shaker or initial buffet is the upper pitch attitude limit. Flight at intermittent stick shaker may be required to obtain positive terrain separation. Smooth, steady control will avoid a pitch attitude overshoot and stall.

Note: * Maximum thrust can be obtained by advancing the thrust levers full forward if the EECs are in the normal mode. If terrain contact is imminent, advance thrust levers full forward.

Note: ** If TO/GA is not available, disconnect autopilot and autothrottle(s) and fly manually.

WARNING: *** Severe windshear may exceed the performance capability of the AFDS. The pilot flying must be prepared to disconnect the autopilot and autothrottle(s) and fly manually.

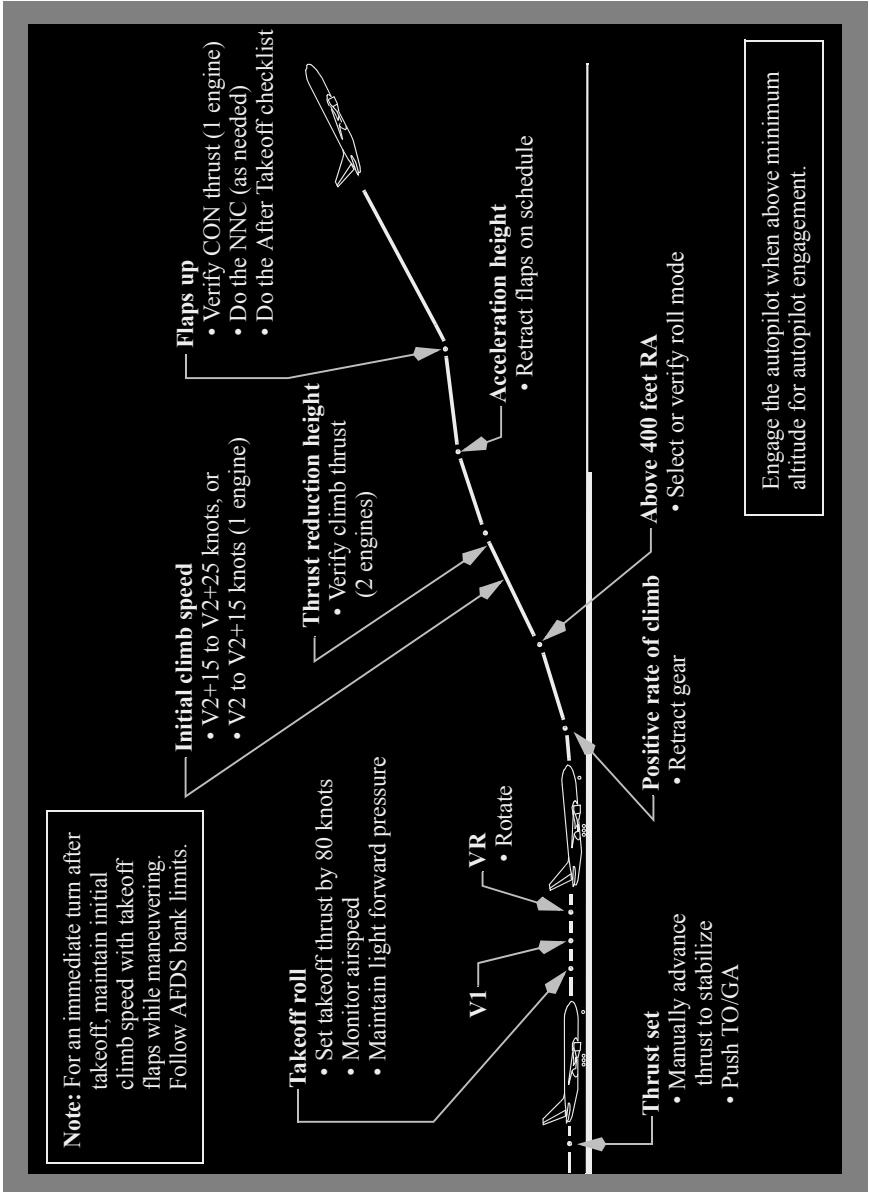
Maneuvers

Flight Patterns

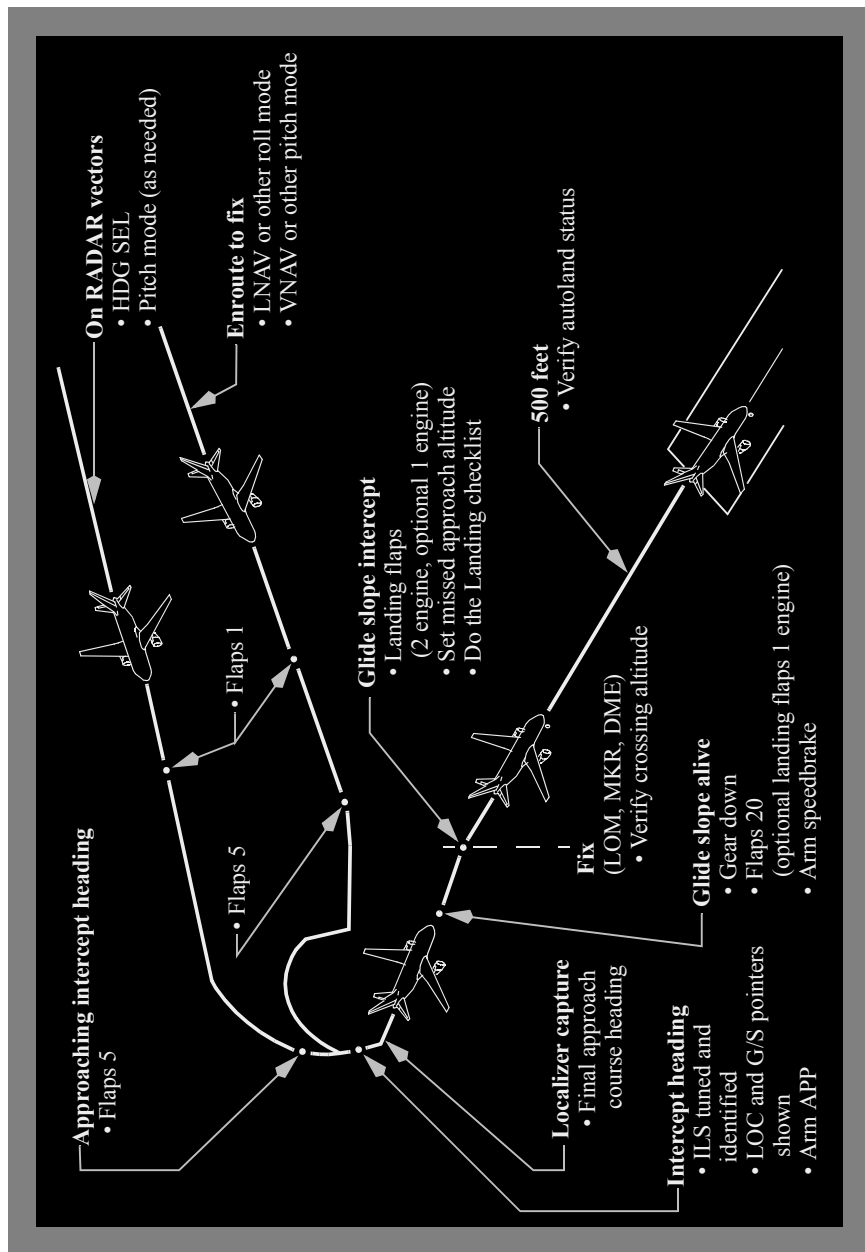
Chapter MAN

Section 2

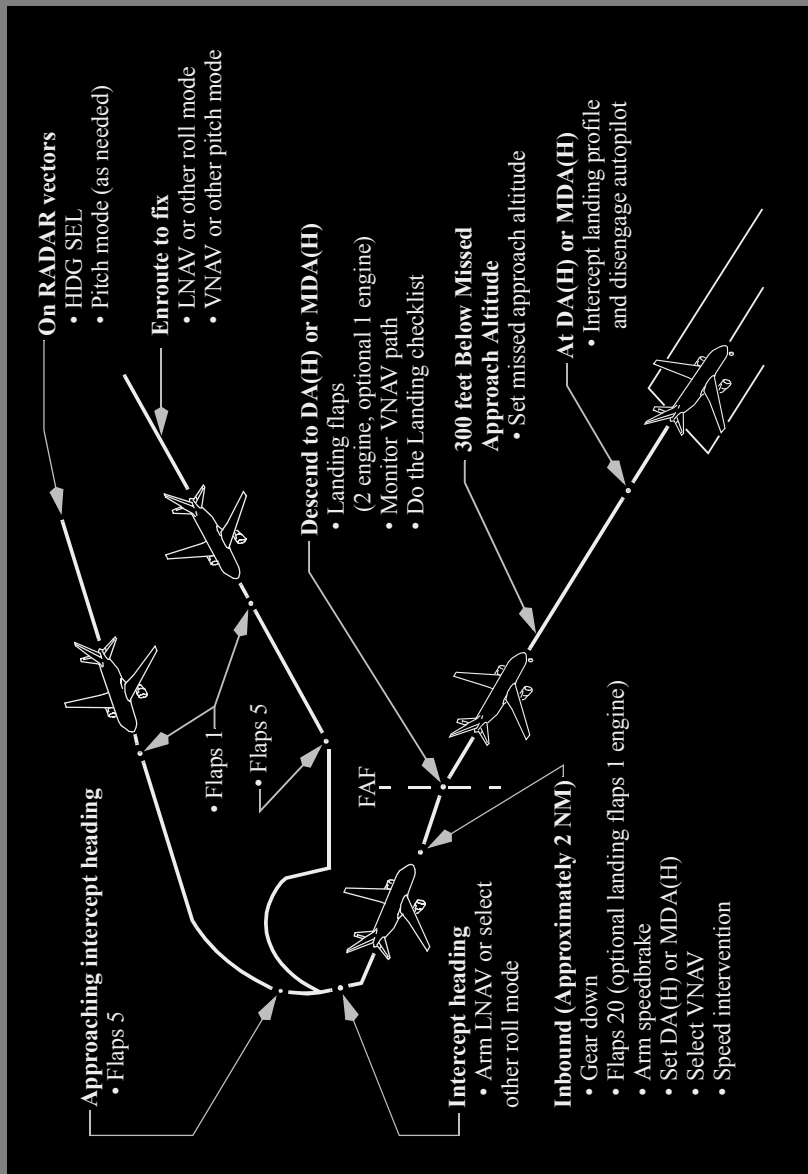
Takeoff



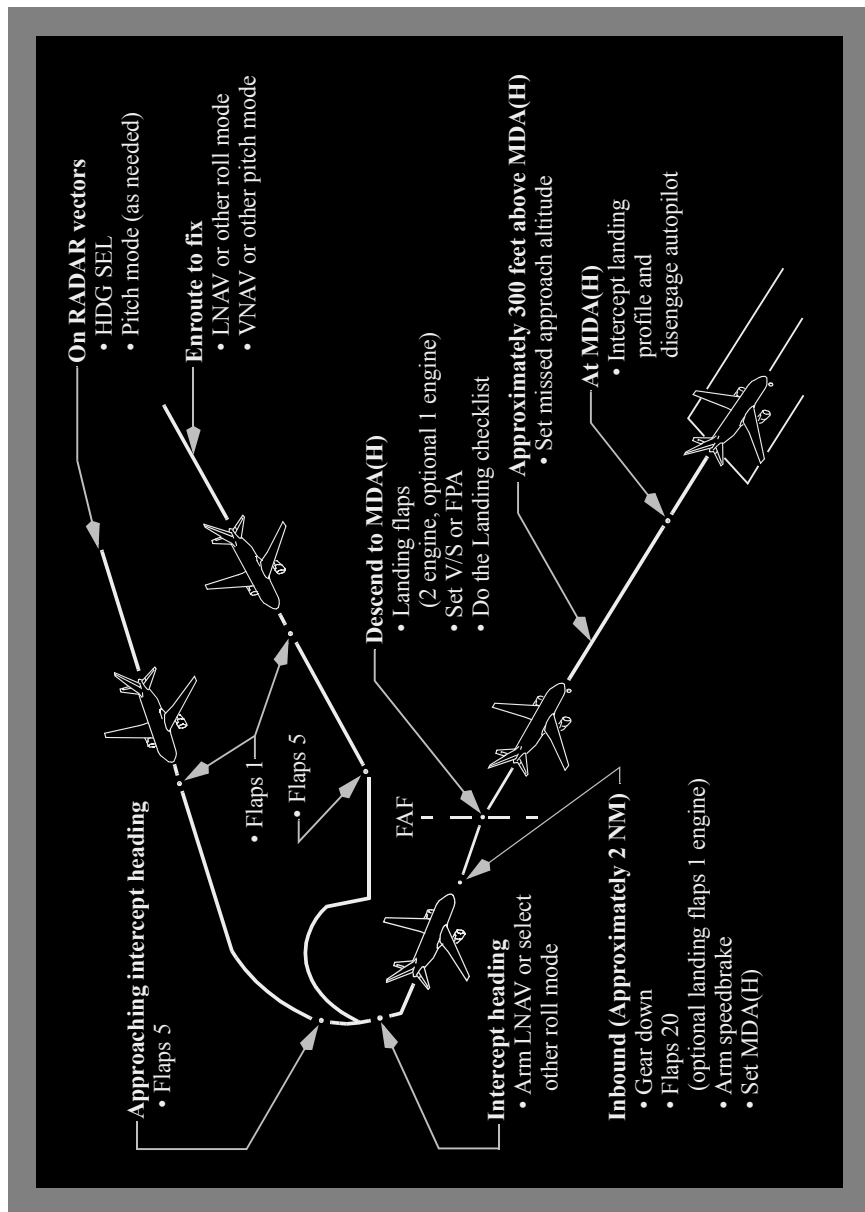
ILS Approach



Instrument Approach Using VNAV



Instrument Approach Using V/S or FPA



Circling Approach

If a missed approach is needed at any time while circling, make an initial climbing turn toward the landing runway and intercept the missed approach course.

Configuration at MDA(H)

- Gear down
- Flaps 20 (landing flaps optional)
- Arm speedbrake

Turning base

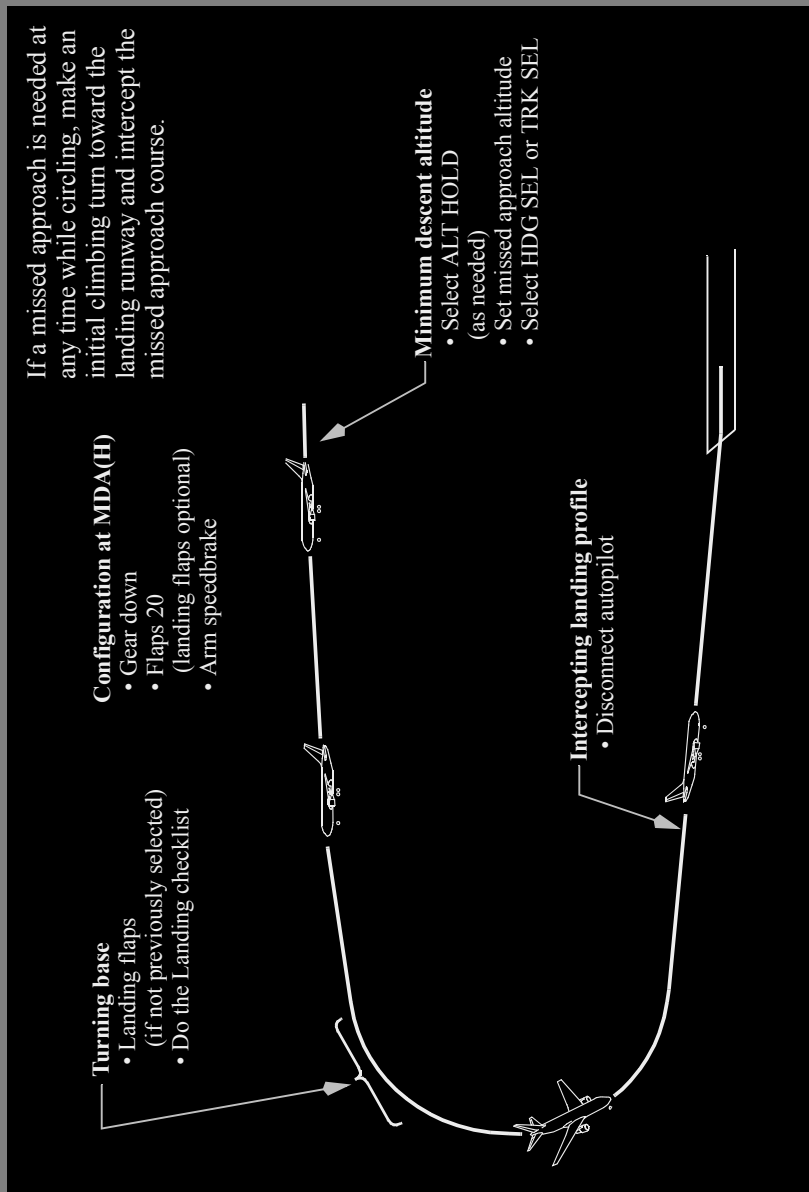
- Landing flaps (if not previously selected)
- Do the Landing checklist

Minimum descent altitude

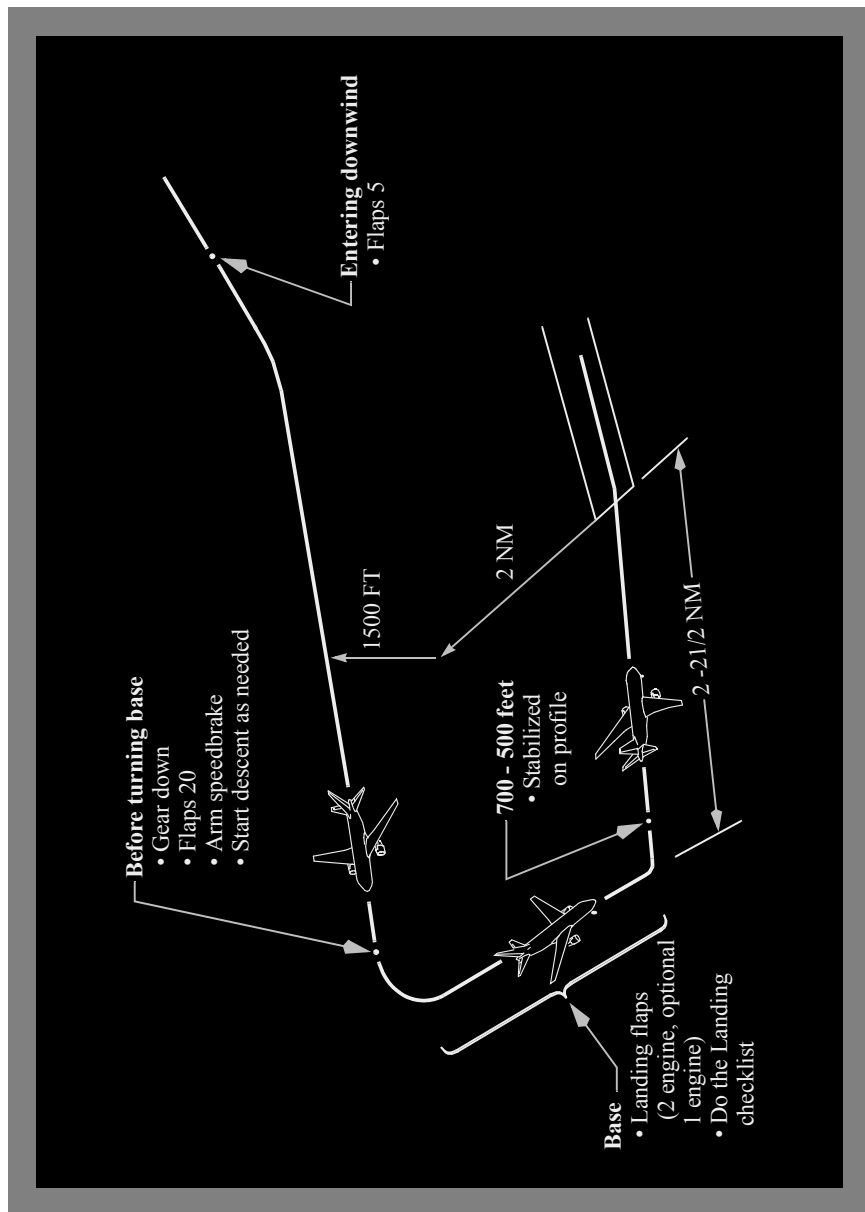
- Select ALT HOLD (as needed)
- Set missed approach altitude
- Select HDG SEL or TRK SEL

Intercepting landing profile

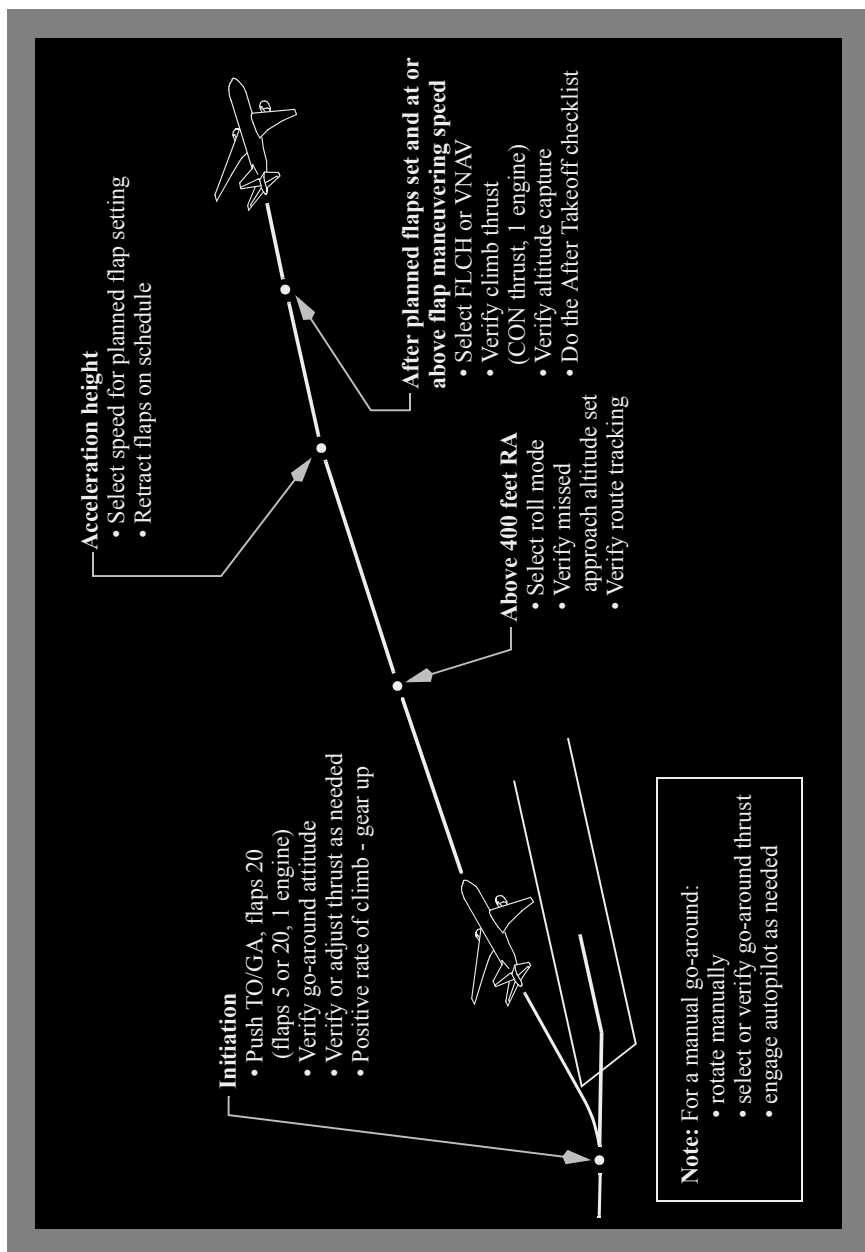
- Disconnect autopilot



Visual Traffic Pattern



Go-Around and Missed Approach



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Performance Inflight A
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777-200ER TRENT892 LB FAA

Flight With Unreliable Airspeed /

Turbulent Air Penetration PI.A12.1

Max Climb EPR	PI.A12.1
VREF	PI.A12.1



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Performance Inflight A

Advisory Information

Chapter PI

Section A10

ADVISORY INFORMATION

Flight With Unreliable Airspeed / Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Climb

Flaps Up, Set Max Climb Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 LB)			
		350	450	550	650
40000 (.82M)	PITCH ATT	4.0	3.5		
	V/S (FT/MIN)	1600	700		
30000 (280 KIAS)	PITCH ATT	4.5	4.0	4.0	4.5
	V/S (FT/MIN)	2300	1600	1000	500
20000 (270 KIAS)	PITCH ATT	7.5	6.5	6.5	6.5
	V/S (FT/MIN)	3700	2700	1900	1400
10000 (270 KIAS)	PITCH ATT	10.0	8.5	8.0	8.0
	V/S (FT/MIN)	4500	3300	2500	1900
SEA LEVEL (270 KIAS)	PITCH ATT	12.5	10.5	9.5	9.0
	V/S (FT/MIN)	5100	3800	3000	2300

Cruise

Flaps Up, Set Thrust for Level Flight

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 LB)			
		350	450	550	650
40000 (.82 M)	PITCH ATT	2.0	2.5		
	EPR	1.127	1.267		
	(Alt Mode %N1)	(78.9)	(84.0)		
35000 (280 KIAS)	PITCH ATT	1.5	2.0	2.5	
	EPR	1.064	1.124	1.229	
	(Alt Mode %N1)	(76.6)	(79.2)	(83.2)	
30000 (280 KIAS)	PITCH ATT	1.5	2.0	3.0	3.5
	EPR	1.020	1.064	1.134	1.234
	(Alt Mode %N1)	(73.0)	(75.5)	(78.9)	(83.2)
25000 (280 KIAS)	PITCH ATT	1.5	2.5	3.0	4.0
	EPR	0.996	1.027	1.080	1.146
	(Alt Mode %N1)	(69.6)	(72.0)	(75.1)	(78.8)
20000 (270 KIAS)	PITCH ATT	1.5	2.5	3.5	4.5
	EPR	0.985	1.011	1.049	1.098
	(Alt Mode %N1)	(65.1)	(68.0)	(71.2)	(74.8)
15000 (270 KIAS)	PITCH ATT	1.5	2.5	3.5	4.5
	EPR	0.976	0.996	1.025	1.060
	(Alt Mode %N1)	(61.7)	(64.3)	(67.6)	(71.1)

ADVISORY INFORMATION

Flight With Unreliable Airspeed / Turbulent Air Penetration
Altitude and/or vertical speed indications may also be unreliable.

Descent

Flaps Up, Set Idle Thrust

PRESSURE ALTITUDE (FT) (SPEED)		WEIGHT (1000 LB)			
		350	450	550	650
40000 (.82M)	PITCH ATT	-1.0	-0.5		
	V/S (FT/MIN)	-2500	-2400		
30000 (280 KIAS)	PITCH ATT	-1.5	-0.5	0.5	1.0
	V/S (FT/MIN)	-2200	-1900	-1800	-1900
20000 (270 KIAS)	PITCH ATT	-1.0	0.0	1.0	2.0
	V/S (FT/MIN)	-1800	-1600	-1600	-1600
10000 (270 KIAS)	PITCH ATT	-1.5	-0.5	1.0	1.5
	V/S (FT/MIN)	-1700	-1500	-1400	-1400
SEA LEVEL (270 KIAS)	PITCH ATT	-1.5	-0.5	0.5	1.5
	V/S (FT/MIN)	-1500	-1400	-1300	-1300

Holding

Flaps Up, Set Thrust for Level Flight

PRESSURE ALTITUDE (FT)		WEIGHT (1000 LB)			
		350	450	550	650
10000	PITCH ATT	4.0	4.5	5.0	5.0
	EPR	1.003	1.018	1.031	1.045
	(Alt Mode %N1)	(51.1)	(57.3)	(62.8)	(67.3)
	KIAS	201	218	238	259
5000	PITCH ATT	4.0	4.5	5.0	5.0
	EPR	0.999	1.010	1.021	1.030
	(Alt Mode %N1)	(47.3)	(53.1)	(58.5)	(63.3)
	KIAS	202	218	237	258

ADVISORY INFORMATION

Flight With Unreliable Airspeed / Turbulent Air Penetration

Altitude and/or vertical speed indications may also be unreliable.

Terminal Area (5000 FT)**Set Thrust for Level Flight**

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 LB)			
		350	450	550	650
FLAPS 0 (GEAR UP) (VREF30 + 80)	PITCH ATT	4.5	5.0	5.5	6.0
	EPR	1.000	1.010	1.030	1.040
	(Alt Mode %N1)	(48.2)	(54.4)	(59.8)	(65.0)
	KIAS	201	217	232	245
FLAPS 1 (GEAR UP) (VREF30 + 60)	PITCH ATT	6.0	6.5	7.0	7.5
	EPR	1.020	1.030	1.050	1.070
	(Alt Mode %N1)	(49.3)	(56.2)	(61.9)	(66.7)
	KIAS	181	197	212	225
FLAPS 5 (GEAR UP) (VREF30 + 40)	PITCH ATT	5.0	5.5	6.0	6.0
	EPR	1.030	1.050	1.070	1.090
	(Alt Mode %N1)	(49.9)	(57.7)	(63.5)	(68.2)
	KIAS	161	177	192	205
FLAPS 15 (GEAR UP) (VREF30 + 20)	PITCH ATT	6.0	6.0	6.5	6.5
	EPR	1.040	1.070	1.090	1.120
	(Alt Mode %N1)	(51.2)	(59.4)	(65.5)	(70.3)
	KIAS	141	157	172	185
FLAPS 20 (GEAR DOWN) (VREF30 + 20)	PITCH ATT	4.0	4.5	5.0	5.0
	EPR	1.070	1.100	1.140	1.180
	(Alt Mode %N1)	(58.4)	(66.2)	(71.5)	(76.4)
	KIAS	141	157	172	185

Final Approach (1500 FT)**Gear Down, Set Thrust for 3° Glideslope**

FLAP POSITION (VREF + INCREMENT)		WEIGHT (1000 LB)			
		350	450	550	650
FLAPS 20 (VREF20 + 10)	PITCH ATT	1.0	1.5	1.5	1.5
	EPR	1.010	1.010	1.020	1.020
	(Alt Mode %N1)	(35.2)	(41.2)	(46.4)	(50.9)
	KIAS	141	159	175	189
FLAPS 25 (VREF25 + 10)	PITCH ATT	0.0	0.5	0.5	0.5
	EPR	1.030	1.040	1.060	1.080
	(Alt Mode %N1)	(43.9)	(51.7)	(57.7)	(62.2)
	KIAS	137	154	169	183
FLAPS 30 (VREF30 + 10)	PITCH ATT	-0.5	0.0	0.0	0.0
	EPR	1.050	1.070	1.090	1.120
	(Alt Mode %N1)	(50.3)	(57.9)	(63.8)	(68.3)
	KIAS	131	147	162	175

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 30

Dry Runway

	LANDING DISTANCE AND ADJUSTMENTS (FT)											
	REF DIST*	WT ADJ	ALT ADJ	WIND ADJ PER 10 KTS		SLOPE ADJ PER 1%		TEMP ADJ PER 10°C		VREF ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	400000 LB LANDING WT	PER 10000 LB ABOVE 40000 LB	PER 1000 FT ABOVE S.L.	HEAD WIND	TAIL WIND	DN HILL	UP HILL	ABV ISA	BLW ISA	PER 10 KTS ABOVE VREF30	ONE REV	NO REV
MAX MANUAL	2740	+70/-40	50	-110	420	30	-30	50	-50	220	30	80
MAX AUTO	3620	+60/-40	80	-160	580	0	0	80	-80	380	0	0
AUTOBRAKE 4	4440	+80/-60	110	-220	800	0	-20	110	-110	480	0	0
AUTOBRAKE 3	5340	+100/-80	140	-270	1000	40	-40	140	-140	580	10	20
AUTOBRAKE 2	5970	+120/-100	170	-320	1160	50	-110	160	-160	530	50	50
AUTOBRAKE 1	6380	+140/-120	190	-350	1300	130	-170	170	-170	510	350	360

Good Reported Braking Action

MAX MANUAL	3730	+70/-60	90	-190	720	90	-80	80	-90	310	160	380
MAX AUTO	4090	+70/-70	100	-190	740	70	-60	90	-90	360	180	410
AUTOBRAKE 4	4460	+80/-80	110	-220	830	20	-30	110	-110	480	20	80
AUTOBRAKE 3	5340	+100/-100	140	-270	1000	40	-40	140	-140	580	10	20

Medium Reported Braking Action

MAX MANUAL	4970	+100/-100	140	-290	1190	230	-170	120	-120	390	480	1220
MAX AUTO	5250	+100/-100	140	-290	1170	220	-140	120	-120	450	460	1210
AUTOBRAKE 4	5250	+110/-100	140	-290	1190	220	-140	130	-130	440	510	1300
AUTOBRAKE 3	5600	+110/-110	150	-320	1260	160	-100	150	-150	580	240	890

Poor Reported Braking Action

MAX MANUAL	6320	+150/-130	200	-430	1880	570	-330	160	-160	450	1060	2970
MAX AUTO	6700	+150/-130	200	-430	1860	570	-340	160	-160	440	1070	3000
AUTOBRAKE 4	6700	+150/-140	200	-430	1870	550	-340	160	-160	440	1090	3050
AUTOBRAKE 3	6700	+150/-130	200	-430	1880	560	-280	170	-170	560	1050	3020

*Reference distance is for sea level, standard day, no wind or slope, VREF30 approach speed and 2 engine reverse thrust.

Max Manual braking data valid for auto speedbrakes. For manual speedbrakes, increase reference landing distance by 200 ft.

Autobrake data valid for both auto and manual speedbrakes.

Actual (unfactored) distances are shown.

Includes distance from 50 ft above threshold (1000 ft of air distance).



ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 25

Dry Runway

BRAKING CONFIGURATION	LANDING DISTANCE AND ADJUSTMENTS (FT)											
	REF DIST*	WT ADJ	ALT ADJ	WIND ADJ PER 10 KTS		SLOPE ADJ PER 1%		TEMP ADJ PER 10°C		VREF ADJ	REVERSE THRUST ADJ	
	400000 LB LANDING WT	PER 10000 LB ABOVE 400000 LB	PER 1000 FT ABOVE S.L.	HEAD WIND	TAIL WIND	DN HILL	UP HILL	ABV ISA	BLW ISA	PER 10 KTS ABOVE VREF25	ONE REV	NO REV
MAX MANUAL	2860	+80/-40	60	-120	430	30	-30	80	-60	220	40	90
MAX AUTO	3860	+70/-40	90	-170	600	0	0	90	-90	400	0	0
AUTOBRAKE 4	4770	+90/-70	120	-230	830	0	-20	130	-130	500	0	0
AUTOBRAKE 3	5760	+110/-90	150	-290	1040	40	-60	160	-160	580	10	20
AUTOBRAKE 2	6420	+130/-110	190	-330	1200	90	-140	170	-170	500	140	140
AUTOBRAKE 1	6750	+150/-130	210	-360	1330	170	-190	180	-180	500	530	560

Good Reported Braking Action

MAX MANUAL	3920	+70/-70	100	-190	730	100	-80	90	-90	310	190	460
MAX AUTO	4330	+80/-70	100	-200	760	70	-50	100	-100	370	210	500
AUTOBRAKE 4	4790	+90/-90	120	-230	860	20	-30	130	-130	500	20	80
AUTOBRAKE 3	5760	+110/-110	150	-290	1040	40	-60	160	-160	580	10	20

Medium Reported Braking Action

MAX MANUAL	5210	+110/-100	150	-300	1210	240	-180	130	-130	390	550	1430
MAX AUTO	5520	+110/-100	150	-300	1200	220	-140	130	-130	450	530	1410
AUTOBRAKE 4	5520	+110/-100	150	-300	1210	210	-130	130	-130	490	540	1480
AUTOBRAKE 3	6020	+120/-120	160	-330	1300	150	-120	160	-160	580	240	960

Poor Reported Braking Action

MAX MANUAL	6610	+150/-140	210	-440	1910	580	-340	170	-170	450	1190	3450
MAX AUTO	7030	+150/-140	210	-440	1890	580	-340	170	-170	440	1200	3490
AUTOBRAKE 4	7030	+160/-140	210	-440	1900	560	-340	170	-170	430	1220	3540
AUTOBRAKE 3	7030	+150/-140	210	-450	1920	550	-280	180	-180	570	1110	3440

*Reference distance is for sea level, standard day, no wind or slope, VREF25 approach speed and 2 engine reverse thrust.

Max Manual braking data valid for auto speedbrakes. For manual speedbrakes, increase reference landing distance by 200 ft.

Autobrake data valid for both auto and manual speedbrakes.

Actual (unfactored) distances are shown.

Includes distance from 50 ft above threshold (1000 ft of air distance).

ADVISORY INFORMATION

Normal Configuration Landing Distance

Flaps 20

Dry Runway

	LANDING DISTANCE AND ADJUSTMENTS (FT)											
	REF DIST*	WT ADJ	ALT ADJ	WIND ADJ PER 10 KTS		SLOPE ADJ PER 1%		TEMP ADJ PER 10°C		VREF ADJ	REVERSE THRUST ADJ	
BRAKING CONFIGURATION	400000 LB LANDING WT	PER 10000 LB ABOVE / BELOW 400000 LB	PER 1000 FT ABOVE S.L.	HEAD WIND	TAIL WIND	DN HILL	UP HILL	ABV ISA	BLW ISA	PER 10 KTS ABOVE VREF20	ONE REV	NO REV
MAX MANUAL	2980	+80/-50	60	-120	420	40	-30	60	-60	230	50	110
MAX AUTO	4040	+70/-70	90	-170	590	0	0	100	-100	410	0	0
AUTOBRAKE 4	5030	+100/-100	130	-240	820	0	-20	130	-130	540	0	0
AUTOBRAKE 3	6000	+120/-120	170	-300	1030	30	-60	170	-170	610	10	20
AUTOBRAKE 2	6610	+140/-150	200	-340	1180	90	-150	180	-180	570	150	150
AUTOBRAKE 1	6980	+160/-160	230	-380	1310	170	-210	200	-200	540	560	570

Good Reported Braking Action

MAX MANUAL	4110	+80/-70	110	-200	710	100	-90	100	-100	320	220	530
MAX AUTO	4310	+80/-80	110	-210	730	70	-60	100	-100	390	240	570
AUTOBRAKE 4	5040	+100/-100	130	-240	850	20	-30	130	-130	540	20	90
AUTOBRAKE 3	6000	+120/-120	170	-300	1030	30	-60	170	-170	610	10	20

Medium Reported Braking Action

MAX MANUAL	5500	+120/-110	160	-310	1160	240	-190	140	-140	410	640	1680
MAX AUTO	5500	+120/-110	160	-310	1150	220	-160	140	-140	460	610	1660
AUTOBRAKE 4	5560	+120/-110	160	-310	1160	200	-130	140	-140	510	620	1730
AUTOBRAKE 3	6260	+130/-130	170	-340	1260	130	-120	170	-170	610	260	1160

Poor Reported Braking Action

MAX MANUAL	7000	+170/-160	230	-460	1800	540	-360	180	-180	470	1370	4080
MAX AUTO	7000	+170/-160	230	-450	1790	550	-360	180	-180	460	1380	4130
AUTOBRAKE 4	7000	+170/-160	230	-450	1790	520	-360	180	-180	470	1400	4180
AUTOBRAKE 3	7100	+170/-160	230	-460	1820	500	-300	190	-190	590	1280	4080

*Reference distance is for sea level, standard day, no wind or slope, VREF20 approach speed and 2 engine reverse thrust.

Max Manual braking data valid for auto speedbrakes. For manual speedbrakes, increase reference landing distance by 210 ft.

Autobrake data valid for both auto and manual speedbrakes.

Actual (unfactored) distances are shown.

Includes distance from 50 ft above threshold (1000 ft of air distance).

ADVISORY INFORMATION**Non-Normal Configuration Landing Distance****Dry Runway**

EICAS MESSAGE	VREF	LANDING DISTANCE AND ADJUSTMENTS (FT)							
		REFERENCE DISTANCE FOR 400000 LB LANDING WT	WT ADJUST PER 10000 LB ABOVE/BELOW 400000 LB	ALT ADJUST PER 1000 FT ABOVE S.L.	WIND ADJ PER 10 KTS		SLOPE ADJ PER 1%		APPROACH SPEED
					HEAD WIND	TAIL WIND	DOWN HILL	UP HILL	PER 10 KTS ABOVE VREF
ANTISKID	VREF30	4970	+100/-100	150	-290	1140	230	-170	390
ENGINE SHUTDOWN (FLAPS 20)	VREF20	3010	+80/-50	70	-120	430	40	-40	240
ENGINE SHUTDOWN (FLAPS 30)	VREF30	2750	+70/-40	60	-120	410	40	-30	230
FLAPS DRIVE (FLAPS BETWEEN 5 AND 20)	VREF30+20	3160	+80/-40	70	-120	450	40	-40	220
FLAPS DRIVE (FLAPS GREATER THAN OR EQUAL TO 20)	VREF20	2980	+80/-40	70	-120	430	40	-30	220
FLAPS DRIVE (FLAPS LESS THAN OR EQUAL TO 5)	VREF30+40	3460	+110/-40	90	-140	480	50	-40	250
FLAPS PRIMARY FAIL	VREF20	3340	+70/-50	70	-130	460	40	-40	280
FLAP/SLAT CONTROL	VREF20	2960	+80/-40	60	-120	430	40	-30	220
FLIGHT CONTROL MODE	VREF20	3420	+70/-50	80	-140	470	40	-40	300
HYD PRESS SYS C	VREF20	3340	+70/-50	70	-130	460	40	-40	280
HYD PRESS SYS L+C	VREF30+20	3840	+70/-60	90	-150	520	60	-50	360
HYD PRESS SYS L+R	VREF30+20	3710	+80/-50	90	-150	530	70	-60	370
HYD PRESS SYS R+C	VREF30+20	3950	+70/-60	90	-160	540	70	-60	400
PITCH UP AUTHORITY (FLAPS 5)	VREF30+40	3480	+100/-40	80	-140	480	40	-40	210
PITCH UP AUTHORITY (FLAPS 20)	VREF30+20	3080	+80/-40	70	-120	440	40	-40	210
PRI FLIGHT COMPUTERS	VREF20	3420	+70/-50	80	-140	470	40	-40	300
SLATS DRIVE	VREF30+30	3460	+80/-50	80	-130	460	40	-40	240
STABILIZER	VREF30+20	3210	+80/-50	70	-120	440	40	-40	230

Actual (unfactored) distances are shown.

Includes distance from 50 ft above threshold to stop (1000 ft of air distance).

Assumes maximum manual braking and maximum available reverse thrust.

ADVISORY INFORMATION**Non-Normal Configuration Landing Distance****Good Reported Braking Action**

EICAS MESSAGE	VREF	LANDING DISTANCE AND ADJUSTMENTS (FT)							
		REFERENCE DISTANCE FOR 400000 LB LANDING WT	WT ADJUST PER 10000 LB ABOVE/ BELOW 400000 LB	ALT ADJUST PER 1000 FT ABOVE S.L.	WIND ADJ PER 10 KTS		SLOPE ADJ PER 1%		APPROACH SPEED
					HEAD WIND	TAIL WIND	DOWN HILL	UP HILL	PER 10 KTS ABOVE VREF
ANTISKID	VREF30	4970	+100/-100	150	-290	1140	230	-170	390
ENGINE SHUTDOWN (FLAPS 20)	VREF20	4280	+80/-80	110	-210	740	120	-100	350
ENGINE SHUTDOWN (FLAPS 30)	VREF30	3840	+70/-70	100	-200	720	100	-90	330
FLAPS DRIVE (FLAPS BETWEEN 5 AND 20)	VREF30+ 20	4400	+70/-70	120	-200	740	100	-90	320
FLAPS DRIVE (FLAPS GREATER THAN OR EQUAL TO 20)	VREF20	4110	+80/-80	110	-200	720	100	-80	310
FLAPS DRIVE (FLAPS LESS THAN OR EQUAL TO 5)	VREF30+ 40	4840	+80/-80	140	-210	770	110	-90	300
FLAPS PRIMARY FAIL	VREF20	4540	+80/-80	120	-210	770	110	-100	380
FLAP/SLAT CONTROL	VREF20	4070	+70/-70	110	-200	710	100	-80	310
FLIGHT CONTROL MODE	VREF20	4670	+90/-90	120	-220	780	120	-100	410
HYD PRESS SYS C	VREF20	4540	+80/-80	120	-210	770	110	-100	380
HYD PRESS SYS L+C	VREF30+ 20	5510	+90/-90	160	-260	910	180	-150	510
HYD PRESS SYS L+R	VREF30+ 20	5740	+90/-90	170	-290	990	250	-200	580
HYD PRESS SYS R+C	VREF30+ 20	5650	+100/-100	160	-260	920	190	-160	550
PITCH UP AUTHORITY (FLAPS 5)	VREF30+ 40	4860	+80/-80	130	-210	760	100	-90	290
PITCH UP AUTHORITY (FLAPS 20)	VREF30+ 20	4260	+70/-70	110	-200	720	100	-80	290
PRI FLIGHT COMPUTERS	VREF20	4670	+90/-90	120	-220	780	120	-100	410
SLATS DRIVE	VREF30+ 30	4780	+80/-80	130	-220	780	110	-100	330
STABILIZER	VREF30+ 20	4440	+80/-80	120	-200	750	110	-90	320

Actual (unfactored) distances are shown.**Includes distance from 50 ft above threshold to stop (1000 ft of air distance).****Assumes maximum manual braking and maximum available reverse thrust.**

ADVISORY INFORMATION**Non-Normal Configuration Landing Distance****Medium Reported Braking Action**

EICAS MESSAGE	VREF	LANDING DISTANCE AND ADJUSTMENTS (FT)							
		REFERENCE DISTANCE FOR 400000 LB LANDING WT	WT ADJUST PER 10000 LB ABOVE/BELOW 400000 LB	ALT ADJUST PER 1000 FT ABOVE S.L.	WIND ADJ PER 10 KTS		SLOPE ADJ PER 1%		APPROACH SPEED PER 10 KTS ABOVE VREF
					HEAD WIND	TAIL WIND	DOWN HILL	UP HILL	
ANTISKID	VREF30	4970	+100/-100	150	-290	1140	230	-170	390
ENGINE SHUTDOWN (FLAPS 20)	VREF20	6000	+130/-130	180	-340	1260	320	-240	460
ENGINE SHUTDOWN (FLAPS 30)	VREF30	5320	+110/-110	150	-320	1220	280	-210	430
FLAPS DRIVE (FLAPS BETWEEN 5 AND 20)	VREF30+20	5890	+120/-120	180	-310	1200	240	-190	400
FLAPS DRIVE (FLAPS GREATER THAN OR EQUAL TO 20)	VREF20	5500	+120/-120	170	-300	1180	240	-190	400
FLAPS DRIVE (FLAPS LESS THAN OR EQUAL TO 5)	VREF30+40	6530	+130/-130	220	-330	1260	260	-210	400
FLAPS PRIMARY FAIL	VREF20	5990	+130/-130	180	-320	1240	270	-210	470
FLAP/SLAT CONTROL	VREF20	5430	+110/-110	160	-300	1160	230	-180	390
FLIGHT CONTROL MODE	VREF20	6180	+130/-130	190	-330	1260	290	-220	510
HYD PRESS SYS C	VREF20	5990	+130/-130	180	-320	1240	270	-210	470
HYD PRESS SYS L+C	VREF30+20	7740	+150/-150	260	-420	1540	480	-350	660
HYD PRESS SYS L+R	VREF30+20	9010	+160/-160	260	-510	1850	840	-550	800
HYD PRESS SYS R+C	VREF30+20	7900	+150/-150	250	-420	1560	500	-370	690
PITCH UP AUTHORITY (FLAPS 5)	VREF30+40	6490	+120/-120	200	-320	1240	250	-200	370
PITCH UP AUTHORITY (FLAPS 20)	VREF30+20	5690	+110/-110	170	-310	1180	230	-180	370
PRI FLIGHT COMPUTERS	VREF20	6180	+130/-130	190	-330	1260	290	-220	510
SLATS DRIVE	VREF30+30	6340	+120/-120	200	-330	1240	260	-210	410
STABILIZER	VREF30+20	5910	+120/-120	180	-320	1210	250	-200	400

Actual (unfactored) distances are shown.

Includes distance from 50 ft above threshold to stop (1000 ft of air distance).

Assumes maximum manual braking and maximum available reverse thrust.

ADVISORY INFORMATION**Non-Normal Configuration Landing Distance****Poor Reported Braking Action**

EICAS MESSAGE	VREF	LANDING DISTANCE AND ADJUSTMENTS (FT)							
		REFERENCE DISTANCE FOR 400000 LB LANDING WT	WT ADJUST PER 10000 LB ABOVE/ BELOW 400000 LB	ALT ADJUST PER 1000 FT ABOVE S.L.	WIND ADJ PER 10 KTS		SLOPE ADJ PER 1%		APPROACH SPEED
					HEAD WIND	TAIL WIND	DOWN HILL	UP HILL	PER 10 KTS ABOVE VREF
ANTISKID	VREF30	6320	+140/-130	210	-430	1800	570	-340	440
ENGINE SHUTDOWN (FLAPS 20)	VREF20	8040	+190/-190	270	-520	2040	830	-500	560
ENGINE SHUTDOWN (FLAPS 30)	VREF30	7080	+160/-160	220	-480	1980	730	-440	510
FLAPS DRIVE (FLAPS BETWEEN 5 AND 20)	VREF30+ 20	7480	+160/-160	250	-460	1910	600	-360	470
FLAPS DRIVE (FLAPS GREATER THAN OR EQUAL TO 20)	VREF20	7000	+160/-160	230	-450	1840	560	-350	460
FLAPS DRIVE (FLAPS LESS THAN OR EQUAL TO 5)	VREF30+ 40	8320	+180/-180	300	-480	1960	610	-390	480
FLAPS PRIMARY FAIL	VREF20	7530	+180/-180	250	-470	1910	610	-390	530
FLAP/SLAT CONTROL	VREF20	6900	+160/-160	230	-440	1820	550	-350	450
FLIGHT CONTROL MODE	VREF20	7790	+190/-190	270	-480	1950	650	-410	570
HYD PRESS SYS C	VREF20	7530	+180/-180	250	-470	1910	610	-390	530
HYD PRESS SYS L+C	VREF30+ 20	10430	+220/-220	360	-650	2560	1300	-730	760
HYD PRESS SYS L+R	VREF30+ 20	13990	+240/-240	400	-920	3550	3300	-1450	1000
HYD PRESS SYS R+C	VREF30+ 20	10570	+230/-230	370	-660	2580	1340	-750	780
PITCH UP AUTHORITY (FLAPS 5)	VREF30+ 40	8180	+170/-170	280	-470	1950	570	-370	430
PITCH UP AUTHORITY (FLAPS 20)	VREF30+ 20	7220	+160/-160	240	-450	1880	550	-350	430
PRI FLIGHT COMPUTERS	VREF20	7790	+190/-190	270	-480	1950	650	-410	570
SLATS DRIVE	VREF30+ 30	7960	+170/-170	270	-480	1920	600	-390	460
STABILIZER	VREF30+ 20	7450	+160/-160	250	-460	1880	580	-370	460

Actual (unfactored) distances are shown.

Includes distance from 50 ft above threshold to stop (1000 ft of air distance).

Assumes maximum manual braking and maximum available reverse thrust.



ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Reference Brake Energy (Millions of Foot Pounds)

		BRAKES ON SPEED (KIAS)																							
		80				100				120				140				160				180			
WEIGHT (1000 LB.)	OAT (°C)	PRESSURE ALTITUDE (1000 FT)																							
		0	4	8	0	4	8	0	4	8	0	4	8	0	4	8	0	4	8	0	4	8			
660	0	17.3	19.0	21.1	26.1	29.0	32.5	36.3	40.7	45.9	47.8	53.8	60.8	60.3	68.0	77.0	73.1	82.3	92.8						
	10	17.8	19.6	21.8	26.9	30.0	33.6	37.5	42.1	47.4	49.4	55.6	62.9	62.3	70.2	79.4	75.4	84.9	95.6						
	15	18.1	20.0	22.2	27.4	30.5	34.2	38.2	42.8	48.2	50.2	56.6	63.9	63.4	71.4	80.7	76.7	86.2	97.0						
	20	18.4	20.3	22.5	27.8	31.0	34.7	38.8	43.5	49.0	51.1	57.5	65.0	64.4	72.5	81.9	77.9	87.5	98.4						
	30	18.8	20.7	23.0	28.5	31.8	35.6	39.8	44.7	50.4	52.5	59.1	66.8	66.2	74.6	84.2	80.0	89.8	100.9						
40	19.0	21.0	23.3	29.0	32.3	36.3	40.6	45.6	51.4	53.6	60.4	68.3	67.7	76.3	86.1	81.9	91.9	103.2							
620	0	16.5	18.1	20.0	24.8	27.5	30.8	34.4	38.6	43.4	45.3	50.9	57.6	57.1	64.8	72.9	69.3	78.1	88.2						
	10	17.0	18.6	20.7	25.6	28.4	31.8	35.6	39.8	44.9	46.8	52.6	59.5	59.0	66.5	75.2	71.6	80.6	90.9						
	15	17.2	18.9	21.0	26.0	28.9	32.4	36.2	40.5	45.6	47.6	53.5	60.5	60.0	67.6	76.4	72.7	81.8	92.3						
	20	17.5	19.2	21.3	26.4	29.4	32.9	36.8	41.2	46.4	48.4	54.4	61.5	61.0	68.7	77.5	73.9	83.1	93.6						
	30	17.9	19.7	21.8	27.1	30.1	33.7	37.7	42.3	47.7	49.7	55.9	63.2	62.7	70.6	79.8	75.9	85.4	96.1						
40	18.1	19.9	22.1	27.5	30.6	34.3	38.4	43.1	48.6	50.7	57.1	64.6	64.1	72.2	81.6	77.7	87.3	98.3							
580	0	15.6	17.1	19.0	23.5	26.0	29.1	32.5	36.4	40.9	42.7	48.0	54.2	53.8	60.7	68.6	65.4	73.7	83.4						
	10	16.1	17.7	19.6	24.2	26.9	30.0	33.6	37.6	42.3	44.1	49.6	56.0	55.6	62.7	70.9	67.5	76.1	86.0						
	15	16.4	18.0	19.9	24.6	27.3	30.6	34.2	38.2	43.0	44.9	50.5	57.0	56.6	63.7	72.1	68.7	77.4	87.3						
	20	16.6	18.2	20.2	25.0	27.8	31.0	34.7	38.9	43.7	45.6	51.3	57.9	57.5	64.8	73.2	69.8	78.6	88.6						
	30	17.0	18.6	20.7	25.6	28.4	31.8	35.6	39.9	44.9	46.9	52.7	59.6	59.1	66.6	75.3	71.7	80.7	91.0						
40	17.1	18.8	20.9	26.0	28.9	32.4	36.3	40.6	45.8	47.8	53.8	60.8	60.4	68.1	77.0	73.4	82.6	93.1							
540	0	14.8	16.2	17.9	22.1	24.5	27.4	30.6	34.2	38.4	40.1	45.0	50.8	50.5	56.9	64.3	61.4	69.2	78.3						
	10	15.2	16.7	18.4	22.8	25.3	28.3	31.6	35.3	39.7	41.3	46.5	52.5	52.2	58.8	66.5	63.4	71.5	80.8						
	15	15.5	17.0	18.8	23.2	25.7	28.7	32.2	35.9	40.4	42.2	47.4	53.4	53.1	59.8	67.6	64.5	72.7	82.1						
	20	15.7	17.2	19.0	23.6	26.1	29.2	32.7	36.5	41.1	42.8	48.1	54.3	53.9	60.8	68.7	65.5	73.8	83.4						
	30	16.1	17.6	19.5	24.1	26.8	29.9	33.5	37.5	42.2	44.0	49.5	55.8	55.4	62.5	70.6	67.4	75.9	85.6						
40	16.2	17.8	19.7	24.5	27.2	30.4	34.1	38.2	43.0	44.9	50.5	57.0	56.6	63.8	72.2	68.9	77.6	87.6							
500	0	14.0	15.3	16.8	20.8	23.0	25.7	28.7	32.0	35.9	37.5	42.0	47.4	47.1	53.0	59.9	57.3	64.6	73.1						
	10	14.4	15.7	17.3	21.5	23.7	26.3	29.6	33.0	37.1	38.7	43.4	49.0	48.7	54.8	61.9	59.2	66.7	75.4						
	15	14.6	16.0	17.6	21.8	24.1	26.9	30.1	33.6	37.7	39.4	44.2	49.8	49.5	55.7	63.0	60.2	67.8	76.7						
	20	14.8	16.2	17.9	22.2	24.5	27.4	30.6	34.1	38.3	40.0	44.9	50.7	50.3	56.6	64.0	61.2	68.9	77.9						
	30	15.1	16.6	18.3	22.7	25.1	28.0	31.4	35.0	39.4	41.1	46.1	52.1	51.7	58.2	65.8	62.9	70.9	80.0						
40	15.3	16.7	18.5	23.0	25.5	28.5	31.9	35.6	40.1	41.9	47.1	53.2	52.8	59.5	67.3	64.3	72.5	81.9							
460	0	13.1	14.3	15.8	19.5	21.5	23.9	26.7	29.8	33.4	34.8	39.0	43.9	43.6	49.1	55.4	53.0	59.8	67.6						
	10	13.5	14.8	16.3	20.1	22.2	24.7	27.6	30.7	34.5	36.0	40.3	45.4	45.1	50.7	57.3	54.8	61.8	69.8						
	15	13.7	15.0	16.5	20.3	22.5	25.1	28.1	31.3	35.1	36.6	41.0	46.2	45.9	51.6	58.3	55.7	62.8	71.0						
	20	14.0	15.2	16.8	20.7	22.9	25.5	28.5	31.8	35.6	37.2	41.7	46.9	46.6	52.4	59.2	56.6	63.8	72.1						
	30	14.2	15.6	17.1	21.2	23.4	26.1	29.2	32.6	36.6	38.2	42.8	48.2	47.9	53.9	60.9	58.2	65.6	74.1						
40	14.3	15.7	17.3	21.5	23.8	26.5	29.7	33.1	37.2	38.9	43.6	49.2	48.9	55.0	62.2	59.5	67.1	75.8							
420	0	12.3	13.4	14.7	18.1	20.0	22.2	24.8	27.5	30.8	32.1	35.9	40.4	40.1	45.1	50.9	48.7	54.8	62.0						
	10	12.7	13.8	15.2	18.7	20.6	22.9	25.6	28.4	31.8	33.2	37.1	41.8	41.5	46.6	52.6	50.3	56.6	64.0						
	15	12.9	14.0	15.4	19.0	20.9	23.3	26.0	28.9	32.4	33.8	37.7	42.5	42.2	47.4	53.5	51.1	57.6	65.1						
	20	13.1	14.2	15.7	19.3	21.3	23.7	26.4	29.4	32.9	34.3	38.4	43.2	42.9	48.2	54.4	52.0	58.5	66.2						
	30	13.3	14.5	16.0	19.7	21.8	24.2	27.1	30.1	33.7	35.2	39.4	44.3	44.0	49.5	55.9	53.4	60.2	68.0						
40	13.4	14.7	16.1	20.0	22.0	24.6	27.5	30.6	34.3	35.8	40.1	45.2	44.9	50.5	57.1	54.6	61.5	69.6							

To correct for wind, enter table with the brakes on speed minus one half the headwind or plus 1.5 times the tailwind.

If ground speed is used for brakes on speed, ignore wind and enter table with sea level, 15°C.

ADVISORY INFORMATION

Recommended Brake Cooling Schedule

Event Adjusted Brake Energy (Millions of Foot Pounds)

No Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	6.3	15.6	24.9	34.0	43.2	52.3	61.6	70.9	80.4
	MAX AUTO	6.1	14.4	22.8	31.3	40.0	49.0	58.2	67.7	77.6
	AUTOBRAKE 4	5.9	13.6	21.2	29.1	37.1	45.5	54.2	63.3	73.0
	AUTOBRAKE 3	5.6	12.7	19.7	26.9	34.2	41.9	49.8	58.3	67.3
	AUTOBRAKE 2	5.3	11.8	18.2	24.8	31.5	38.4	45.6	53.3	61.5
	AUTOBRAKE 1	5.2	11.0	16.8	22.8	28.9	35.2	41.9	48.9	56.4

2 Engine Reverse Thrust

		REFERENCE BRAKE ENERGY PER BRAKE (MILLIONS OF FOOT POUNDS)								
EVENT		10	20	30	40	50	60	70	80	90
RTO MAX MAN		10	20	30	40	50	60	70	80	90
LANDING	MAX MAN	5.8	14.7	23.4	32.0	40.4	48.8	57.2	65.6	74.2
	MAX AUTO	4.3	11.6	18.9	26.4	34.1	42.0	50.2	58.9	68.0
	AUTOBRAKE 4	3.6	9.2	15.0	21.1	27.5	34.4	41.7	49.6	58.2
	AUTOBRAKE 3	2.5	6.6	11.1	15.9	21.0	26.7	32.8	39.5	46.8
	AUTOBRAKE 2	1.4	4.4	7.6	11.3	15.3	19.7	24.5	29.8	35.6
	AUTOBRAKE 1	1.0	3.0	5.3	7.8	10.6	13.9	17.5	21.7	26.4

Cooling Time (Minutes)

		EVENT ADJUSTED BRAKE ENERGY (MILLIONS OF FOOT POUNDS)									
		16 & BELOW	17	18	20	24	28	32	35	36 TO 44	45 & ABOVE
GEAR DOWN	NO SPECIAL		1	2	3	4	6	7	7	CAUTION	FUSE PLUG MELT ZONE
INFLIGHT	PROCEDURE										
GROUND	REQUIRED		11	18	26	42	55	66	73		
BTMS	UP TO 2.4		2.4	2.6	2.9	3.4	4.0	4.5	4.9	5.0 TO 6.3	6.3 & ABOVE

Observe maximum quick turnaround limit.

Table shows energy per brake added by a single stop with all brakes operating. Energy is assumed to be equally distributed among the operating brakes. Total energy is the sum of residual energy plus energy added.

Add 1.0 million foot pounds for each taxi mile.

For one brake deactivated, increase brake energy by 10 percent.

For two brakes deactivated, increase brake energy by 20 percent.

When in caution zone, wheel fuse plugs may melt. Delay takeoff and inspect after one hour. If overheat occurs after takeoff, extend gear soon for at least 8 minutes.

When in fuse plug melt zone, clear runway immediately. Unless required, do not set parking brake. Do not attempt to taxi for one hour. Tire, wheel and brake replacement may be required. If overheat occurs after takeoff, extend gear soon for at least 12 minutes.

Brake temperature monitor system (BTMS) indication on EICAS may be used 10 to 15 minutes after airplane has come to a complete stop, or inflight with gear retracted, to determine recommended cooling schedule.

(When inflight with gear extended, the BTMS indications may vary between individual brakes, due to air-stream effects.)

ADVISORY INFORMATION**Landing Climb Limit Weight****Valid for approach with flaps 20 and landing with flaps 25 or 30**

AIRPORT OAT		LANDING CLIMB LIMIT WEIGHT (1000 LB)					
		AIRPORT PRESSURE ALTITUDE (FT)					
(°C)	(°F)	-2000	0	2000	4000	6000	8000
54	129	570.0	537.3				
52	126	583.3	550.2				
50	122	597.2	562.9	520.9			
48	118	611.1	575.4	533.6			
46	115	625.2	587.8	545.7	502.4		
44	111	638.9	601.5	557.7	512.0		
42	108	651.9	615.9	569.4	521.5	484.3	
40	104	663.8	629.6	580.7	531.0	492.5	
38	100	673.3	641.1	592.5	540.0	500.1	462.2
36	97	682.2	658.2	604.0	549.1	507.7	469.4
34	93	691.0	669.1	613.7	558.0	515.2	476.4
32	90	691.2	678.9	623.1	566.4	523.0	483.5
30	86	691.3	686.2	632.1	574.8	531.1	490.5
28	82	691.5	688.7	640.7	583.1	539.0	497.9
26	79	691.6	688.8	650.1	590.8	546.8	505.5
24	75	691.7	688.9	655.3	598.0	554.4	511.9
22	72	691.9	689.0	655.5	603.9	559.0	516.1
20	68	692.0	689.1	655.6	606.3	561.3	519.4
18	64	692.0	689.2	655.9	606.4	563.6	523.0
16	61	692.0	689.3	656.1	606.4	564.8	526.9
14	57	692.1	689.4	656.3	606.6	564.9	529.7
12	54	692.2	689.6	656.5	606.7	564.9	530.6
10	50	692.3	689.7	656.7	606.8	565.0	530.7
-40	-40	694.9	692.7	659.6	609.5	567.5	532.9

Based on engine bleed for 2 packs on, engine anti-ice on or off and wing anti-ice off.

With engine bleed for packs off, increase weight by 2000 lb.

With engine and wing anti-ice on, decrease weight by 3400 lb.

When operating in icing conditions during any part of the flight with forecast landing temperature below 10°C, decrease weight by 45400 lb.

Intentionally
Blank

Performance Inflight A

One Engine Inoperative

Chapter PI

Section A11

ENGINE INOP

Initial Max Continuous EPR

Based on .84M, engine bleed for packs on and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)								
	27	29	31	33	35	37	39	41	43
20 & ABOVE	1.243	1.234	1.226	1.211	1.192	1.183	1.181	1.175	1.170
15	1.284	1.275	1.263	1.250	1.232	1.222	1.220	1.216	1.212
10	1.327	1.319	1.307	1.288	1.273	1.264	1.262	1.259	1.257
5	1.364	1.366	1.354	1.337	1.314	1.306	1.304	1.302	1.300
0	1.364	1.402	1.406	1.388	1.368	1.355	1.354	1.351	1.349
-5	1.364	1.402	1.439	1.444	1.424	1.412	1.411	1.408	1.406
-10	1.364	1.402	1.439	1.473	1.482	1.471	1.469	1.467	1.465
-15	1.364	1.402	1.439	1.473	1.506	1.523	1.523	1.521	1.518
-20 & BELOW	1.364	1.402	1.439	1.473	1.506	1.523	1.523	1.521	1.518

ENGINE INOP

Max Continuous EPR

Based on engine bleed for packs on or off and anti-ice off

37000 FT to 29000 FT Pressure Altitudes

37000 FT PRESS ALT			TAT (°C)										
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
200	0.63	1.531	1.531	1.531	1.531	1.531	1.489	1.436	1.385	1.341	1.302	1.263	1.233
240	0.74	1.534	1.534	1.534	1.534	1.534	1.534	1.517	1.455	1.402	1.349	1.308	1.268
280	0.86	1.506	1.506	1.506	1.506	1.506	1.506	1.506	1.506	1.469	1.413	1.356	1.304
35000 FT PRESS ALT			TAT (°C)										
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
200	0.60	1.517	1.517	1.517	1.517	1.517	1.486	1.433	1.385	1.341	1.304	1.267	1.234
240	0.71	1.514	1.514	1.514	1.514	1.514	1.514	1.502	1.444	1.394	1.344	1.305	1.266
280	0.82	1.519	1.519	1.519	1.519	1.519	1.519	1.519	1.519	1.476	1.417	1.362	1.311
33000 FT PRESS ALT			TAT (°C)										
CIAS	M	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5
200	0.58	1.499	1.499	1.499	1.499	1.499	1.499	1.447	1.401	1.358	1.323	1.290	1.257
240	0.68	1.491	1.491	1.491	1.491	1.491	1.491	1.491	1.452	1.402	1.355	1.314	1.278
280	0.79	1.489	1.489	1.489	1.489	1.489	1.489	1.489	1.489	1.473	1.414	1.363	1.314
320	0.89	1.427	1.427	1.427	1.427	1.427	1.427	1.427	1.427	1.427	1.427	1.389	1.338
31000 FT PRESS ALT			TAT (°C)										
CIAS	M	-45	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10
200	0.55	1.480	1.480	1.480	1.480	1.480	1.461	1.414	1.372	1.332	1.303	1.274	1.245
240	0.66	1.471	1.471	1.471	1.471	1.471	1.471	1.461	1.410	1.365	1.322	1.291	1.260
280	0.76	1.459	1.459	1.459	1.459	1.459	1.459	1.459	1.459	1.415	1.366	1.321	1.281
320	0.85	1.426	1.426	1.426	1.426	1.426	1.426	1.426	1.426	1.426	1.406	1.353	1.306
29000 FT PRESS ALT			TAT (°C)										
CIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
200	0.53	1.493	1.493	1.493	1.493	1.493	1.454	1.411	1.369	1.334	1.302	1.270	1.252
240	0.63	1.475	1.475	1.475	1.475	1.475	1.475	1.441	1.395	1.352	1.315	1.286	1.255
280	0.73	1.446	1.446	1.446	1.446	1.446	1.446	1.446	1.426	1.378	1.334	1.293	1.259
320	0.82	1.413	1.413	1.413	1.413	1.413	1.413	1.413	1.413	1.413	1.362	1.317	1.273
360	0.91	1.351	1.351	1.351	1.351	1.351	1.351	1.351	1.351	1.351	1.351	1.330	1.284

EPR Adjustments for Engine Bleed

ANTI-ICE CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	37	35	33	31	29
ENGINE ONLY	-0.004	-0.003	-0.003	-0.003	-0.003
ENGINE & WING*	-0.015	-0.014	-0.013	-0.012	-0.012
ENGINE & WING**	-0.026	-0.024	-0.022	-0.021	-0.020

* Wing anti-ice on, packs on.

**Wing anti-ice on, packs off.

ENGINE INOP

Max Continuous EPR

Based on engine bleed for packs on or off and anti-ice off

27000 FT to 20000 Pressure Altitudes

27000 FT PRESS ALT		TAT (°C)											
CIAS	M	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15
200	0.51	1.507	1.507	1.507	1.507	1.507	1.499	1.452	1.409	1.368	1.336	1.302	1.268
240	0.60	1.488	1.488	1.488	1.488	1.488	1.488	1.483	1.434	1.391	1.349	1.317	1.286
280	0.70	1.443	1.443	1.443	1.443	1.443	1.443	1.443	1.443	1.398	1.355	1.314	1.279
320	0.79	1.395	1.395	1.395	1.395	1.395	1.395	1.395	1.395	1.395	1.368	1.323	1.282
360	0.88	1.330	1.330	1.330	1.330	1.330	1.330	1.330	1.330	1.330	1.330	1.325	1.281
25000 FT PRESS ALT		TAT (°C)											
CIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
200	0.49	1.521	1.521	1.521	1.521	1.521	1.496	1.450	1.407	1.366	1.337	1.303	1.278
240	0.58	1.502	1.502	1.502	1.502	1.502	1.502	1.477	1.431	1.389	1.350	1.316	1.282
280	0.67	1.450	1.450	1.450	1.450	1.450	1.450	1.450	1.430	1.384	1.344	1.305	1.273
320	0.76	1.386	1.386	1.386	1.386	1.386	1.386	1.386	1.386	1.380	1.335	1.296	1.257
360	0.85	1.319	1.319	1.319	1.319	1.319	1.319	1.319	1.319	1.319	1.319	1.288	1.249
24000 FT PRESS ALT		TAT (°C)											
CIAS	M	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20
200	0.48	1.523	1.523	1.523	1.523	1.523	1.512	1.465	1.426	1.379	1.346	1.312	1.278
240	0.57	1.502	1.502	1.502	1.502	1.502	1.502	1.490	1.443	1.399	1.357	1.324	1.289
280	0.66	1.455	1.455	1.455	1.455	1.455	1.455	1.455	1.447	1.400	1.358	1.316	1.283
320	0.75	1.392	1.392	1.392	1.392	1.392	1.392	1.392	1.392	1.392	1.350	1.309	1.269
360	0.83	1.324	1.324	1.324	1.324	1.324	1.324	1.324	1.324	1.324	1.324	1.300	1.259
22000 FT PRESS ALT		TAT (°C)											
CIAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
200	0.46	1.523	1.523	1.523	1.523	1.523	1.496	1.449	1.404	1.364	1.330	1.295	1.271
240	0.55	1.505	1.505	1.505	1.505	1.505	1.505	1.473	1.426	1.380	1.341	1.307	1.273
280	0.63	1.463	1.463	1.463	1.463	1.463	1.463	1.463	1.433	1.387	1.342	1.302	1.269
320	0.72	1.407	1.407	1.407	1.407	1.407	1.407	1.407	1.407	1.385	1.340	1.297	1.255
360	0.80	1.336	1.336	1.336	1.336	1.336	1.336	1.336	1.336	1.336	1.329	1.287	1.246
20000 FT PRESS ALT		TAT (°C)											
CIAS	M	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25
200	0.44	1.519	1.519	1.519	1.519	1.519	1.519	1.477	1.429	1.383	1.346	1.311	1.275
240	0.53	1.512	1.512	1.512	1.512	1.512	1.512	1.509	1.461	1.413	1.366	1.329	1.294
280	0.61	1.469	1.469	1.469	1.469	1.469	1.469	1.469	1.465	1.417	1.368	1.321	1.287
320	0.69	1.422	1.422	1.422	1.422	1.422	1.422	1.422	1.422	1.422	1.376	1.329	1.283
360	0.77	1.350	1.350	1.350	1.350	1.350	1.350	1.350	1.350	1.350	1.350	1.317	1.274

EPR Adjustments for Engine Bleed

ANTI-ICE CONFIGURATION	PRESSURE ALTITUDE (1000 FT)				
	27	25	24	22	20
ENGINE ONLY	-0.004	-0.005	-0.006	-0.007	-0.006
ENGINE & WING*	-0.012	-0.012	-0.012	-0.013	-0.012
ENGINE & WING**	-0.019	-0.019	-0.019	-0.019	-0.018

* Wing anti-ice on, packs on.

**Wing anti-ice on, packs off.

ENGINE INOP

Max Continuous EPR

Based on engine bleed for packs on or off and anti-ice off

18000 FT to 5000 FT Pressure Altitudes

18000 FT PRESS ALT													TAT (°C)	
KLAS	M	-25	-20	-15	-10	-5	0	5	10	15	20	25	30	
200	0.42	1.511	1.511	1.511	1.511	1.511	1.500	1.455	1.411	1.369	1.337	1.305	1.275	
240	0.51	1.515	1.515	1.515	1.515	1.515	1.515	1.495	1.449	1.403	1.362	1.328	1.293	
280	0.59	1.472	1.472	1.472	1.472	1.472	1.472	1.472	1.447	1.400	1.354	1.313	1.282	
320	0.67	1.424	1.424	1.424	1.424	1.424	1.424	1.424	1.424	1.402	1.356	1.311	1.269	
360	0.75	1.362	1.362	1.362	1.362	1.362	1.362	1.362	1.362	1.362	1.349	1.305	1.262	
16000 FT PRESS ALT													TAT (°C)	
KLAS	M	-25	-20	-15	-10	-5	0	5	10	15	20	25	30	
200	0.41	1.504	1.504	1.504	1.504	1.504	1.504	1.479	1.438	1.396	1.360	1.332	1.304	
240	0.49	1.511	1.511	1.511	1.511	1.511	1.511	1.511	1.477	1.432	1.389	1.354	1.322	
280	0.57	1.474	1.474	1.474	1.474	1.474	1.474	1.474	1.474	1.432	1.387	1.344	1.310	
320	0.64	1.427	1.427	1.427	1.427	1.427	1.427	1.427	1.427	1.427	1.384	1.340	1.296	
360	0.72	1.372	1.372	1.372	1.372	1.372	1.372	1.372	1.372	1.372	1.372	1.337	1.294	
14000 FT PRESS ALT													TAT (°C)	
KLAS	M	-20	-15	-10	-5	0	5	10	15	20	25	30	35	
200	0.39	1.488	1.488	1.488	1.488	1.488	1.488	1.452	1.413	1.373	1.343	1.315	1.288	
240	0.47	1.491	1.491	1.491	1.491	1.491	1.491	1.484	1.444	1.402	1.361	1.330	1.298	
280	0.54	1.463	1.463	1.463	1.463	1.463	1.463	1.463	1.449	1.407	1.365	1.324	1.293	
320	0.62	1.417	1.417	1.417	1.417	1.417	1.417	1.417	1.417	1.400	1.357	1.315	1.275	
360	0.69	1.368	1.368	1.368	1.368	1.368	1.368	1.368	1.368	1.368	1.354	1.313	1.272	
12000 FT PRESS ALT													TAT (°C)	
KLAS	M	-15	-10	-5	0	5	10	15	20	25	30	35	40	
200	0.38	1.475	1.475	1.475	1.475	1.475	1.468	1.431	1.393	1.355	1.329	1.302	1.276	
240	0.45	1.473	1.473	1.473	1.473	1.473	1.473	1.453	1.415	1.375	1.339	1.308	1.277	
280	0.52	1.452	1.452	1.452	1.452	1.452	1.452	1.452	1.424	1.385	1.344	1.308	1.275	
320	0.60	1.407	1.407	1.407	1.407	1.407	1.407	1.407	1.407	1.373	1.333	1.292	1.257	
360	0.67	1.359	1.359	1.359	1.359	1.359	1.359	1.359	1.359	1.359	1.327	1.288	1.248	
10000 FT PRESS ALT													TAT (°C)	
KLAS	M	-15	-10	-5	0	5	10	15	20	25	30	35	40	
200	0.36	1.462	1.462	1.462	1.462	1.462	1.462	1.444	1.408	1.371	1.338	1.311	1.284	
240	0.43	1.452	1.452	1.452	1.452	1.452	1.452	1.452	1.421	1.383	1.345	1.313	1.283	
280	0.51	1.438	1.438	1.438	1.438	1.438	1.438	1.438	1.433	1.397	1.358	1.318	1.286	
320	0.58	1.392	1.392	1.392	1.392	1.392	1.392	1.392	1.392	1.382	1.344	1.305	1.265	
360	0.65	1.344	1.344	1.344	1.344	1.344	1.344	1.344	1.344	1.344	1.332	1.295	1.258	
5000 FT PRESS ALT													TAT (°C)	
KLAS	M	-10	-5	0	5	10	15	20	25	30	35	40	45	
200	0.33	1.422	1.422	1.422	1.422	1.422	1.422	1.422	1.399	1.367	1.334	1.306	1.282	
240	0.40	1.400	1.400	1.400	1.400	1.400	1.400	1.400	1.394	1.361	1.328	1.295	1.269	
280	0.46	1.382	1.382	1.382	1.382	1.382	1.382	1.382	1.382	1.365	1.330	1.295	1.263	
320	0.53	1.349	1.349	1.349	1.349	1.349	1.349	1.349	1.349	1.349	1.323	1.288	1.253	
360	0.59	1.303	1.303	1.303	1.303	1.303	1.303	1.303	1.303	1.303	1.303	1.270	1.237	

EPR Adjustments for Engine Bleed

ANTI-ICE CONFIGURATION	PRESSURE ALTITUDE (1000 FT)					
	18	16	14	12	10	5
ENGINE ONLY	-0.010	-0.013	-0.015	-0.015	-0.015	-0.010
ENGINE & WING*	-0.015	-0.018	-0.019	-0.019	-0.018	-0.012
ENGINE & WING**	-0.021	-0.024	-0.024	-0.023	-0.022	-0.014

* Wing anti-ice on, packs on.

**Wing anti-ice on, packs off.



ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown Speed/Level Off Altitude

100 ft/min residual rate of climb

Includes APU fuel burn

WEIGHT (1000 LB)		OPTIMUM DRIFTDOWN SPEED (KIAS)	LEVEL OFF PRESSURE ALTITUDE (FT)		
START DRIFT DOWN	LEVEL OFF		ISA+10°C & BELOW	ISA+15°C	ISA+20°C
660	639	282	17600	16200	14700
620	600	274	19300	17900	16400
580	562	265	21000	19700	18200
540	523	256	22900	21600	20100
500	485	247	24800	23500	22100
460	446	237	26500	25400	24200
420	407	227	28400	27300	26100
380	367	216	30400	29200	28100
340	328	204	33200	31000	30200

ENGINE INOP

MAX CONTINUOUS THRUST

Driftdown/LRC Cruise Range Capability

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
128	121	115	110	105	100	96	92	88	85	82
258	244	231	220	209	200	191	183	176	169	163
389	367	348	330	314	300	287	275	264	254	244
521	491	465	441	419	400	382	366	351	337	325
653	615	582	552	525	500	478	457	438	421	405
785	739	699	662	630	600	573	548	526	505	486
918	864	816	773	735	700	668	639	613	588	566
1051	989	934	884	840	800	764	730	700	672	646
1184	1114	1051	996	945	900	859	821	787	755	726
1318	1239	1169	1107	1051	1000	954	912	874	838	806
1452	1364	1287	1218	1156	1100	1049	1003	960	921	885
1586	1490	1405	1329	1261	1200	1144	1094	1047	1005	965
1720	1615	1523	1441	1367	1300	1239	1184	1134	1088	1045
1854	1741	1641	1552	1472	1400	1335	1275	1221	1171	1125
1988	1867	1759	1663	1577	1500	1430	1366	1307	1254	1204
2122	1992	1877	1775	1683	1600	1525	1457	1394	1337	1284
2256	2118	1995	1886	1788	1700	1620	1547	1481	1420	1364
2390	2243	2113	1997	1893	1800	1715	1638	1568	1503	1444

Driftdown/Cruise Fuel and Time

AIR DIST (NM)	FUEL REQUIRED (1000 LB)									TIME (HR:MIN)
	WEIGHT AT START OF DRIFT DOWN (1000 LB)									
	340	380	420	460	500	540	580	620	660	
100	2.1	2.2	2.4	2.6	2.7	2.9	3.1	3.2	3.4	0:13
200	4.7	5.0	5.4	5.9	6.3	6.8	7.4	7.6	8.1	0:27
300	7.5	8.0	8.7	9.5	10.3	11.1	11.9	12.4	13.2	0:41
400	10.2	11.0	12.0	13.1	14.2	15.2	16.4	17.2	18.4	0:56
500	12.8	13.9	15.2	16.5	17.9	19.3	20.7	21.9	23.4	1:10
600	15.4	16.8	18.4	20.0	21.7	23.4	25.1	26.6	28.3	1:25
700	17.9	19.7	21.5	23.4	25.4	27.4	29.4	31.2	33.2	1:40
800	20.5	22.5	24.6	26.9	29.1	31.4	33.7	35.8	38.1	1:55
900	23.0	25.3	27.7	30.2	32.8	35.3	37.9	40.3	43.0	2:10
1000	25.5	28.1	30.8	33.6	36.4	39.3	42.2	44.9	47.8	2:25
1100	28.0	30.9	33.9	37.0	40.1	43.2	46.4	49.4	52.6	2:40
1200	30.5	33.7	36.9	40.3	43.7	47.1	50.6	53.8	57.4	2:55
1300	33.0	36.4	40.0	43.6	47.3	51.0	54.7	58.3	62.2	3:10
1400	35.4	39.2	43.0	46.9	50.8	54.8	58.9	62.7	66.9	3:26
1500	37.9	41.9	46.0	50.2	54.4	58.7	63.0	67.1	71.6	3:41
1600	40.3	44.6	49.0	53.4	57.9	62.5	67.1	71.5	76.3	3:56
1700	42.7	47.3	51.9	56.7	61.4	66.3	71.2	75.9	80.9	4:11
1800	45.1	49.9	54.9	59.9	64.9	70.0	75.2	80.2	85.5	4:27

APU fuel included.

Driftdown at optimum driftdown speed and cruise at LRC speed.



ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Altitude Capability
100 ft/min residual rate of climb

WEIGHT (1000 LB)	PRESSURE ALTITUDE (FT)		
	ISA+10°C & BELOW	ISA+15°C	ISA+20°C
660	15000	12700	10400
640	15600	13500	11100
620	16300	14300	11900
600	16900	15100	12700
580	17700	16100	14100
560	18700	17100	15400
540	19800	18200	16900
520	20700	19300	17500
500	21700	20300	18600
480	22700	21400	19700
460	23700	22400	20900
440	24800	23500	22100
420	25900	24700	23300
400	27100	25800	24600
380	28300	27000	25800
360	29500	28200	26900
340	30800	29500	28100
320	32400	30800	29400
300	34100	32500	30700

ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Control

WEIGHT (1000 LB)		PRESSURE ALTITUDE (1000 FT)							
		10	15	17	19	21	23	25	27
660	EPR	1.258	1.365						
	MACH	.585	.641						
	KIAS	325	325						
	FF/ENG	19637	20013						
620	EPR	1.237	1.335	1.385					
	MACH	.585	.641	.665					
	KIAS	325	325	325					
	FF/ENG	18893	19190	19415					
580	EPR	1.217	1.302	1.349	1.408				
	MACH	.580	.630	.655	.684				
	KIAS	322	320	320	322				
	FF/ENG	18013	18059	18238	18696				
540	EPR	1.194	1.268	1.308	1.356	1.420			
	MACH	.563	.611	.633	.659	.689			
	KIAS	313	310	309	310	312			
	FF/ENG	16696	16656	16733	16922	17428			
500	EPR	1.172	1.237	1.271	1.312	1.362	1.428		
	MACH	.545	.592	.612	.635	.661	.692		
	KIAS	302	300	298	298	299	301		
	FF/ENG	15394	15334	15331	15411	15601	16141		
460	EPR	1.151	1.209	1.238	1.272	1.314	1.365	1.433	
	MACH	.526	.572	.592	.612	.635	.661	.693	
	KIAS	291	289	288	287	286	287	289	
	FF/ENG	14112	14046	14023	14021	14093	14273	14753	
420	EPR	1.131	1.181	1.207	1.236	1.270	1.312	1.363	1.433
	MACH	.505	.551	.570	.590	.610	.633	.660	.692
	KIAS	279	278	277	276	275	274	275	277
	FF/ENG	12840	12771	12750	12729	12724	12782	12940	13370
380	EPR	1.111	1.155	1.177	1.203	1.232	1.265	1.307	1.358
	MACH	.482	.527	.546	.566	.586	.606	.629	.656
	KIAS	267	266	265	264	263	262	261	262
	FF/ENG	11570	11516	11489	11470	11450	11441	11479	11605
340	EPR	1.092	1.133	1.150	1.171	1.196	1.224	1.257	1.297
	MACH	.459	.502	.520	.539	.559	.579	.600	.623
	KIAS	254	253	252	251	251	250	249	248
	FF/ENG	10328	10268	10248	10224	10206	10188	10171	10187
300	EPR	1.075	1.106	1.123	1.142	1.162	1.185	1.213	1.244
	MACH	.434	.474	.492	.510	.530	.550	.570	.591
	KIAS	240	238	238	238	237	236	235	234
	FF/ENG	9133	9031	9011	8995	8976	8955	8938	8918



ENGINE INOP

MAX CONTINUOUS THRUST

Long Range Cruise Diversion Fuel and Time

Ground to Air Miles Conversion

AIR DISTANCE (NM)					GROUND DISTANCE (NM)	AIR DISTANCE (NM)				
HEADWIND COMPONENT (KTS)						TAILWIND COMPONENT (KTS)				
100	80	60	40	20		20	40	60	80	100
282	261	242	226	213	200	191	182	174	167	160
564	522	484	452	425	400	381	364	348	334	321
846	783	727	680	638	600	572	546	522	501	481
1130	1045	970	906	850	800	763	728	696	667	641
1415	1309	1214	1133	1063	1000	953	909	870	833	801
1700	1572	1459	1361	1276	1200	1144	1091	1044	1000	961
1987	1837	1703	1589	1489	1400	1334	1273	1217	1166	1120
2275	2102	1948	1817	1703	1600	1525	1455	1391	1332	1279
2564	2368	2194	2045	1916	1800	1715	1636	1564	1498	1438

Reference Fuel and Time Required at Check Point

AIR DIST (NM)	PRESSURE ALTITUDE (1000 FT)							
	10		14		18		22	
	FUEL (1000 LB)	TIME (HR:MIN)	FUEL (1000 LB)	TIME (HR:MIN)	FUEL (1000 LB)	TIME (HR:MIN)	FUEL (1000 LB)	TIME (HR:MIN)
200	7.9	0:38	6.8	0:37	6.1	0:36	5.4	0:35
400	16.8	1:13	15.3	1:10	14.1	1:07	13.1	1:04
600	25.7	1:48	23.6	1:44	22.0	1:39	20.7	1:34
800	34.4	2:24	31.9	2:17	29.8	2:11	28.2	2:04
1000	43.0	2:59	40.1	2:51	37.6	2:43	35.6	2:34
1200	51.6	3:35	48.1	3:25	45.3	3:15	42.9	3:04
1400	60.0	4:12	56.1	3:59	52.9	3:47	50.2	3:35
1600	68.4	4:48	64.1	4:24	60.4	4:20	57.4	4:05
1800	76.6	5:25	71.9	5:08	67.8	4:53	64.5	4:36

Fuel Required Adjustment (1000 LB)

REFERENCE FUEL REQUIRED (1000 LB)	WEIGHT AT CHECK POINT (1000 LB)				
	300	400	500	600	700
5	-0.7	-0.4	0.0	0.6	1.5
10	-2.1	-1.1	0.0	1.7	3.5
15	-3.4	-1.7	0.0	2.7	5.5
20	-4.6	-2.3	0.0	3.7	7.4
25	-5.8	-2.9	0.0	4.6	9.4
30	-7.1	-3.5	0.0	5.6	11.3
35	-8.3	-4.1	0.0	6.5	13.2
40	-9.5	-4.7	0.0	7.4	15.1
45	-10.8	-5.3	0.0	8.3	16.9
50	-12.0	-5.9	0.0	9.2	18.7
55	-13.2	-6.5	0.0	10.1	20.5
60	-14.5	-7.1	0.0	10.9	22.3
65	-15.7	-7.7	0.0	11.7	24.0
70	-16.9	-8.3	0.0	12.5	25.8
75	-18.1	-9.0	0.0	13.3	27.5
80	-19.4	-9.6	0.0	14.1	29.1

APU fuel included.

ENGINE INOP

MAX CONTINUOUS THRUST

**Holding
Flaps up**

WEIGHT (1000 LB)		PRESSURE ALTITUDE (FT)					
		1500	5000	10000	15000	20000	25000
660	EPR	1.162	1.199	1.270	1.362		
	KIAS	259	260	261	277		
	FF/ENG	17770	17790	17960	18700		
620	EPR	1.148	1.180	1.244	1.329		
	KIAS	251	252	253	262		
	FF/ENG	16570	16570	16680	17150		
580	EPR	1.134	1.162	1.219	1.299	1.409	
	KIAS	243	244	245	248	269	
	FF/ENG	15390	15400	15440	15690	16690	
540	EPR	1.120	1.145	1.195	1.268	1.364	
	KIAS	234	235	236	237	252	
	FF/ENG	14230	14230	14240	14380	15010	
500	EPR	1.107	1.130	1.173	1.237	1.324	1.464
	KIAS	226	226	227	228	235	259
	FF/ENG	13100	13080	13080	13150	13510	14800
460	EPR	1.093	1.113	1.151	1.207	1.287	1.397
	KIAS	219	219	219	219	220	238
	FF/ENG	12030	11980	11970	11980	12120	12860
420	EPR	1.081	1.097	1.130	1.177	1.246	1.344
	KIAS	213	213	213	213	213	220
	FF/ENG	11010	10930	10890	10890	10950	11290
380	EPR	1.069	1.083	1.110	1.150	1.208	1.294
	KIAS	207	207	207	207	207	207
	FF/ENG	10270	9920	9850	9840	9850	9970
340	EPR	1.058	1.070	1.092	1.125	1.173	1.245
	KIAS	200	200	200	200	200	200
	FF/ENG	9270	9170	8840	8820	8810	8850
300	EPR	1.047	1.057	1.076	1.102	1.141	1.201
	KIAS	192	192	192	192	192	192
	FF/ENG	8290	8200	8010	7920	7900	7810

This table includes 5% additional fuel for holding in a racetrack pattern.



Performance Inflight A

General

Chapter PI

Section A12

Max Climb EPR

Based on engine bleed for packs on and anti-ice off

TAT (°C)	PRESSURE ALTITUDE (1000 FT)/SPEED (IAS OR MACH)									
	0	5	10	15	20	25	30	35	40	43
	310	310	310	310	310	310	310	0.84	0.84	0.84
60	1.132	1.122	1.127	1.146	1.166	1.159	1.152	1.174	1.178	1.170
50	1.168	1.161	1.147	1.146	1.166	1.159	1.152	1.174	1.178	1.170
40	1.209	1.205	1.191	1.184	1.173	1.159	1.152	1.174	1.178	1.170
30	1.217	1.253	1.243	1.237	1.228	1.196	1.157	1.174	1.178	1.170
20	1.217	1.253	1.284	1.298	1.292	1.259	1.228	1.191	1.178	1.170
10	1.217	1.253	1.284	1.329	1.364	1.337	1.307	1.273	1.260	1.257
0	1.217	1.253	1.284	1.329	1.374	1.399	1.402	1.367	1.351	1.348
-10	1.217	1.253	1.284	1.329	1.374	1.399	1.434	1.482	1.468	1.465
-15 & BELOW	1.217	1.253	1.284	1.329	1.374	1.399	1.434	1.506	1.521	1.518

EPR Adjustments for Engine Bleed

BLEED CONFIGURATION	PRESSURE ALTITUDE (1000 FT)									
	0	5	10	15	20	25	30	35	40	43
ENGINE ONLY	-0.008	-0.010	-0.015	-0.015	-0.006	-0.005	-0.003	-0.003	-0.004	-0.005
ENGINE & WING*	-0.010	-0.012	-0.018	-0.020	-0.012	-0.012	-0.011	-0.014	-0.017	-0.019
ENGINE & WING**	-0.012	-0.014	-0.021	-0.025	-0.018	-0.019	-0.020	-0.024	-0.029	-0.033

*Wing anti-ice on, packs on.

**Wing anti-ice on, single bleed source and both packs off.

VREF

WEIGHT (1000 LB)	FLAPS		
	40	25	20
680	169	177	183
600	159	166	172
560	153	161	166
520	148	155	160
480	142	149	154
440	136	142	147
400	129	136	140
360	123	129	133
320	116	121	126

Intentionally
Blank

**Non-Normal Checklists****Chapter NNC****Index****Section Index****A**

ABORTED ENGINE START L, R	0.1,7.1
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U EVACUATION

Condition: **Evacuation is needed.**

PARKING brakeSet

OUTFLOW VALVE switches (Both)..... **MAN**

Outflow valve MANUAL switches (Both) **OPEN**

Hold until the outflow valves are fully open.

FUEL CONTROL switches (Both) **CUTOFF**

Advise the cabin to evacuate.

Advise the tower.

Engine fire switches (Both) **Pull**

APU fire switch**Override and pull**

If an engine or APU fire warning light is illuminated:

Related fire switch **Rotate and hold**

Rotate to the stop and hold for 1 second.





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